

Factor-Adjusted Knockoff Inference for High-Dimensional Time Series

Jie Wei

School of Economics, Huazhong University of Science and Technology

Lan Gao

Department of Business Analytics and Statistics

University of Tennessee, Knoxville

Yuan Ke

Department of Statistics, University of Georgia

Yonghui Zhang

School of Economics, Renmin University of China

Abstract

This paper proposes FAR-Knockoff, a factor-adjusted knockoff procedure for variable selection in high-dimensional time series. The method uses weighted principal components to estimate latent factors, constructs Fixed-X knockoffs for the idiosyncratic components, and estimates a whitening transformation to accommodate serially dependent errors. We establish asymptotic false discovery rate (FDR) control under estimated whitening, imperfect screening, and causal martingale innovations, including conditional heteroskedasticity, without requiring Gaussianity or symmetry. The proofs also provide finite-sample inequalities that isolate the FDR costs of whitening estimation, imperfect screening, and distributional approximation. In the simulation designs examined, FAR-Knockoff has favorable FDR–power performance relative to the implemented competitors. In an application to U.S. Treasury bond risk premia, it produces sparse models and evidence of maturity-specific incremental predictability beyond yield-curve and return-forecasting factors.

Keywords: factor-augmented model, false discovery rate, variable selection, temporal dependence

1 Introduction

The increasing availability of large-scale datasets in economics and finance, such as FRED-MD and FRED-QD (McCracken and Ng, 2016, 2020), as well as the expansive set of firm-level characteristics in empirical asset pricing (the so-called “factor zoo”), has greatly enhanced researchers’ ability to study regression relationships comprehensively. However, given that such data are typically collected for general purposes, it remains essential to identify which variables are truly relevant for a specific task. Accurate variable selection not only simplifies models and improves interpretability but also enhances predictive performance.

A critical aspect of high-dimensional variable selection is reproducibility—ensuring that selected variables reflect genuine population-level importance rather than sample-specific noise. False discovery rate (FDR) control provides a practical solution by limiting the proportion of spurious selections, and a more concise model may improve out-of-sample prediction relative to dense or weakly regularized alternatives, as illustrated in our empirical study. FDR control also offers a less conservative and more interpretable alternative to family-wise error rate control. Traditional FDR procedures, such as that of Benjamini and Hochberg (1995), rely on valid p -values, which are often unavailable in high dimensions or under complex dependence structures (e.g. Candès et al., 2018; Fan et al., 2019).

Knockoff inference offers a flexible and powerful framework for FDR control without relying on p -values. By generating artificial “knockoff” variables that mimic the dependence structure of the original covariates but are constructed to be unrelated to the response, the method enables valid testing through negative controls. Key variants include Fixed-X (Barber and Candès, 2015) and Model-X knockoffs (Candès et al., 2018), whose implementations differ in whether moment matching is performed at the sample or population level.

However, their validity relies on an exchangeability condition, which generally breaks down under serial dependence—a hallmark of time series data. For instance, Chi et al. (2025) show that applying Model-X to time series models with lagged covariates can invalidate the inference.

We propose FAR-Knockoff, a method for variable selection with FDR control in high-dimensional time series. By integrating serial-dependence adjustment and factor modeling, the procedure addresses two limitations of existing knockoff methods in this setting. Our first contribution is to accommodate serial dependence through a generalized least squares (GLS)-style whitening transformation. Because its covariance matrix must be estimated in practice, we derive FDR bounds that explicitly retain the resulting approximation error.

Our principal guarantees are asymptotic. The nonasymptotic inequalities used in their proofs quantify the separate effects of screening, factor and whitening estimation, and score approximation; they should not be read as a general exact finite-sample guarantee under estimated nuisance quantities.

A second challenge is the loss of power in knockoff inference under strong collinearity, which is common in macroeconomic and financial datasets. For example, employment measures across sectors or similar equity factors tend to exhibit high correlations. This hampers Lasso-based statistics, which are sensitive to the penalty level, especially when the so-called irrepresentable condition fails. Intuitively, high correlation can cause knockoff statistics to shrink true signals, reducing power. This issue is also confirmed in our simulations.

To improve power, our second contribution is to incorporate factor adjustment into knockoff inference. We decompose the covariates into common latent factors and idiosyncratic components, constructing knockoffs only for the latter while including the estimated factors as controls in the regression. This strategy reduces collinearity among covariates

and improves power. Unlike previous approaches (e.g. Fan et al., 2020a), our method extracts factors via weighted principal component analysis (PCA), using pre-whitened data to ensure compatibility with the knockoff framework under time series dependence.

To establish FDR control for FAR-Knockoff, we develop guarantees directly under causal martingale innovations and imperfect screening. Building on the insights of Barber et al. (2020), we work with the conditional distribution of the 2κ -dimensional sufficient score (Chen et al., 2024), rather than the full data. The theory accounts jointly for whitening error, martingale coupling error, and the drift generated by signals missed in screening. It also shows that PCA and factor-number estimation do not directly enter conditional Fixed-X exchangeability, although they remain relevant for rates and power.

The rest of this paper is organized as follows. Section 2 introduces the FAR model and the adjustment for serial dependence using weighted PCA. Section 3 presents the FAR-Knockoff procedure for variable selection, including the construction of knockoffs and the associated variable selection strategy. Section 4 establishes the theoretical properties of FAR-Knockoff. Section 5 evaluates the finite-sample performance of the proposed method through simulations. Section 6 illustrates the practical utility of FAR-Knockoff through an empirical application to U.S. bond risk premia prediction. Section 7 concludes. Proofs of the main theorems, additional technical details, and supplementary numerical results are provided in the supplementary material.

1.1 Related Work

Recent work has extended knockoffs beyond i.i.d. data. TSKI uses subsampling and e-value aggregation to accommodate temporal dependence (Chi et al., 2025). FAR-Knockoff instead conditions on the transformed inference design and gives explicit approximation-

sensitive FDR bounds, building on Barber et al. (2020).

Approximate Model-X methods allow a misspecified working covariate law but, in the cited time-series settings, retain temporal independence (Fan et al., 2025a,b). FAR-Knockoff conditions on the realized design and handles serially dependent errors by whitening and score coupling. Factor structure also has a different role than in IPAD, which generates Model-X copies by resampling i.i.d. idiosyncratic errors (Fan et al., 2020c). We use an approximate factor model (Bai, 2003) only to decorrelate the design before Fixed-X construction, following Fan et al. (2020a) and Fan et al. (2022), with FDR control rather than selection consistency as the target.

1.2 Notations

Uppercase Roman letters generally denote matrices (such as X , F , and U), except that Y denotes the response vector. Lowercase Greek letters denote scalars or parameter and error vectors, whose dimensions are stated at first use; a subscripted entry, such as β_j or ε_t , is scalar. Given an index set $\mathcal{S}$ and a matrix X , we use $X_{\mathcal{S}}$ to denote the submatrix of X consisting of columns indexed by $\mathcal{S}$. Define the projection matrix $M_X = I - X(X'X)^{-1}X'$, where I denotes the identity matrix. We write $X \succeq 0$ if X is positive semi-definite. For a set A , let $|A|$ denote its cardinality. For a square matrix A , let $\sigma_{\min}(A)$ and $\sigma_{\max}(A)$ denote its smallest and largest eigenvalues, respectively. For a general matrix B , we use $\|B\|_F = [\text{Tr}(BB')]^{1/2}$ to denote its Frobenius norm, $\|B\| = \sigma_{\max}^{1/2}(B'B)$ to denote its spectral norm, and $\|B\|_{\max} = \max_{i,j} |B_{ij}|$ for its entrywise maximum norm. We denote by $a \vee b$ the maximum of a and b . Throughout the paper, we use indices s and t to denote time periods, and indices i and j to denote cross-sectional units, where the total number of units is N .

2 FAR Model and Serial Dependence Adjustment

2.1 Problem Setup

Consider the high-dimensional linear regression model

$$Y_t = X_t' \beta + \varepsilon_t, \quad \text{for } t = 1, \dots, T, \quad (2.1)$$

where $Y_t \in \mathbb{R}$ is the dependent variable, $X_t = (X_{t1}, \dots, X_{tN})' \in \mathbb{R}^N$ is the vector of covariates (which may include a constant term), $\beta = (\beta_1, \dots, \beta_N)' \in \mathbb{R}^N$ is the coefficient vector, and ε_t is the error term.

The model in (2.1) can also be written in matrix form as

$$Y = X\beta + \varepsilon, \quad (2.2)$$

where $Y = (Y_1, \dots, Y_T)'$, $X = (X_1, \dots, X_T)' \in \mathbb{R}^{T \times N}$, and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_T)'$. We assume that the coefficient vector β is sparse with an unknown support set $\mathcal{S}_0 = \{i : \beta_i \neq 0, 1 \leq i \leq N\}$, and denote its cardinality by $\kappa_0 = |\mathcal{S}_0|$. In addition, we assume that ε is independent of X , satisfies $\mathbb{E}(\varepsilon) = \mathbf{0}$, and has an unknown covariance matrix $\text{Cov}(\varepsilon) = \Sigma_\varepsilon^0$.

When the covariates in X exhibit strong cross-sectional correlations, it becomes challenging to consistently recover the true support set $\mathcal{S}_0$. To address this issue, Fan et al. (2020a) propose decomposing the covariates using a latent factor model:

$$X = F\Lambda' + U, \quad (2.3)$$

where $F = (F_1, \dots, F_T)' \in \mathbb{R}^{T \times r}$ is the matrix of r latent factors, $\Lambda = (\Lambda_1, \dots, \Lambda_N)' \in \mathbb{R}^{N \times r}$ is the factor loading matrix, and $U = (U_1, \dots, U_T)' \in \mathbb{R}^{T \times N}$ is the matrix of idiosyncratic

components, with each $U_t = (U_{t1}, \dots, U_{tN})'$. Under mild regularity conditions and suitable identification restrictions (e.g., $F'F/T = I$ and $\Lambda'\Lambda$ being diagonal), the latent factors F and the idiosyncratic components U can be consistently estimated (Bai, 2003).

Combining the model (2.2) with the factor decomposition (2.3) yields the following factor-augmented regression (FAR) model:

$$Y = F\Lambda'\beta + U\beta + \varepsilon. \quad (2.4)$$

Rather than performing variable selection directly on the covariate matrix X in model (2.1), we focus on the idiosyncratic component U in the FAR model (2.4), which generally exhibits weaker cross-sectional dependence.

Another important challenge that has not been thoroughly addressed in the existing literature is how to control the FDR in variable selection based on the FAR model. If the error terms ε_t were serially uncorrelated, one could extend the seminal knockoff framework (e.g. Barber and Candès, 2015; Candès et al., 2018; Liu et al., 2024) to the FAR setting by constructing knockoff copies of the idiosyncratic components U . However, in many applications in economics, engineering, finance, and public health, the assumption of serial independence is unrealistic. As a result, directly applying existing knockoff constructions may fail to control the FDR, even when F and U were treated as observed. This failure arises because the key *pairwise exchangeability* condition, established in Lemma 3 of Barber and Candès (2015), is violated in the presence of serially correlated errors.

The sparse regression (2.1) is a motivating special case. Our general framework is the factor-augmented sparse regression model (Fan et al., 2024, FARM):

$$Y = F\gamma^* + U\beta^* + \varepsilon \quad \text{and} \quad X = F\Lambda' + U, \quad (2.5)$$

where $\gamma^* \in \mathbb{R}^r$ measures the factor contribution and the sparse vector $\beta^* \in \mathbb{R}^N$ measures the contributions of the idiosyncratic covariates. Equivalently, $Y = F\varphi + X\beta^* + \varepsilon$, where $\varphi = \gamma^* - \Lambda'\beta^*$. Thus FARM augments sparse regression by factor directions and nests the diffusion-index model when $\beta^* = 0$ (Stock and Watson, 2002; Bai and Ng, 2006); it bridges factor-based dimension reduction and sparse regression while adjusting for latent confounding.

Because β_j^* is unchanged in these two representations, selection of U_j targets the coefficient of the original predictor X_j conditional on the factors and the other covariates, as in Fan et al. (2020a). Selected variables therefore identify incremental predictive information beyond the common factors estimated from the full panel, whereas unselected variables may still be informative through those factors.

2.2 Estimation of Error Covariance Matrix

The covariance matrix of the error term (i.e. Σ_ε^0) plays a crucial role in the knockoff construction process, as it helps account for the impact of serial dependence. However, it is well known that the sample covariance matrix estimator performs poorly in high-dimensional settings, and becomes infeasible when the dimensionality exceeds the sample size. In this subsection, we propose to estimate Σ_ε^0 using a factor-augmented screening procedure followed by a banded covariance matrix estimator. This two-step approach improves estimation accuracy by reducing dimensionality and mitigating the effects of strong temporal and cross-sectional dependence.

Denote by $\hat{F}$, $\hat{\Lambda}$, and $\hat{U}$ the estimators of F , Λ , and U , respectively. Following the ideas of Wang (2012) and Fan et al. (2020a), we formulate a series of factor-augmented marginal

regressions:

$$\tilde{\beta}_j = \arg \min_{\beta_j \in \mathbb{R}} \frac{1}{2T} \left\| M_{\hat{F}} Y - \hat{U}_{\cdot,j} \beta_j \right\|_2^2, \quad \text{for } j = 1, 2, \dots, N, \quad (2.6)$$

where $M_{\hat{F}} = I_T - \hat{F}(\hat{F}'\hat{F})^{-1}\hat{F}'$ is the projection matrix onto the orthogonal complement of the column space of $\hat{F}$, and $\hat{U}_{\cdot,j}$ denotes the j th column of $\hat{U}$.

For a prescribed screening threshold ξ , define the screened set

$$\tilde{\mathcal{S}} = \left\{ j : \left| \tilde{\beta}_j \right| \geq \xi \right\}. \quad (2.7)$$

The threshold balances missed signals against over-selection. We do not assume sure screening for FDR control. Let

$$\beta_j^M := \frac{\mathbb{E}\{U_{tj}(U_t'\beta)\}}{\mathbb{E}(U_{tj}^2)}, \quad \mathcal{A} := \mathcal{S}_0 \setminus \tilde{\mathcal{S}}, \quad s := |\mathcal{S}_0|,$$

where β_j^M is the population factor-adjusted marginal coefficient targeted by (2.6).

Assumption 1 (Screening regularity). *There are deterministic $a_{NT} \rightarrow 0$ and $\delta_{NT}^{\text{scr}} \rightarrow 0$ such that, with probability at least $1 - \delta_{NT}^{\text{scr}}$,*

$$\max_{j \leq N} |\tilde{\beta}_j - \beta_j^M| \leq a_{NT}. \quad (2.8)$$

Moreover, $\min_j \mathbb{E}(U_{tj}^2) \geq c > 0$ and $\max_{j \leq N} |\sum_{k \neq j} \mathbb{E}(U_{tj}U_{tk})\beta_k| \leq Ca_{NT}$. Finally, $\xi \geq 2a_{NT}$ and $K_{NT} := |\{j : |\beta_j^M| \geq \xi/2\}| = o(T/m)$, where m is the covariance-banding bandwidth introduced below.

The weighted condition allows nonzero individual cross-sectional correlations; only their beta-weighted aggregate contribution to each marginal coefficient must be small. Over-selection is allowed if the retained model is not too large, while under-selection is allowed

if the missed signals are weak. On the event (2.8), this means

$$\kappa := |\tilde{\mathcal{S}}| \leq K_{NT}, \quad \|\beta_{\mathcal{A}}\|_{\infty} \leq b_{NT}, \quad \|\beta_{\mathcal{A}}\|_1 \leq sb_{NT}, \quad \|\beta_{\mathcal{A}}\|_2 \leq \sqrt{s} b_{NT},$$

where $b_{NT} := \xi + Ca_{NT}$. None of the results below assumes $\mathcal{A} = \emptyset$.

Based on the selected variables, the exact post-screening model is

$$Y = X_{\tilde{\mathcal{S}}} \beta_{\tilde{\mathcal{S}}} + d_0 + \varepsilon, \quad d_0 := X_{\mathcal{A}} \beta_{\mathcal{A}}. \quad (2.9)$$

where $X_{\tilde{\mathcal{S}}}$ is the submatrix of X consisting of the columns whose indices are in $\tilde{\mathcal{S}}$. Let $(\hat{\alpha}, \hat{\beta}'_{\tilde{\mathcal{S}}})'$ be the ordinary least-squares coefficient from regressing Y on the post-screening refit design

$$Z_{\tilde{\mathcal{S}}} = [\mathbf{1}, X_{\tilde{\mathcal{S}}}], \quad P_{\tilde{\mathcal{S}}} = Z_{\tilde{\mathcal{S}}} (Z'_{\tilde{\mathcal{S}}} Z_{\tilde{\mathcal{S}}})^{-1} Z'_{\tilde{\mathcal{S}}}.$$

We estimate the error term by the corresponding OLS residual. Equivalently,

$$\hat{\varepsilon} = Y - \hat{\alpha} \mathbf{1} - X_{\tilde{\mathcal{S}}} \hat{\beta}_{\tilde{\mathcal{S}}} = (I - P_{\tilde{\mathcal{S}}})Y = (I - P_{\tilde{\mathcal{S}}})(\varepsilon + d_0).$$

Thus under-selection enters the covariance estimator through an explicit missed-signal term rather than through an unspecified “modified error.”

To estimate the error covariance matrix Σ_{ε}^0 , we exploit the weak dependence structure of the error terms and consider the following banded estimator:

$$\hat{\Sigma}_{\varepsilon}(r, t) = \hat{\sigma}_{|t-r|}^{(m)}, \quad \text{where} \quad \hat{\sigma}_h^{(m)} = \begin{cases} \frac{1}{T-|h|} \sum_{t=1}^{T-|h|} \hat{\varepsilon}_t \hat{\varepsilon}_{t+h}, & \text{if } |h| \leq m, \\ 0, & \text{if } |h| > m, \end{cases} \quad (2.10)$$

with $m = O(T^\gamma)$ for some $\gamma \in (0, 1)$. Here, $\widehat{\varepsilon}_t$ denotes the residual at time t obtained from (2.9). In our numerical studies, we set $\gamma = 1/5$ so that m grows slowly with the sample size T . The estimator in (2.10) is closely related to the Newey–West (1987) estimator for weakly dependent time series, and can also be combined with thresholding techniques under approximate sparsity as discussed in Fan et al. (2011).

We first impose the high-level conditions needed for covariance estimation. They do not restrict the errors to be Gaussian or symmetric.

Assumption 2 (Covariance regularity). *The error process is stationary, $|\mathbb{E}(\varepsilon_t \varepsilon_{t+h})| \leq C\rho^{|h|}$ for some $\rho \in (0, 1)$, and*

$$\max_{|h| \leq m} \left| \frac{1}{T - |h|} \sum_{t=1}^{T-|h|} \{\varepsilon_t \varepsilon_{t+|h|} - \mathbb{E}(\varepsilon_t \varepsilon_{t+|h|})\} \right| = O_p(T^{-1/2} \ln T).$$

Uniformly over $|S| \leq K_{NT}$, the refit design $Z_S = [\mathbf{1}, X_S]$ is balanced, with $\sigma_{\min}(Z_S' Z_S / T) \geq c$ and $\sigma_{\max}(Z_S' Z_S / T) \leq C$ with probability approaching one. Moreover, uniformly over columns Z_l of $[\mathbf{1}, X]$ and $|h| \leq m$, $T^{-1} |\langle \varepsilon, L_h Z_l \rangle| = O_p(\sqrt{\ln(NT)/T})$, where L_h is the lag- h alignment operator.

Remark 2.1. *Assumption 2 separates the covariance-estimation requirements from the distributional form of the innovations. Its autocovariance concentration condition is satisfied by many weakly dependent causal processes; other covariance estimators may be used if they deliver the same operator-norm rate. The refit condition is a standard sparse-eigenvalue requirement.*

Theorem 2.2 (Whitening under imperfect screening). *Suppose Assumptions 1 and 2 hold. Then*

$$\left\| \widehat{\Sigma}_\varepsilon - \Sigma_\varepsilon^0 \right\| = O_p(\omega_{NT}),$$

where

$$\omega_{NT} := C \left[\rho^m + mT^{-1/2} \ln T + m \frac{(K_{NT} + 1) \ln(NT)}{T} + R_{\text{miss}} \right]$$

and

$$R_{\text{miss}} := m \|\beta_{\mathcal{A}}\|_1 \sqrt{\frac{(K_{NT} + 1) \ln(NT)}{T}} + m \|\beta_{\mathcal{A}}\|_1 \sqrt{\frac{\ln(NT)}{T}} + m \|\beta_{\mathcal{A}}\|_1^2.$$

The four displayed channels are banding bias, covariance sampling error, selection-uniform projection error, and missed-signal contamination. The bound $\|\beta_{\mathcal{A}}\|_1 \leq sb_{NT}$ recovers the simpler but coarser rate stated in terms of sparsity and screening resolution. When $\mathcal{A} = \emptyset$, the last channel vanishes as a special case; it is not assumed in the theory.

2.3 FAR with Serial Dependence Adjustment

Let $\widehat{\Sigma}_\varepsilon$ be the error covariance matrix estimated using the procedure described in Section 2.2. To account for serial dependence in the time series, we propose a transformation by left-multiplying both sides of the post-screening model (2.9) with $\widehat{\Sigma}_\varepsilon^{-1/2}$. Specifically, the exact transformed model is

$$Y^\dagger = X_{\widetilde{\mathcal{S}}}^\dagger \beta_{\widetilde{\mathcal{S}}} + d^\dagger + \varepsilon^\dagger, \quad (2.11)$$

where $Y^\dagger = \widehat{\Sigma}_\varepsilon^{-1/2} Y$, $X_{\widetilde{\mathcal{S}}}^\dagger = \widehat{\Sigma}_\varepsilon^{-1/2} X_{\widetilde{\mathcal{S}}}$, $d^\dagger = \widehat{\Sigma}_\varepsilon^{-1/2} d_0$, and $\varepsilon^\dagger = \widehat{\Sigma}_\varepsilon^{-1/2} \varepsilon$.

Let $\check{F}$ and $\check{\Lambda}_{\widetilde{\mathcal{S}}}$ denote the estimators of the latent factors and factor loadings in the factor model for $X_{\widetilde{\mathcal{S}}}^\dagger$, given by

$$X_{\widetilde{\mathcal{S}}}^\dagger = F^\dagger \Lambda'_{\widetilde{\mathcal{S}}} + U_{\widetilde{\mathcal{S}}}^\dagger,$$

subject to the identification conditions $F^{\dagger'} F^\dagger / T = I_r$ and $\Lambda'_{\widetilde{\mathcal{S}}} \Lambda_{\widetilde{\mathcal{S}}}$ being diagonal.* We then define the estimated idiosyncratic component as $\check{U}_{\widetilde{\mathcal{S}}} = X_{\widetilde{\mathcal{S}}}^\dagger - \check{F} \check{\Lambda}'_{\widetilde{\mathcal{S}}}$. It is worth noting that

*Here we slightly abuse notation: $\Lambda_{\widetilde{\mathcal{S}}}$ refers to the submatrix of Λ consisting of the rows indexed by $\widetilde{\mathcal{S}}$.

$\check{\check{U}}_{\tilde{\mathcal{S}}}$ generally differs from $\widehat{\Sigma}_{\varepsilon}^{-1/2}\widehat{U}_{\tilde{\mathcal{S}}}$, where $\widehat{U}_{\tilde{\mathcal{S}}}$ is obtained from (unweighted) PCA applied directly to X . This discrepancy arises because PCA, in general, does not commute with the transformation induced by left-multiplication with $\widehat{\Sigma}_{\varepsilon}^{-1/2}$.

By substituting $\check{\check{F}}$, $\check{\check{\Lambda}}_{\tilde{\mathcal{S}}}$, and $\check{\check{U}}_{\tilde{\mathcal{S}}}$ into equation (2.11), we obtain the exact factor-adjusted representation

$$Y^{\dagger} = \check{\check{F}}\check{\check{\Lambda}}'_{\tilde{\mathcal{S}}}\beta_{\tilde{\mathcal{S}}} + \check{\check{U}}_{\tilde{\mathcal{S}}}\beta_{\tilde{\mathcal{S}}} + d^{\dagger} + \varepsilon^{\dagger}. \quad (2.12)$$

The procedure below fits the same factor-augmented Lasso as before and does not estimate $d^{\dagger}$ separately. In the validity analysis, $d^{\dagger}$ is a design-measurable drift; Lemma C.2 quantifies its effect. By construction, $\check{\check{F}}$ and $\check{\check{U}}_{\tilde{\mathcal{S}}}$ are orthogonal, i.e., $\check{\check{F}}'\check{\check{U}}_{\tilde{\mathcal{S}}} = 0$. Following the normalization scheme in Barber and Candès (2015), we further normalize each column of $\check{\check{U}}_{\tilde{\mathcal{S}}}$ such that $\|\check{\check{U}}_{\tilde{\mathcal{S}},j}\|_2^2/T = 1$ for every $j \in \tilde{\mathcal{S}}$.

If the true covariance matrix Σ_{ε}^0 were known and used in the transformation, the error in model (2.12) would be serially whitened, simplifying the theoretical justification of false discovery rate (FDR) control via knockoffs. However, in practice, we must estimate Σ_{ε}^0 using $\widehat{\Sigma}_{\varepsilon}$, and the resulting estimation error may inflate the FDR. Fortunately, as we show later, the consistency of $\widehat{\Sigma}_{\varepsilon}$ (in spectral norm) translates into the robustness of FDR control.[†]

3 FAR-Knockoff Procedure for Variable Selection

3.1 Construction of Knockoffs for FAR Model

Since both cross-sectional and serial dependencies have been adjusted in the FAR model (2.12), we can adopt the “Fixed-X” knockoff framework of Barber and Candès (2015) to construct knockoff copies of the covariates $\check{\check{U}}_{\tilde{\mathcal{S}}}$.

[†]In a simulation study not reported here, we implement FAR-Knockoff by replacing $\widehat{\Sigma}_{\varepsilon}$ with the infeasible Σ_{ε}^0 , resulting in similar findings.

By selecting the screening model size to satisfy $\kappa < (T - \hat{r})/2$, where $\hat{r}$ is the realized estimated factor number, we define $\tilde{U}_{\tilde{\mathcal{S}}}$ as the knockoff copy of $\check{U}_{\tilde{\mathcal{S}}}$, such that

$$\frac{1}{T} \begin{bmatrix} \check{U}_{\tilde{\mathcal{S}}} & \tilde{U}_{\tilde{\mathcal{S}}} \end{bmatrix}' \begin{bmatrix} \check{U}_{\tilde{\mathcal{S}}} & \tilde{U}_{\tilde{\mathcal{S}}} \end{bmatrix} = \begin{bmatrix} \Sigma_{\check{U}_{\tilde{\mathcal{S}}}} & \Sigma_{\check{U}_{\tilde{\mathcal{S}}}} - \text{diag}(\nu) \\ \Sigma_{\check{U}_{\tilde{\mathcal{S}}}} - \text{diag}(\nu) & \Sigma_{\check{U}_{\tilde{\mathcal{S}}}} \end{bmatrix} := \Psi, \quad (3.13)$$

where $\Sigma_{\check{U}_{\tilde{\mathcal{S}}}} = \check{U}_{\tilde{\mathcal{S}}}^L \check{U}_{\tilde{\mathcal{S}}} / T$, ν is a nonnegative vector of length κ , $\text{diag}(\nu) \in \mathbb{R}^{\kappa \times \kappa}$ is a diagonal matrix with entries from ν and satisfies $2\Sigma_{\check{U}_{\tilde{\mathcal{S}}}} - \text{diag}(\nu) \succeq 0$.

Equation (3.13), together with the normalization, implies

$$\left\| \check{U}_{\tilde{\mathcal{S}},j} \right\|_2^2 / T = \left\| \tilde{U}_{\tilde{\mathcal{S}},j} \right\|_2^2 / T = 1.$$

Specifically, the knockoff copy $\tilde{U}_{\tilde{\mathcal{S}}}$ can be constructed as

$$\tilde{U}_{\tilde{\mathcal{S}}} = \check{U}_{\tilde{\mathcal{S}}} \left(I - \Sigma_{\check{U}_{\tilde{\mathcal{S}}}}^{-1} \text{diag}(\nu) \right) + VO,$$

where O is a $\kappa \times \kappa$ matrix satisfying $O'O/T = 2\text{diag}(\nu) - \text{diag}(\nu)\Sigma_{\check{U}_{\tilde{\mathcal{S}}}}^{-1}\text{diag}(\nu)$, and V is a $T \times \kappa$ orthonormal matrix such that

$$V'\check{U}_{\tilde{\mathcal{S}}} = 0 \quad \text{and} \quad V'\tilde{U}_{\tilde{\mathcal{S}}} = 0. \quad (3.14)$$

Note that in the original construction of knockoffs, Barber and Candès (2015) require only the orthogonality condition $V'\check{U}_{\tilde{\mathcal{S}}} = 0$. In contrast, our factor-augmented model imposes an additional restriction, namely $V'\tilde{U}_{\tilde{\mathcal{S}}} = 0$, thereby adapting the original Fixed-X knockoff framework to our specific setting. This additional condition is essential for deriving a simple yet informative sufficient statistic, as introduced in Section 4, and is easy

to satisfy because the estimated factor dimension $\hat{r}$ is small. As a result of the double orthogonality condition in (3.14), both the original covariates $\check{U}_{\tilde{s}}$ and their knockoff copies $\tilde{U}_{\tilde{s}}$ are orthogonal to the estimated factors $\check{F}$:

$$\check{F}'\check{U}_{\tilde{s}} = 0 \quad \text{and} \quad \check{F}'\tilde{U}_{\tilde{s}} = 0. \quad (3.15)$$

Remark 3.1. *The construction of $\tilde{U}_{\tilde{s}}$ as the knockoff copy of $\check{U}_{\tilde{s}}$ basically follows the Fixed-X paradigm in matching the empirical moments shown in (3.13). Alternatively, one may consider the Model-X paradigm, given that it can accommodate both linear and nonlinear models. However, Model-X knockoff requires the covariates, $\check{U}_{t,\tilde{s}}$ in our case, are i.i.d. over time, whereas in an approximate factor model (Bai and Ng, 2002; Bai, 2003) as in (2.3), it is too restrictive to assume U_{it} are independent over time for all i .*

Remark 3.2 (Estimated factor number). *We choose IC_{p1} of Bai and Ng (2002) to estimate the number of factors. For FDR validity, the realized factor number and PCA output are part of the conditioned-on design; consistency of $\hat{r}$ is therefore not needed for the null-swap argument itself. It is needed to verify the stated PCA rates and power conditions. This separation is made precise in Section 4.1 and Lemma C.5.*

Remark 3.3 (Always-included and predetermined controls). *Let $Q_t \in \mathbb{R}^{q_0}$, with fixed q_0 , collect controls that are always included but not tested:*

$$Y_t = Q_t'\alpha + X_t'\beta + \varepsilon_t.$$

Under strict exogeneity, after whitening one may partial out $Q^\dagger = \hat{\Sigma}_\varepsilon^{-1/2}Q$ from the response and screened covariates before PCA, and impose $V'Q^\dagger = 0$ in addition to (3.14). The FDR

analysis then applies to the high-dimensional block X after replacing the factor rank r by $r + q_0$.

If Q_t is only predetermined, for example $Q_t = Y_{t-1}$, ordinary full-sample residualization need not preserve the conditioning argument. A valid implementation uses nuisance coefficients estimated on an earlier fold, or a one-sided predictable residualizer, and requires the resulting Q -projected score to satisfy the score-coupling and threshold-stability requirements in Appendix Assumptions C.2–C.3. With the lower-triangular whitening root and $Q_t \in \mathcal{F}_{t-1}$, this preserves the martingale-score route. Appendix C.3 records the score calculation. This distinction is relevant when yield-curve or lagged-response controls are partialled out in the empirical analysis.

3.2 FAR-Knockoff Procedure

In this subsection, we introduce a variable selection strategy to leverage model (2.4) using knockoffs. As implied by Lemma 3.1 of Fan et al. (2020a), it is straightforward to show that estimating β in (2.4), restricted to $\tilde{\mathcal{S}}$, is equivalent to estimating $\beta_{\tilde{\mathcal{S}}}$ in the following reparameterized model:

$$Y^\dagger = \check{F}\theta + \check{U}_{\tilde{\mathcal{S}}}\beta_{\tilde{\mathcal{S}}} + \varepsilon^\dagger,$$

where $\theta \in \mathbb{R}^r$ is treated as a nuisance parameter. We then consider the following augmented Lasso regression with regularization parameter λ :

$$(\hat{\beta}, \hat{\theta}) = \arg \min_{b \in \mathbb{R}^{2\kappa}, \theta \in \mathbb{R}^r} \left\{ \frac{1}{2T} \left\| Y^\dagger - \check{F}\theta - [\check{U}_{\tilde{\mathcal{S}}}, \tilde{U}_{\tilde{\mathcal{S}}}]b \right\|_2^2 + \lambda \|b\|_1 \right\}. \quad (3.16)$$

To measure variable importance, we compute the knockoff statistics in the form of Lasso

coefficient-difference (LCD),

$$W_j = |\widehat{\beta}_j| - |\widehat{\beta}_{j+\kappa}|, \quad j = 1, \dots, \kappa. \quad (3.17)$$

Alternatively, one may also consider other importance statistics such as the Lasso signed matrix (LSM) and Bayesian variable selection (BVS). We choose LCD as it is shown to possess higher power, in line with the finding of Candès et al. (2018).

Given a prescribed target level of false discovery rate (FDR) q , we define the selection threshold G as

$$G = \min \left\{ g \in \mathcal{W} : \frac{1 + \left| \left\{ j \in \tilde{\mathcal{S}} : W_j \leq -g \right\} \right|}{\left| \left\{ j \in \tilde{\mathcal{S}} : W_j \geq g \right\} \right| \vee 1} \leq q \right\}, \quad (3.18)$$

where $\mathcal{W} = \{|W_j| : 1 \leq j \leq \kappa\} \setminus \{0\}$ is the set of unique, nonzero absolute values of the knockoff statistics. Based on this threshold, the set of selected variables is given by

$$\widehat{\mathcal{S}} = \left\{ j \in \tilde{\mathcal{S}} : W_j \geq G \right\}.$$

Note that our variable selection strategy is conducted conditional on the screened set $\tilde{\mathcal{S}}$. The screening step is essential not only for estimating Σ_ε^0 but also for constructing knockoffs when $N \geq T$. Moreover, a smaller $\tilde{\mathcal{S}}$ can improve FDR control in our method, as will be discussed later. However, conditioning on $\tilde{\mathcal{S}}$ introduces potential bias: null covariates retained in $\tilde{\mathcal{S}}$ are more likely to be selected, potentially inflating the FDR. This post-selection inference issue has been studied in Lee et al. (2016), Kuchibhotla et al. (2022), and Rasines and Young (2023), among others.

To address the potential bias introduced by pre-screening, we use sequential sample

splitting. The estimation and inference blocks are separated by a gap of length $2l_T$, where $l_T = \lfloor T^{1/5} \rfloor$; for equal outer blocks, their lengths are $(T - 2l_T)/2$. The complete procedure is summarized below.

Algorithm 1 FAR-Knockoff

Input: Time series (X, Y) , target FDR q , screening rule, covariance bandwidth m , factor-selection rule, and Lasso penalty λ .

- 1: Split the observations in time order into an estimation block, a middle block gap, and an inference block.
 - 2: On the estimation block, estimate factors and idiosyncratic components, obtain $\tilde{\mathcal{S}}$, and estimate $\hat{\Sigma}_\varepsilon$ from the post-screening residuals.
 - 3: Whiten the inference block by $\hat{\Sigma}_\varepsilon^{-1/2}$; re-estimate its factor space and form $\check{U}_{\tilde{\mathcal{S}}}$.
 - 4: Construct $\tilde{U}_{\tilde{\mathcal{S}}}$ satisfying the Fixed-X Gram identities and orthogonality to the estimated factors.
 - 5: Fit (3.16), compute $W_j = |\hat{\beta}_j| - |\hat{\beta}_{j+\kappa}|$, and apply (3.18) to obtain the selected original-variable indices.
-

For simplicity of presentation, we refer to the proposed knockoff-based variable selection procedure for the FAR model, along with the sample splitting scheme, as **FAR-Knockoff**. To streamline notation, we suppress fold indices such as $\langle 1 \rangle$ from expressions like $\tilde{\mathcal{S}}^{\langle 1 \rangle}$ when no confusion arises from the context.

Practical tuning. The theory suggests choosing ξ at the order of the uniform first-stage error, with $\xi \geq 2a_{NT}$, while keeping the screened dimension small enough for $K_{NT} = o(T/m)$. The bandwidth should increase slowly, with $m/T \rightarrow 0$, to balance covariance-estimation variance against truncation bias; our numerical choice $m \asymp T^{1/5}$ has this form. The augmented-Lasso penalty is chosen to dominate the empirical score fluctuation appearing in the power conditions, and the factor number is selected by the information criterion in Remark 3.2. Appendix A.1 records the numerical implementation.

Remark 3.4 (Data recycling). *To recover some power lost to sample splitting, the estimation fold may be reused in the final Lasso as in Barber and Candès (2019): on each recycled*

row, set the knockoff column equal to its original column, and retain the ordinary knockoff construction on the inference fold. If b_j and $\tilde{b}_j$ are the two coefficients, the recycled rows inform only their common direction $c_j = (b_j + \tilde{b}_j)/2$, not the contrast $d_j = (b_j - \tilde{b}_j)/2$. They can therefore help a true pair cross the Lasso entry threshold, while the inference fold alone decides whether the original beats its knockoff. Because the recycled score shift is identical within each pair, pair-swap invariance is preserved. The explicit algorithm and the distinction between the stacked Lasso Gram matrix and the random inference-score covariance are given in Lemma C.7.

4 Theoretical Properties of FAR-Knockoff

4.1 Conditional score reduction

Let $\hat{\mathcal{S}}$ be the final selected set and let $\mathcal{S}_0 = \{j : \beta_j \neq 0\}$, with $s = |\mathcal{S}_0|$. We use

$$\text{FDP} = \frac{|\hat{\mathcal{S}} \cap \mathcal{S}_0^c|}{|\hat{\mathcal{S}}| \vee 1}, \quad \text{FDR} = \mathbb{E}(\text{FDP}), \quad \text{Power} = \mathbb{E} \left(\frac{|\hat{\mathcal{S}} \cap \mathcal{S}_0|}{s \vee 1} \right).$$

Let $\mathcal{G}$ contain the estimation-fold information, the inference-fold covariates, the screened set, the covariance estimator, the PCA output (including the estimated number of factors), and the randomization used to construct the knockoffs. Write

$$\mathbf{U} = [\check{\mathbf{U}}_{\tilde{\mathcal{S}}}, \tilde{\mathbf{U}}_{\tilde{\mathcal{S}}}], \quad \mathbf{\Psi} = T^{-1} \mathbf{U}' \mathbf{U}, \quad \kappa = |\tilde{\mathcal{S}}|.$$

Conditional on $\mathcal{G}$, these matrices are fixed.

The exact inference-fold model, allowing imperfect screening, is

$$Y^\dagger = \check{F}\theta + \check{U}_{\tilde{\mathcal{S}}}\beta_{\tilde{\mathcal{S}}} + g + \varepsilon^\dagger, \quad g = X_{\mathcal{A}}^\dagger\beta_{\mathcal{A}}, \quad \mathcal{A} = \mathcal{S}_0 \setminus \tilde{\mathcal{S}}.$$

Lemma 4.1 (Conditional score reduction). *Conditional on $\mathcal{G}$, the factor-adjusted Lasso and its knockoff statistics depend on the error process only through the 2κ -dimensional score*

$$\zeta = T^{-1/2}\mathbb{U}'\varepsilon^\dagger, \quad D = T^{-1/2}\mathbb{U}'g. \quad (4.19)$$

The common-component part of each missed covariate is orthogonal to both members of every knockoff pair. Hence only the idiosyncratic omitted signal contributes to the antisymmetric drift $D_j - D_{j+\kappa}$.

The reduction in Lemma 4.1, rather than Gaussianity, is the organizing principle of the theory. It separates three sources of approximation: distributional approximation of ζ , antisymmetric drift from missed signals, and covariance mismatch from estimated whitening.

Imperfect screening and estimated factors. Over-selection enlarges κ and therefore the design, covariance, and distributional-approximation rates. Under-selection produces the explicit term g in (4.19); Assumption 1 makes every missed coefficient no larger than the screening resolution b_{NT} . Estimated factors and the estimated factor number belong to $\mathcal{G}$. For every realized PCA decomposition, the identity

$$X_{\tilde{\mathcal{S}}}^\dagger\beta_{\tilde{\mathcal{S}}} = \check{F}\check{\Lambda}'_{\tilde{\mathcal{S}}}\beta_{\tilde{\mathcal{S}}} + \check{U}_{\tilde{\mathcal{S}}}\beta_{\tilde{\mathcal{S}}}$$

holds exactly. Thus factor-estimation error does not itself break the conditional null-swap identity (Lemma C.5). For FDR, it enters through the screening and design failure probabilities, the whitening rate, and the antisymmetric missed-signal drift. In particular, (C.15) bounds its contribution to the original-design omitted score by $\|\beta_{\mathcal{A}}\|_1(2Cr_{U,NT} + r_{U,NT}^2)$, where $r_{U,NT}$ measures the error in the transformed idiosyncratic columns. Lemma C.3 transfers this score bound to the drift and hence to the mean-comparison cost below. Exact conditional exchangeability therefore does not mean that estimation accuracy is irrelevant to the unconditional FDR bound.

Assumption 3 (Estimated-design regularity). *The process $\{(F_t, U_t, \varepsilon_t)\}$ is stationary and beta-mixing, and the separation between the estimation and inference folds has total variation error $c_{NT} \rightarrow 0$. On a $\mathcal{G}$ -measurable event $\mathcal{B}_{\mathcal{G}}$ with $\mathbb{P}(\mathcal{B}_{\mathcal{G}}^c) \rightarrow 0$,*

$$\sigma_{\min}(\Psi) \geq c_0, \quad \|\Xi - \Psi\| \leq C\omega_{NT}, \quad \omega_{NT} = o(1),$$

where $\Xi = \text{Cov}(\zeta \mid \mathcal{G})$ under the corresponding independent-fold reference law. The relevant idiosyncratic Gram matrices have eigenvalues uniformly bounded away from zero and infinity for realized screened models of size at most K_{NT} . On the same event,

$$\max_{j \in \tilde{\mathcal{S}} \cap \mathcal{S}_0^c} T^{-1}|(\check{U}_j - \tilde{U}_j)'g| \leq \vartheta_{NT},$$

where the maximum is zero if no retained null exists.

The last requirement controls the beta-weighted omitted signal; it does not require every pairwise idiosyncratic correlation to vanish. For each retained null pair, replace $(D_j, D_{j+\kappa})$

by their average and call the resulting vector $\bar{D}$. Then

$$\|D - \bar{D}\|_2^2 \leq \frac{1}{2} K_{NT} \bar{\omega}_{NT}^2, \quad \bar{\omega}_{NT} = \sqrt{T} \vartheta_{NT}.$$

Let $\epsilon_{NT}^{\text{mean}}$ denote the resulting Gaussian mean-comparison cost and let $\epsilon_{NT}^{\text{cov}}$ denote the covariance-comparison cost generated by $\|\Xi - \Psi\|$. Their explicit bounds are

$$\begin{aligned} \epsilon_{NT}^{\text{mean}} &= C \{ \sqrt{K_{NT}} \bar{\omega}_{NT} \sqrt{\ln(NT)} + K_{NT} \bar{\omega}_{NT}^2 \}, \\ \epsilon_{NT}^{\text{cov}} &= C \{ K_{NT} \omega_{NT} + \omega_{NT} \sqrt{K_{NT} \ln(NT)} + \omega_{NT} \ln(NT) \}, \quad \bar{\epsilon}_{NT} = \epsilon_{NT}^{\text{mean}} + \epsilon_{NT}^{\text{cov}}. \end{aligned}$$

Conditional Gaussian reduction. Let P^{ind} denote the reference law obtained by decoupling the estimation and inference folds while preserving the inference block's joint law. This is a proof device; the total variation transfer back to the actual time-series law is priced by $2c_{NT}$.

Lemma 4.2 (Gaussian error implies a Gaussian conditional score). *Suppose that, under P^{ind} , the unwhitened inference-fold error is conditionally Gaussian given its covariates,*

$$\varepsilon \mid X \sim N(0, \Sigma_\varepsilon^0).$$

Then, conditional on $\mathcal{G}$, $\zeta \sim N(0, \Xi)$.

A primitive specification is $\varepsilon = (\Sigma_\varepsilon^0)^{1/2} e$ with $e \sim N(0, I)$ independent of the relevant design information.

The common proof pipeline. All probability measures below are conditional laws of the same 2κ -dimensional sufficient score, not laws of the original $T \times N$ data array:

$$P_D^a \xrightarrow[\eta_{NT}]{\rho_{NT}} P_D^\star \xrightarrow{\epsilon_{NT}^{\text{mean}}} P_D^m \xrightarrow{\epsilon_{NT}^{\text{cov}}} P_D^\dagger \xrightarrow{\text{exact null signs}} \text{FDR} \leq q + o(1).$$

Here P_D^a is the actual score law, $P_D^\star = N(D, \Xi)$ is its covariance-matched Gaussian law, $P_D^m = N(\bar{D}, \Xi)$ symmetrizes the missed-signal drift within null pairs, and $P_D^\dagger = N(\bar{D}, \Psi)$ is swap invariant. Under the Gaussian condition in Lemma 4.2, the first arrow is an identity. Under causal martingale innovations, Appendix C supplies that arrow by coupling the low-dimensional score.

4.2 FDR control

We now use the score comparisons to obtain two FDR guarantees under imperfect screening. The Gaussian result makes the effects of screening, missed-signal drift, and estimated whitening explicit. The causal-martingale result uses the same argument with an additional score-coupling step.

Theorem 4.3 (Gaussian score). *Suppose Assumptions 1, 2, and 3 hold, and suppose the Gaussian condition in Lemma 4.2 holds. If $\epsilon_{NT}^{\text{mean}} = o(1)$ and $\epsilon_{NT}^{\text{cov}} = o(1)$, then*

$$\begin{aligned} \text{FDR} &\leq q \exp(\bar{\epsilon}_{NT}) + \delta_{NT}^{\text{scr}} + \mathbb{P}(\mathcal{B}_G^c) + 2c_{NT} \\ &\quad + \frac{2\{1 + \exp(\epsilon_{NT}^{\text{mean}})\}}{NT} + o(1) = q + o(1). \end{aligned}$$

Theorem 4.4 (Causal-martingale score). *Suppose Assumptions 1, 2, and 3, together with*

Appendix Assumptions C.1–C.3, hold. If $\epsilon_{NT}^{\text{mean}} = o(1)$ and $\epsilon_{NT}^{\text{cov}} = O(1)$, then

$$\text{FDR} \leq q + o(1).$$

The causal representation is driven by martingale-difference innovations, without Gaussianity or symmetry. It allows ARCH/GARCH-type conditional heteroskedasticity and asymmetric heavy tails under the stated regularity conditions.

The two theorems use the same conditional-score architecture. The Gaussian case starts at P_D^* and transfers the bounded, score-measurable FDP directly through the two Gaussian comparisons. The martingale case adds only the first coupling arrow; because a coupling perturbs the discontinuous selection threshold, Appendix Assumption C.3 controls null statistics lying near that threshold. Appendix Proposition C.12 gives the full finite-sample martingale bound and the primitive sufficient rate $K_{NT} = o\{(T/\ln T)^{1/5}\}$.

How screening enters the FDR guarantee. Screening acts through three routes. First, failure of the regularity event (2.8) enters additively as δ_{NT}^{scr} . Second, over-selection enlarges K_{NT} , thereby increasing the selection-uniform projection error in ω_{NT} and the covariance-comparison cost $\epsilon_{NT}^{\text{cov}}$. Third, under-selection creates both the residual-covariance contamination R_{miss} in Theorem 2.2 and the antisymmetric score drift measured by $\epsilon_{NT}^{\text{mean}}$. Under sure screening, $R_{\text{miss}} = \epsilon_{NT}^{\text{mean}} = 0$, although the cost of over-selection may remain.

For completeness, a standard beta-min condition recovers sure screening as a special case. If

$$\min_{j \in \mathcal{S}_0} |\beta_j^M| \geq \xi + a_{NT},$$

then (2.8) implies $\mathcal{S}_0 \subseteq \tilde{\mathcal{S}}$ and hence $\mathbb{P}(\mathcal{S}_0 \subseteq \tilde{\mathcal{S}}) \geq 1 - \delta_{NT}^{\text{scr}}$. Under Assumption 1, the primitive condition $\min_{j \in \mathcal{S}_0} |\beta_j| \geq \xi + (1 + C_0)a_{NT}$ is sufficient, where $C_0 a_{NT}$ bounds $|\beta_j^M -$

$|\beta_j|$. The sure-screening specialization in Appendix C.1 records the simplified covariance rate.

4.3 Power

For power, factor-estimation error matters through the conditioning of the augmented design and the full omitted-signal score, not merely its null-pair contrast. Equations (C.21)–(C.22) propagate the same transformed-design error into χ_{NT} below. Keeping this score at the Lasso tuning scale preserves signal separation; screening mistakes add the explicit loss π_{NT} in Theorem 4.5. These requirements explain why conditional FDR validity alone does not guarantee high power.

Assumption 4 (Power regularity). *Let λ_{NT} denote the tuning scale, let $\mathcal{S}_R = \mathcal{S}_0 \cap \tilde{\mathcal{S}}$, and put $\pi_{NT} = |\mathcal{A}|/(s \vee 1)$. The retained model contains a nonvanishing fraction of strong signals: for constants $c > 0$ and $\varkappa_T \rightarrow \infty$,*

$$|\{j \in \mathcal{S}_R : |\beta_j| \gg \sqrt{\kappa} \lambda_{NT}\}| \geq c\kappa, \quad \min_{j \in \mathcal{S}_R} |\beta_j| \geq \varkappa_T \lambda_{NT}.$$

The augmented design satisfies the restricted-eigenvalue condition used for the Lasso, the knockoff conditional-variance matrix is bounded away from singularity, and the omitted-signal score satisfies

$$\chi_{NT} := \|T^{-1}\mathbb{U}'g\|_\infty = O_p(\lambda_{NT}).$$

Theorem 4.5 (Gaussian power under imperfect screening). *Suppose Assumptions 1, 2, 3, and 4 hold, together with the Gaussian condition in Lemma 4.2. Set*

$$\lambda_{NT} = \sqrt{\frac{\ln(2K_{NT} \vee 2)}{T}}, \quad \lambda = C_2 \lambda_{NT}$$

for a sufficiently large constant C_2 . Then, with $\varphi = (1 + \sqrt{5})/2$,

$$\frac{|\widehat{\mathcal{S}} \cap \mathcal{S}_0|}{s \vee 1} \geq 1 - \pi_{NT} - C(\varphi + 1)\varkappa_T^{-1} - o_p(1), \quad \text{Power} \geq 1 - \mathbb{E}(\pi_{NT}) - o(1).$$

On the screening event, Lemma C.1 gives $\mathcal{A} \subseteq \{j \in \mathcal{S}_0 : |\beta_j| \leq b_{NT}\}$ and hence

$$\mathbb{E}(\pi_{NT}) \leq \frac{1}{s \vee 1} \sum_{j \in \mathcal{S}_0} \mathbf{1}\{|\beta_j| \leq b_{NT}\} + \delta_{NT}^{\text{scr}}.$$

Thus the power loss caused by screening is governed by the fraction of active coefficients below the screening resolution. Appendix Theorem C.15 gives the corresponding result for causal martingale innovations.

5 Simulations

In this section, we consider two simulated experiments to evaluate the finite-sample performance of FAR-Knockoff and several well-established competitive methods. The first experiment is a synthetic model, while the second is a calibrated model based on real-world financial data. For each simulated experiment, we consider four combinations of sample sizes: $N \in \{200, 400\}$ and $T \in \{200, 400\}$. The main text reports 500 replications at $q = 0.2$.[‡]

[‡]Results for $q = 0.1$ are reported in Appendix A.2. Under TSKI's within-subsample level $q/2$, a nonempty selection requires at least 20 positive exceedances, although the designs contain only ten signals. We therefore retain this mechanically low-discovery configuration in the appendix rather than as the main comparison.

5.1 Competitive Methods and Evaluation Criteria

We compare FAR-Knockoff with Lasso and FARMSELECT (Fan et al., 2020a), Model-X and Fixed-X knockoffs (Candès et al., 2018), and TSKI (Chi et al., 2025). Lasso and FarmSelect are selection benchmarks, not FDR-controlling procedures. Model-X uses estimated covariate moments; Fixed-X matches sample moments and is evaluated when $T > N$. TSKI uses two interlaced subsamples, within-subsample level $q/2$, and e-value aggregation at level q . Appendix A.1 specifies the fitting and threshold choices; the comparisons concern these configurations, not a ranking over all possible tunings.

We report empirical FDR, power, accuracy $(TP + TN)/(N_1 + N_0)$, $F_1 = 2TP/(2TP + FP + FN)$, and false-positive rate FP/N_0 , where N_1 and N_0 denote the numbers of active and inactive variables and TP, TN, FP, FN have their usual meanings. Selections from a screened design are mapped back to the original covariate indices before evaluation. Empty selections contribute zero FDP and power; unavailable methods are marked N.A., not scored as empty selections. Appendix A.6 reports Monte Carlo standard deviations for FAR-Knockoff and the pre-screened benchmarks.

5.2 Experiment 1: Synthetic Model

The first experiment is based on the following factor-augmented model:

$$Y_t = X_t' \beta + \varepsilon_t, \quad X_{ti} = \lambda_i' F_t + U_{ti}, \quad \text{for } t = 1, \dots, T, \quad i = 1, \dots, N.$$

We set the number of latent factors to $r = 3$. The factor loadings λ_{ik} are independently drawn from $\mathcal{N}(1, 1)$, and the factors F_{tk} are generated from $\mathcal{N}(0, 1)$ independently across t

and k .[§] The coefficient vector is specified as $\beta = (\beta_1, \dots, \beta_{10}, 0'_{(N-10) \times 1})'$, where the nonzero entries $\beta_1, \dots, \beta_{10}$ are drawn i.i.d. from $\text{Uniform}(1, 3)$.

The idiosyncratic components U_{ti} are drawn independently from $\mathcal{N}(0, 0.25)$, and the error terms ε_t follow a rescaled t -distribution: $\varepsilon_t \sim \sqrt{3/5} \times t(5)$, to introduce moderate heavy-tailed behavior. Notably, ε_t is temporally independent in this experiment. As a robustness check, we have also experimented with $\varepsilon_t \sim \mathcal{N}(0, 1)$ or $(\chi^2(3) - 3)/\sqrt{6}$, and observed similar performance. The additional results are reported in Appendix A.5.

Results for Experiment 1 at $q = 0.2$ are reported in Table 1. Lasso and FarmSelect attain high power but do not control FDR. Fixed-X is conservative, and Model-X is also conservative when $(N, T) = (200, 400)$. TSKI has low power in these factor designs under the specified tuning; this is not evidence of an intrinsic weakness. FAR-Knockoff combines low empirical FDR with higher power as T increases. The non-factor results in Appendix A.5 and the weak-factor experiment in Appendix A.8 assess sensitivity to the factor structure.

Table 1: Simulation results for Experiment 1 with target FDR $q = 0.2$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.703	0.910	0.446	0.884	0.117	LASSO	0.717	0.997	0.438	0.866	0.141
FARMSELECT	0.337	1.000	0.776	0.964	0.038	FARMSELECT	0.212	1.000	0.858	0.976	0.025
MODEL-X	0.170	0.998	0.898	0.987	0.013	MODEL-X	0.051	0.347	0.346	0.964	0.004
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.057	0.357	0.351	0.964	0.004
TSKI	0.057	0.397	0.377	0.966	0.004	TSKI	0.083	0.546	0.505	0.972	0.006
FAR-KNOCKOFF	0.047	0.919	0.919	0.993	0.003	FAR-KNOCKOFF	0.050	0.984	0.962	0.996	0.003
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.769	0.877	0.364	0.921	0.078	LASSO	0.768	0.920	0.369	0.919	0.081
FARMSELECT	0.410	1.000	0.712	0.972	0.028	FARMSELECT	0.247	1.000	0.831	0.985	0.015
MODEL-X	0.232	0.990	0.853	0.990	0.010	MODEL-X	0.173	0.993	0.892	0.993	0.007
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.019	0.141	0.135	0.978	0.001	TSKI	0.041	0.425	0.408	0.984	0.001
FAR-KNOCKOFF	0.038	0.892	0.903	0.996	0.001	FAR-KNOCKOFF	0.069	0.991	0.954	0.998	0.002

[§]In additional simulation designs and results in Appendix A.5, we also experiment with non-factor-driven predictors: Specifically, $X_t \sim \mathcal{N}(0, \Sigma_X)$ with $\Sigma_X(i, j) = 0.5^{|i-j|}$.

5.3 Experiment 2: Calibrated Model

The second experiment follows the calibrated setting of Fan et al. (2020a), based on real financial data. We compute centered monthly returns for stocks in the S&P 500 index with complete data from January 1970 to December 2022. This dataset, obtained from CRSP, includes returns for 256 stocks over 636 months.

The calibration details and parameter estimates used in this experiment are summarized in Table 2. For completeness, the full calibration procedure and data generation steps are described in Appendix A.4 in the supplementary material.[¶]

Table 2: Parameters calibrated for Experiment 2

Σ_Λ			$\hat{\Phi}$			Σ_η			σ_e^2
1.6921	0	0	0.0912	-0.0315	-0.0426	0.9889	-0.0002	0.0075	0.9748
0	1.0920	0	-0.0426	0.1468	-0.0325	-0.0002	0.9760	-0.0098	
0	0	0.6584	0.0048	0.0920	0.1193	0.0075	-0.0098	0.9773	

Table 3 reports Experiment 2 at $q = 0.2$. Lasso has high false-discovery rates, whereas FarmSelect achieves high power and mostly low FDR in this calibrated design. The full-sample knockoff benchmarks exhibit either low power or inflated FDR in several configurations; TSKI makes few discoveries under the stated tuning. FAR-Knockoff’s empirical FDR is below the target in all four configurations, with power at least 0.97. These comparisons concern the implementations studied, not all possible tunings.^{||}

The supplementary material reports replication-level standard deviations of FDP and power for FAR-Knockoff and the pre-screened benchmarks, and adds two theory-targeted stress tests: GARCH errors and a data-driven screen calibrated so that its omissions are overwhelmingly confined to three deliberately weak signals. Appendices A.6 and A.7 show

[¶]We further stress-test FAR-Knockoff with stronger temporal correlation for ε_t in Experiment 2. The results, reported in Appendix A.5, support its robust performance.

^{||}Appendix A.9 examines raw-covariate screening before Model-X and Fixed-X inference. Screening alone does not uniformly improve their FDR–power trade-off. These variants differ from FAR-Knockoff in both preprocessing and screening size, so they are not a one-component ablation of factor adjustment.

that FDR remains controlled in both departures, while imperfect screening lowers power by making omitted signals unavailable to the second stage.

Table 3: Simulation results for Experiment 2 with target FDR $q = 0.2$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.887	0.094	0.101	0.915	0.041	LASSO	0.856	0.174	0.149	0.908	0.053
FARMSELECT	0.000	1.000	1.000	1.000	0.000	FARMSELECT	0.194	1.000	0.868	0.977	0.024
MODEL-X	0.518	0.331	0.332	0.939	0.029	MODEL-X	0.032	0.048	0.055	0.951	0.001
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.043	0.059	0.067	0.951	0.002
TSKI	0.008	0.006	0.006	0.950	0.000	TSKI	0.001	0.001	0.001	0.950	0.000
FAR-KNOCKOFF	0.146	0.978	0.901	0.989	0.010	FAR-KNOCKOFF	0.154	0.977	0.895	0.988	0.012
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.919	0.084	0.081	0.952	0.026	LASSO	0.924	0.086	0.079	0.950	0.028
FARMSELECT	0.000	1.000	1.000	1.000	0.000	FARMSELECT	0.000	1.000	1.000	1.000	0.000
MODEL-X	0.475	0.234	0.245	0.970	0.011	MODEL-X	0.478	0.329	0.338	0.972	0.012
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.002	0.002	0.002	0.975	0.000	TSKI	0.000	0.002	0.002	0.975	0.000
FAR-KNOCKOFF	0.137	0.977	0.903	0.995	0.005	FAR-KNOCKOFF	0.158	0.985	0.896	0.994	0.006

6 Real Data Application: Unspanned Macroeconomic Signals in Bond Risk Premia

We ask whether a large macroeconomic panel contains stable predictive information for U.S. Treasury excess returns beyond that already summarized by the yield curve. For a compact main presentation, we report two- and three-year bonds here and the corresponding four- and five-year results in Appendix B; all maturities use the same forecast sample, information set, and specification.

6.1 Data, forecasting design, and methods

The targets are one-year log excess holding returns on two- through five-year Fama–Bliss zero-coupon bonds (Ludvigson and Ng, 2009). The January 1970–December 2023 sample contains 118 complete series from the November 2024 FRED-MD vintage (McCracken and

Ng, 2016), transformed by the database codes and standardized within each estimation window.

At each forecast origin τ , a 400-month rolling window supplies 388 fully realized pairs $(X_s, r_{s \rightarrow s+12}^{(n)})$ with $s + 12 \leq \tau$. Every estimation step is repeated at the origin, producing 236 common forecasts from May 2004 through December 2023.

Every specification includes three yield-curve principal components (level, slope, and curvature) (Litterman and Scheinkman, 1991) and the Cochrane and Piazzesi (2005) (CP) return-forecasting factor. Their loadings and coefficients are re-estimated using only information available at each origin, so selected macro variables must add to a level–slope–curvature–CP benchmark.

We compare recycled FAR-Knockoff with LASSO, FARMSELECT (Fan et al., 2020a), MODEL-X, and FIXED-X knockoffs (Barber and Candès, 2015; Candès et al., 2018), using identical observations, controls, and post-selection OLS refits. FAR-Knockoff and Farm-Select choose one to six macro factors by their implemented information criterion. For prediction, FAR-Knockoff uses $q = 0.2$ and averages its LCD statistics over 20 random original–knockoff relabelings to reduce sensitivity to a single labeling. Appendix B reports factor-cap sensitivity.

6.2 Short-end forecast performance

Table 4 reports RMSE; parentheses give mean selected macro-model size among nonempty windows, excluding the four controls. The final rows test FAR-Knockoff against the nested benchmark.

Table 4: Out-of-sample forecast performance at the short end

	2 years	3 years
Yield factors + CP	1.458 (–)	2.812 (–)
LASSO	2.516 (23.33)	4.638 (18.81)
FARMSELECT	2.988 (43.81)	5.322 (41.49)
MODEL-X	1.911 (3.69)	3.042 (3.28)
FIXED-X	1.920 (1.74)	3.208 (2.62)
FAR-KNOCKOFF	1.380 (2.78)	2.763 (2.10)
Clark–West statistic	2.49	1.78
One-sided p -value	0.0065	0.0377

Notes: The first six rows report RMSE, with conditional mean selected size in parentheses. RMSE and tests use all 236 windows, including those with no selected macro variables. Clark–West tests compare FAR-Knockoff with yield factors + CP using Bartlett HAC standard errors with lag 11.

FAR-Knockoff has the lowest RMSE at both short maturities while retaining only two to three macro variables in nonempty windows. At two years, RMSE falls from 1.458 to 1.380, a 10.5% reduction in mean squared forecast error; the one-sided Clark–West p -value is 0.0065. At three years, the corresponding reduction is 3.5% ($p = 0.0377$), but it is sensitive to the macro factor cap.

This maturity localization has a close precedent in the macro-finance literature. Ludvigson and Ng (2009) document macro-factor predictability across two- to five-year bonds, whereas the robust reanalysis of Bauer and Hamilton (2018) finds substantially weaker evidence beyond the yield curve and identifies the two-year bond as the exceptional maturity in the relevant spanning exercises. Our rolling forecasts are consistent with a maturity-specific increment beyond a strong yield benchmark. The selected variables below suggest

a connection to policy and cyclical news, although forecasting associations alone do not identify a causal channel.

The cross-method comparison shows that the gain does not arise from adding many predictors. LASSO and FARMSELECT select much larger models and substantially increase forecast loss. MODEL-X and FIXED-X are more parsimonious but do not improve on the benchmark. FAR-Knockoff combines the smallest short-end forecast loss with a sparse selected block.

6.3 Selected predictors and economic interpretation

FAR-Knockoff selects coordinates of the estimated idiosyncratic component $\hat{U}$, after common macro factors and the always-included yield controls have been partialled out. The equivalent representation in Section 2 retains each original predictor’s coefficient, so a selected series identifies its incremental information beyond the factors and controls, not a structural or causal shock. Table 5 reports frequencies over all 236 windows.

Table 5: Most frequently selected short-end residual macro predictors

Maturity	FRED-MD series	Frequency	Incremental economic channel
2 years	T5YFFM	24.6%	policy-path / term-premium news
	PERMITNE	24.6%	forward-looking housing activity
	TB3SMFFM	13.1%	short-rate spread / policy expectations
3 years	T5YFFM	26.3%	policy-path / term-premium news
	VIXCLSx	17.4%	uncertainty and risk-bearing capacity
	CES0600000007	16.1%	goods-producing workweek

The two-year selections suggest a policy-expectations and interest-sensitive activity channel. T5YFFM and TB3SMFFM are the five-year Treasury and three-month Treasury-bill spreads over the federal funds rate. Their residual components describe relative rate move-

ments not absorbed by the macro factors and yield controls. For a one-year holding return on a two-year bond, revisions to the short end of the yield curve are a natural source of price variation. `PERMITNE`, Northeast housing permits, adds forward-looking information about rate-sensitive demand. Its residual is unusual housing activity relative to the common macro cycle, rather than simply another measure of that cycle. The recurring combination of rate spreads and housing activity is therefore consistent with policy-transmission news beyond the yield benchmark, without establishing a structural mechanism.

At three years, `T5YFFM` remains prominent, while `VIXCLSx` and `CES0600000007`, the goods-producing workweek, indicate uncertainty and labor-utilization channels. The latter is consistent with the real-activity component of bond predictability in Ludvigson and Ng (2009) and models in which output risks affect bond risk premia without being fully spanned by yields (Joslin et al., 2014). The VIX association is also consistent with evidence linking bond-return predictability to macroeconomic conditions and short-rate expectations (Gargano et al., 2019). These interpretations concern residual predictive information, not the signs of causal effects, which selection frequencies do not identify.

7 Conclusion

This paper introduces FAR-Knockoff for variable selection with FDR control in large time-series panels. It combines factor adjustment with serial-dependence correction. By conditioning on the estimated design and coupling a low-dimensional score, we establish asymptotic FDR control under estimated whitening, imperfect screening, and causal martingale innovations without requiring temporally independent covariates, Gaussianity, or symmetry. The power theory makes the loss from screened-out signals explicit. Across the reported designs, the simulations show favorable FDR–power performance. In the bond application,

FAR-Knockoff produces sparse models and lower forecast loss at the short end, with the strongest evidence at the two-year maturity.

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Supplementary Material for “Factor-Adjusted Knockoff Inference for High-Dimensional Time Series”

In the supplementary material, Appendix A contains implementation details and additional numerical results, while Appendix C states the causal-martingale conditions and gives the technical lemmas and proofs supporting the two FDR theorems in Section 4.

A Implementation Details and Additional Results for Numerical Studies

A.1 Implementation Details for Competitive Methods

The methods are evaluated under the same DGPs and sample-size configurations, but differ in their use of factor adjustment, subsampling, and regularization. The following specifies the settings in the retained simulation driver. In particular, a prediction-selected penalty and a knockoff FDR threshold are different tuning parameters; using the same numerical penalty across all methods would not make their comparisons equivalent.

Computational cost. The full-panel PCA and screening path are the only stages whose dimension is N ; after screening, the knockoff construction and augmented Lasso depend on $\kappa = |\tilde{S}|$. With dense linear algebra, factor extraction and screening cost approximately $O(TNr)$ and $O(LTN)$, respectively, for L path values, while Fixed-X construction costs $O(T\kappa^2 + \kappa^3)$. The current whitening implementation uses a dense $T \times T$ eigendecomposition, requiring $O(T^3)$ operations and $O(T^2)$ storage; banded solvers are preferable for very large T . Rolling windows and independent swap draws can be parallelized.

1. **Lasso.** The full-sample fit uses `cv.glmnet` with $\alpha = 1$, an intercept, default column standardization, and ten-fold cross-validation. The penalty minimizes cross-validation error (`lambda.min`); selected variables have nonzero coefficients. The driver specifies $\sqrt{\log N/T}$ only as a fallback if no minimizing penalty is returned. This benchmark has no FDR-target parameter.
2. **FarmSelect.** The call `farm.select(X,Y)` uses robust factor adjustment and the default SCAD penalty, rather than Lasso. The factor count is estimated, not supplied from the DGP. In the retained FarmSelect 1.0.2 implementation, `robust=TRUE`, `tau=2`, and `cv=FALSE` govern the robustification step. The final SCAD penalty is nevertheless selected by `cv.ncvreg` at its minimizing penalty, with $\lceil T/3 \rceil$ folds. Thus `cv=FALSE` does not mean that the regression penalty is chosen without cross-validation. Like Lasso, FarmSelect is not calibrated to an FDR target.
3. **Model-X knockoffs.** The generator is `create.second_order`, not an oracle generator using population moments. It estimates the mean and covariance from the observed predictors; in the retained knockoff 0.3.6 implementation, a non-positive-definite sample covariance triggers shrinkage estimation. Approximate SDP is the default construction. The statistic is the package’s cross-validated Lasso coefficient difference, and the final filter uses offset one at q .

4. **Fixed-X knockoffs.** The full-sample comparator uses `create.fixed` and the package’s default coefficient-difference statistic, with offset one at q . The driver evaluates this comparator only when $T > N$; unavailable cases are not evidence of zero power. Its numerical implementation uses cross-validation for the statistic and should not be identified with an exact fixed-penalty theoretical guarantee.
5. **TSKI.** A precision estimate from `isee(X, "log", 5, 0, 0)` is passed to the Gaussian knockoff generator `gknockoffX`. Odd and even observations form two interlaced subsamples. Within each subsample, the Lasso coefficient-difference statistic uses ten-fold cross-validation over penalties $\{10^{2-0.2\ell} : \ell = 0, \dots, 20\}$. The within-subsample knockoff+ level is $\tau_1 = q/2$; the resulting e-values are averaged and the e-BH rule is applied at q , as in the aggregation framework of Chi et al. (2025). Here q denotes the final FDR target, not the subsampling parameter used in that paper.
6. **FAR-Knockoff.** Screening and covariance estimation use the first half-sample; inference uses the second half. Each stage chooses its factor count from 1 to 8 using the implemented information criterion. The screening target is $\min(0.25\lfloor T/2 \rfloor, 0.7N)$, and the covariance bandwidth is $\lfloor T^{0.2} \rfloor$. The construction uses approximate SDP and the final filter has offset one. The retained statistic uses a narrow two-penalty cross-validation choice between $n^{-0.3}$ and $n^{-0.3} + 10^{-3}$, where $n = \lfloor T/2 \rfloor$; it is not a broad prediction-tuning search. The weak-factor experiment keeps these choices unchanged and maps selections back to their original variable indices.

Screened benchmarks and evaluation. The supplementary screening variants use raw-covariate Lasso screening on the first half-sample, with size target $\min(0.25T, 0.7N)$, followed by Model-X or Fixed-X on the second half. The tables explicitly label these variants as Screening+Model-X and Screening+Fixed-X; they are distinct from the full-sample benchmarks in the main text. Local selections are mapped through the retained column indices before comparison with the true support. The regenerated results use 500 replications per configuration, including empty selections. Computational failures and unavailable fits are kept separate from empty selections. Saved original-coordinate selections permit independent reconstruction of the reported scores.

Interpreting TSKI’s low-discovery configurations. Let $R_+(t)$ and $R_-(t)$ be the positive and negative exceedance counts within a TSKI subsample. A nonempty knockoff+ selection requires $(1 + R_-(t))/R_+(t) \leq \tau_1$, hence necessarily $R_+(t) \geq \lceil 1/\tau_1 \rceil$. With $\tau_1 = q/2$, this lower bound is 20 at $q = 0.1$ and 10 at $q = 0.2$. The designs contain ten true signals: even perfect separation of just those ten signals cannot clear the within-subsample threshold at $q = 0.1$. This is a finite-sample interaction between sparsity and the stated tuning, not an impossibility result for TSKI. Additional positive null statistics can also affect the threshold. If neither subsample passes its threshold, all aggregated e-values are zero and e-BH makes no discoveries. Conversely, in the non-factor design at $q = 0.2$, Table 13 reports TSKI power of 1.000 in all four configurations. We therefore interpret relative performance as design- and tuning-dependent, rather than as a universal method ranking.

A.2 Additional Results at $q = 0.1$

At $q = 0.1$, TSKI applies its knockoff+ filter within each subsample at $q/2 = 0.05$. A nonempty within-subsample selection therefore requires at least 20 positive exceedances, whereas each design has ten true signals. This finite-sample threshold effect makes the comparison mechanically low-discovery, so we retain the results here rather than in the main text.

Table 6: Simulation results for Experiment 1 with target FDR $q = 0.1$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.703	0.910	0.446	0.884	0.117	LASSO	0.717	0.997	0.438	0.866	0.141
FARMSELECT	0.337	1.000	0.776	0.964	0.038	FARMSELECT	0.212	1.000	0.858	0.976	0.025
MODEL-X	0.084	0.962	0.916	0.993	0.006	MODEL-X	0.011	0.118	0.112	0.955	0.001
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.013	0.141	0.134	0.956	0.001
TSKI	0.000	0.000	0.000	0.950	0.000	TSKI	0.000	0.000	0.000	0.950	0.000
FAR-KNOCKOFF	0.033	0.656	0.641	0.981	0.002	FAR-KNOCKOFF	0.034	0.935	0.917	0.995	0.002
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.769	0.877	0.364	0.921	0.078	LASSO	0.768	0.920	0.369	0.919	0.081
FARMSELECT	0.410	1.000	0.712	0.972	0.028	FARMSELECT	0.247	1.000	0.831	0.985	0.015
MODEL-X	0.111	0.927	0.866	0.994	0.004	MODEL-X	0.072	0.958	0.917	0.997	0.002
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.000	0.000	0.000	0.975	0.000	TSKI	0.000	0.000	0.000	0.975	0.000
FAR-KNOCKOFF	0.028	0.602	0.590	0.989	0.001	FAR-KNOCKOFF	0.047	0.976	0.951	0.998	0.001

Table 7: Simulation results for Experiment 2 with target FDR $q = 0.1$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.887	0.094	0.101	0.915	0.041	LASSO	0.856	0.174	0.149	0.908	0.053
FARMSELECT	0.000	1.000	1.000	1.000	0.000	FARMSELECT	0.194	1.000	0.868	0.977	0.024
MODEL-X	0.135	0.101	0.091	0.946	0.009	MODEL-X	0.002	0.002	0.002	0.950	0.000
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.001	0.003	0.003	0.950	0.000
TSKI	0.000	0.000	0.000	0.950	0.000	TSKI	0.000	0.000	0.000	0.950	0.000
FAR-KNOCKOFF	0.070	0.915	0.877	0.991	0.005	FAR-KNOCKOFF	0.059	0.900	0.870	0.991	0.004
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.919	0.084	0.081	0.952	0.026	LASSO	0.924	0.086	0.079	0.950	0.028
FARMSELECT	0.000	1.000	1.000	1.000	0.000	FARMSELECT	0.000	1.000	1.000	1.000	0.000
MODEL-X	0.100	0.055	0.051	0.973	0.003	MODEL-X	0.116	0.114	0.106	0.974	0.004
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.000	0.000	0.000	0.975	0.000	TSKI	0.000	0.000	0.000	0.975	0.000
FAR-KNOCKOFF	0.071	0.938	0.899	0.996	0.002	FAR-KNOCKOFF	0.071	0.941	0.902	0.996	0.002

A.3 Extensions to $\kappa + r < T < 2\kappa + r$

When $\kappa + r < T < 2\kappa + r$, the construction of knockoffs directly via (3.13) is no longer possible. We may instead adapt the extension approach by Barber and Candès (2015) or Liu et al. (2024) to FAR. Specifically, we draw a $(2\kappa + r - T)$ -dimensional vector $Y^\ddagger$ with i.i.d. $\mathcal{N}(0, 1)$ entries. We then augment the response vector $Y^\dagger$ in (2.12) with the new $(2\kappa + r - T) \times 1$ vector $Y^\ddagger$. As for $\tilde{F}$ and $\tilde{U}_{\tilde{S}}$, we augment each with $(2\kappa + r - T)$ rows of

zeros so that

$$\check{F}^{(a)} \equiv \begin{pmatrix} \check{F} \\ 0 \end{pmatrix}, \quad \check{U}_{\tilde{\mathcal{S}}}^{(a)} \equiv \begin{pmatrix} \check{U}_{\tilde{\mathcal{S}}} \\ 0 \end{pmatrix}.$$

Then approximately (if Σ_ε^0 is estimated accurately enough),

$$\begin{bmatrix} Y^\dagger \\ Y^\ddagger \end{bmatrix} \sim \mathcal{N} \left(\check{F}^{(a)} \check{\Lambda}'_{\tilde{\mathcal{S}}} \beta_{\tilde{\mathcal{S}}} + \check{U}_{\tilde{\mathcal{S}}}^{(a)} \beta_{\tilde{\mathcal{S}}}, I \right).$$

The augmented data therefore contain $2\kappa + r$ observations. We can apply the preceding procedure to construct $\tilde{U}_{\tilde{\mathcal{S}}}$ and then conduct inference in the same way.

A.4 Calibration and Data Generation Process for the Second Simulated Experiment

The calibration and data-generation process is outlined below. Let z_t , $t = 1, \dots, 636$, denote the centered monthly excess returns.

1. Fit z_t with a three-factor model. We apply PCA on the sample covariance of $\{z_t\}_{t=1}^{636}$ and denote σ_k and ξ_k , $k = 1, 2, 3$, as the top three eigenvalues and corresponding eigenvectors, respectively. We estimate factor loadings by $\tilde{\Lambda} = (\sqrt{\sigma_1}\xi_1, \sqrt{\sigma_2}\xi_2, \sqrt{\sigma_3}\xi_3)$ and factors by $\tilde{F}_t = (\sigma_1^{-1/2}\xi_1' z_t, \sigma_2^{-1/2}\xi_2' z_t, \sigma_3^{-1/2}\xi_3' z_t)'$, respectively.
2. Obtain the sample covariance of the rows of $\tilde{\Lambda}$: $\Sigma_\Lambda = \text{diag}(\sigma_1, \sigma_2, \sigma_3)$. Generate loadings matrix Λ whose rows are draws from i.i.d. $\mathcal{N}(0, \Sigma_\Lambda)$.
3. Fit a VAR(1) model $\tilde{F}_t = \Phi \tilde{F}_{t-1} + \eta_t$. Denote $\hat{\Phi}$ the estimate of Φ and calculate $\Sigma_\eta = I - \hat{\Phi}\hat{\Phi}'$. Then generate F_t from the VAR(1) model $F_t = \hat{\Phi}F_{t-1} + \eta_t$ with $F_0 = 0$, where η_t is generated from i.i.d. $\mathcal{N}(0, \Sigma_\eta)$.
4. Calculate the residual $\tilde{U}_t = z_t - \tilde{\Lambda}\tilde{F}_t$ and Σ_U the sample covariance matrix of $\tilde{U}_t$. Denote σ_U^2 the average of the diagonal entries of Σ_U . Generate covariates X_t from a factor model $X_t = \Lambda F_t + U_t$, where the entries in U_t are drawn independently from $\mathcal{N}(0, \sigma_U^2)$.
5. Generate the response Y_t from a sparse linear model $Y_t = X_t' \beta + \varepsilon_t$. The true coefficients are $\beta = (\beta_1, \dots, \beta_{10}, 0'_{(N-10) \times 1})'$, and the nonzero coefficients $\beta_1, \dots, \beta_{10}$ are drawn from i.i.d. Uniform(2,5). We draw ε_t from an AR(1) model $\varepsilon_t = 0.5\varepsilon_{t-1} + e_t$ with $e_t \sim i.i.d. \mathcal{N}(0, 0.3)$.

A.5 Robustness Check for Simulation Design

Below we conduct robustness checks for Experiment 1 with (i) different distribution specifications of ε_t , including symmetric and asymmetric distributions, (ii) non-factor-driven predictors, and (iii) under- or over-estimation of the number of latent factors. We then report Monte Carlo variability and two additional stress tests aligned with the revised theory.

Table 8: Simulation results for Experiment 1 of Gaussian errors with target FDR $q = 0.1$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.699	0.913	0.451	0.886	0.116	LASSO	0.715	0.998	0.440	0.867	0.140
FARMSELECT	0.337	1.000	0.774	0.963	0.039	FARMSELECT	0.211	1.000	0.860	0.977	0.024
MODEL-X	0.082	0.973	0.928	0.993	0.005	MODEL-X	0.014	0.125	0.118	0.955	0.001
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.008	0.111	0.107	0.955	0.001
TSKI	0.000	0.000	0.000	0.950	0.000	TSKI	0.000	0.000	0.000	0.950	0.000
FAR-KNOCKOFF	0.024	0.633	0.623	0.980	0.001	FAR-KNOCKOFF	0.036	0.941	0.923	0.995	0.002
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.769	0.877	0.364	0.921	0.078	LASSO	0.768	0.921	0.368	0.919	0.081
FARMSELECT	0.412	1.000	0.714	0.974	0.027	FARMSELECT	0.248	1.000	0.830	0.985	0.016
MODEL-X	0.104	0.953	0.897	0.995	0.004	MODEL-X	0.076	0.979	0.938	0.997	0.002
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.001	0.002	0.001	0.975	0.000	TSKI	0.000	0.000	0.000	0.975	0.000
FAR-KNOCKOFF	0.022	0.551	0.541	0.988	0.001	FAR-KNOCKOFF	0.041	0.970	0.948	0.998	0.001

Table 9: Simulation results for Experiment 1 of Gaussian errors with target FDR $q = 0.2$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.699	0.913	0.451	0.886	0.116	LASSO	0.715	0.998	0.440	0.867	0.140
FARMSELECT	0.337	1.000	0.774	0.963	0.039	FARMSELECT	0.211	1.000	0.860	0.977	0.024
MODEL-X	0.164	0.997	0.902	0.988	0.013	MODEL-X	0.055	0.337	0.331	0.963	0.004
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.056	0.375	0.372	0.965	0.004
TSKI	0.056	0.380	0.358	0.965	0.004	TSKI	0.090	0.619	0.580	0.975	0.006
FAR-KNOCKOFF	0.039	0.922	0.928	0.994	0.002	FAR-KNOCKOFF	0.051	0.983	0.959	0.996	0.003
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.769	0.876	0.364	0.921	0.078	LASSO	0.768	0.921	0.368	0.919	0.081
FARMSELECT	0.412	1.000	0.714	0.974	0.027	FARMSELECT	0.248	1.000	0.830	0.985	0.016
MODEL-X	0.230	0.994	0.855	0.990	0.010	MODEL-X	0.165	0.995	0.899	0.994	0.006
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.019	0.142	0.137	0.978	0.001	TSKI	0.034	0.406	0.392	0.984	0.001
FAR-KNOCKOFF	0.038	0.882	0.895	0.996	0.001	FAR-KNOCKOFF	0.059	0.992	0.962	0.998	0.002

In Tables 8 and 9, we report simulation results with $\varepsilon_t \sim \mathcal{N}(0, 1)$ for Experiment 1.

In Tables 10 and 11, we report simulation results with $\varepsilon_t \sim (\chi^2(3) - 3)/\sqrt{6}$ for Experiment 1.

Table 10: Simulation results for Experiment 1 with $\varepsilon_t \sim (\chi^2(3) - 3)/\sqrt{6}$ and target FDR $q = 0.1$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.696	0.918	0.454	0.886	0.115	LASSO	0.710	0.998	0.446	0.870	0.136
FARMSELECT	0.372	1.000	0.748	0.958	0.045	FARMSELECT	0.210	1.000	0.861	0.977	0.024
MODEL-X	0.074	0.945	0.904	0.993	0.005	MODEL-X	0.012	0.159	0.153	0.957	0.001
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.019	0.164	0.154	0.957	0.001
TSKI	0.000	0.000	0.000	0.950	0.000	TSKI	0.005	0.010	0.006	0.950	0.001
FAR-KNOCKOFF	0.027	0.591	0.579	0.978	0.002	FAR-KNOCKOFF	0.039	0.937	0.917	0.995	0.002
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.767	0.879	0.367	0.922	0.077	LASSO	0.765	0.927	0.372	0.919	0.081
FARMSELECT	0.438	1.000	0.691	0.971	0.030	FARMSELECT	0.257	1.000	0.824	0.983	0.017
MODEL-X	0.109	0.899	0.840	0.994	0.004	MODEL-X	0.073	0.975	0.935	0.997	0.002
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.000	0.000	0.000	0.975	0.000	TSKI	0.000	0.000	0.000	0.975	0.000
FAR-KNOCKOFF	0.029	0.591	0.579	0.989	0.001	FAR-KNOCKOFF	0.047	0.978	0.953	0.998	0.001

Table 11: Simulation results for Experiment 1 with $\varepsilon_t \sim (\chi^2(3) - 3)/\sqrt{6}$ and target FDR $q = 0.2$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.696	0.918	0.454	0.886	0.115	LASSO	0.710	0.998	0.446	0.870	0.136
FARMSELECT	0.372	1.000	0.748	0.958	0.045	FARMSELECT	0.210	1.000	0.861	0.977	0.024
MODEL-X	0.161	0.997	0.903	0.988	0.013	MODEL-X	0.063	0.395	0.384	0.965	0.005
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.061	0.359	0.346	0.964	0.005
TSKI	0.054	0.400	0.380	0.966	0.004	TSKI	0.083	0.588	0.546	0.974	0.006
FAR-KNOCKOFF	0.043	0.900	0.905	0.993	0.003	FAR-KNOCKOFF	0.053	0.985	0.960	0.996	0.003
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.767	0.879	0.367	0.922	0.077	LASSO	0.765	0.927	0.372	0.919	0.081
FARMSELECT	0.438	1.000	0.691	0.971	0.030	FARMSELECT	0.257	1.000	0.824	0.983	0.017
MODEL-X	0.234	0.991	0.850	0.990	0.010	MODEL-X	0.162	0.996	0.902	0.994	0.006
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.019	0.177	0.170	0.979	0.001	TSKI	0.043	0.429	0.411	0.984	0.001
FAR-KNOCKOFF	0.046	0.888	0.894	0.996	0.001	FAR-KNOCKOFF	0.071	0.992	0.954	0.998	0.002

In Tables 12 and 13, we report simulation results with non-factor-driven predictors for Experiment 1. Specifically, $X_t \sim \mathcal{N}(0, \Sigma_X)$ with $\Sigma_X(i, j) = 0.5^{|i-j|}$.

In Tables 14 and 15, we report simulation results of FAR-Knockoff by fixing the estimated number of PCA factors $\hat{r}$ to be smaller than or greater than the true number of factors (3) for Experiment 1.

The corrected original-coordinate evaluation leads to two qualifications. First, under the non-factor design, FAR-Knockoff continues to control the target FDR, with power between 0.871 and 0.898 at $q = 0.2$. Second, underestimating the factor number is more

Table 12: Simulation results for Experiment 1 with non-factor-driven predictors and target FDR $q = 0.1$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.545	1.000	0.609	0.924	0.080	LASSO	0.578	1.000	0.574	0.910	0.095
FARMSELECT	0.074	0.989	0.946	0.989	0.011	FARMSELECT	0.199	1.000	0.861	0.974	0.027
MODEL-X	0.079	1.000	0.955	0.995	0.006	MODEL-X	0.051	0.630	0.601	0.978	0.003
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.053	0.675	0.645	0.980	0.004
TSKI	0.000	0.000	0.000	0.950	0.000	TSKI	0.000	0.000	0.000	0.950	0.000
FAR-KNOCKOFF	0.045	0.668	0.643	0.981	0.003	FAR-KNOCKOFF	0.070	0.733	0.694	0.982	0.005
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.599	1.000	0.554	0.950	0.052	LASSO	0.580	1.000	0.573	0.955	0.046
FARMSELECT	0.032	1.000	0.978	0.998	0.002	FARMSELECT	0.029	1.000	0.976	0.996	0.004
MODEL-X	0.064	1.000	0.965	0.998	0.002	MODEL-X	0.082	1.000	0.954	0.997	0.003
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.000	0.000	0.000	0.975	0.000	TSKI	0.000	0.000	0.000	0.975	0.000
FAR-KNOCKOFF	0.050	0.687	0.660	0.991	0.002	FAR-KNOCKOFF	0.053	0.672	0.642	0.990	0.002

Table 13: Simulation results for Experiment 1 with non-factor-driven predictors and target FDR $q = 0.2$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.545	1.000	0.609	0.924	0.080	LASSO	0.578	1.000	0.574	0.910	0.095
FARMSELECT	0.074	0.989	0.946	0.989	0.011	FARMSELECT	0.199	1.000	0.861	0.974	0.027
MODEL-X	0.152	1.000	0.911	0.989	0.012	MODEL-X	0.153	0.823	0.750	0.980	0.012
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.130	0.859	0.802	0.984	0.010
TSKI	0.102	1.000	0.943	0.993	0.007	TSKI	0.144	1.000	0.918	0.990	0.010
FAR-KNOCKOFF	0.099	0.871	0.836	0.987	0.007	FAR-KNOCKOFF	0.144	0.898	0.830	0.985	0.011
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.599	1.000	0.554	0.950	0.052	LASSO	0.580	1.000	0.573	0.955	0.046
FARMSELECT	0.032	1.000	0.978	0.998	0.002	FARMSELECT	0.029	1.000	0.976	0.996	0.004
MODEL-X	0.143	1.000	0.916	0.995	0.005	MODEL-X	0.161	1.000	0.906	0.994	0.006
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.126	1.000	0.928	0.996	0.004	TSKI	0.129	1.000	0.927	0.996	0.004
FAR-KNOCKOFF	0.104	0.885	0.845	0.994	0.003	FAR-KNOCKOFF	0.130	0.871	0.814	0.992	0.005

Table 14: Simulation results of FAR-Knockoff for Experiment 1 with under- and over-estimated $\hat{r}$ and target FDR $q = 0.1$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
$\hat{r}$	FDR	Power	F_1	Accuracy	FPR	$\hat{r}$	FDR	Power	F_1	Accuracy	FPR
2	0.006	0.122	0.120	0.956	0.000	2	0.018	0.443	0.434	0.971	0.001
4	0.021	0.535	0.525	0.976	0.001	4	0.037	0.929	0.910	0.994	0.002
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
$\hat{r}$	FDR	Power	F_1	Accuracy	FPR	$\hat{r}$	FDR	Power	F_1	Accuracy	FPR
2	0.005	0.060	0.058	0.976	0.000	2	0.018	0.445	0.436	0.986	0.001
4	0.025	0.551	0.541	0.988	0.001	4	0.043	0.961	0.938	0.998	0.001

Table 15: Simulation results of FAR-Knockoff for Experiment 1 with under- and over-estimated $\hat{r}$ and target FDR $q = 0.2$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
$\hat{r}$	FDR	Power	F_1	Accuracy	FPR	$\hat{r}$	FDR	Power	F_1	Accuracy	FPR
2	0.029	0.453	0.493	0.971	0.001	2	0.044	0.749	0.763	0.985	0.003
4	0.036	0.887	0.902	0.992	0.002	4	0.051	0.972	0.949	0.996	0.003
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
$\hat{r}$	FDR	Power	F_1	Accuracy	FPR	$\hat{r}$	FDR	Power	F_1	Accuracy	FPR
2	0.021	0.226	0.255	0.980	0.000	2	0.043	0.743	0.751	0.992	0.001
4	0.039	0.858	0.874	0.995	0.001	4	0.066	0.990	0.956	0.998	0.002

consequential than overestimating it: fixing $\hat{r} = 2$ preserves FDR control but can substantially reduce power, whereas fixing $\hat{r} = 4$ has a much smaller effect. Thus the robustness concerns FDR validity, not invariance of power to factor-number misspecification.

In Tables 16 and 17, we report simulation results with stronger temporal correlation for ε_t for Experiment 2. Specifically, We draw ε_t from an AR(1) model $\varepsilon_t = 0.9\varepsilon_{t-1} + e_t$ with $e_t \sim i.i.d.\mathcal{N}(0, 0.3)$.

Table 16: Simulation results for Experiment 2 with stronger temporal correlation and target FDR $q = 0.1$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.872	0.105	0.113	0.917	0.040	LASSO	0.843	0.189	0.159	0.906	0.057
FARMSELECT	0.020	1.000	0.989	0.999	0.001	FARMSELECT	0.204	1.000	0.862	0.976	0.025
MODEL-X	0.142	0.124	0.111	0.947	0.009	MODEL-X	0.003	0.014	0.013	0.950	0.000
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.000	0.000	0.000	0.950	0.000
TSKI	0.000	0.000	0.000	0.950	0.000	TSKI	0.000	0.000	0.000	0.950	0.000
FAR-KNOCKOFF	0.042	0.719	0.696	0.983	0.003	FAR-KNOCKOFF	0.078	0.769	0.725	0.983	0.006
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.911	0.092	0.089	0.952	0.025	LASSO	0.922	0.087	0.081	0.951	0.027
FARMSELECT	0.039	1.000	0.977	0.999	0.002	FARMSELECT	0.002	1.000	0.999	1.000	0.000
MODEL-X	0.096	0.055	0.051	0.974	0.003	MODEL-X	0.100	0.091	0.085	0.974	0.003
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.000	0.000	0.000	0.975	0.000	TSKI	0.000	0.000	0.000	0.975	0.000
FAR-KNOCKOFF	0.042	0.719	0.696	0.992	0.001	FAR-KNOCKOFF	0.076	0.893	0.849	0.995	0.003

A.6 Monte Carlo variability

Tables 18 and 19 report Monte Carlo means and across-replication standard deviations for the regenerated screened procedures. Each configuration has 500 replications. The means are the same as those in the corresponding simulation tables. Empty selections are included with FDP and power equal to zero; all 500 fits succeeded in these reruns. The standard error of a reported mean is its standard deviation divided by $\sqrt{500}$.

Power dispersion can be large when a procedure alternates between few discoveries and recovery of most signals. Small mean FDR together with small power can reflect inactivity rather than accurate variable recovery. The screened benchmarks in Experiment 2 particularly illustrate why both criteria should be read together.

Table 17: Simulation results for Experiment 2 with stronger temporal correlation and target FDR $q = 0.2$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.872	0.105	0.113	0.917	0.040	LASSO	0.843	0.189	0.159	0.906	0.057
FARMSELECT	0.020	1.000	0.989	0.999	0.001	FARMSELECT	0.204	1.000	0.862	0.976	0.025
MODEL-X	0.487	0.361	0.352	0.940	0.029	MODEL-X	0.030	0.064	0.073	0.952	0.001
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	0.057	0.069	0.080	0.951	0.003
TSKI	0.002	0.003	0.003	0.950	0.000	TSKI	0.002	0.004	0.004	0.950	0.000
FAR-KNOCKOFF	0.095	0.892	0.857	0.989	0.006	FAR-KNOCKOFF	0.185	0.939	0.842	0.982	0.016
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
LASSO	0.911	0.092	0.089	0.952	0.025	LASSO	0.922	0.087	0.081	0.951	0.027
FARMSELECT	0.039	1.000	0.977	0.999	0.002	FARMSELECT	0.002	1.000	0.999	1.000	0.000
MODEL-X	0.460	0.269	0.282	0.970	0.012	MODEL-X	0.450	0.333	0.344	0.973	0.011
FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.	FIXED-X	N.A.	N.A.	N.A.	N.A.	N.A.
TSKI	0.000	0.000	0.000	0.975	0.000	TSKI	0.000	0.000	0.000	0.975	0.000
FAR-KNOCKOFF	0.096	0.893	0.859	0.994	0.003	FAR-KNOCKOFF	0.152	0.968	0.884	0.993	0.006

Table 18: Monte Carlo variability of FAR-Knockoff

Design	Method	q	N	T	FDR (SD)	Power (SD)
Experiment 1	FAR	0.1	200	200	0.033 (0.062)	0.656 (0.472)
			200	400	0.034 (0.062)	0.935 (0.245)
			400	200	0.028 (0.055)	0.602 (0.485)
			400	400	0.047 (0.071)	0.976 (0.153)
Experiment 1	FAR	0.2	200	200	0.047 (0.071)	0.919 (0.176)
			200	400	0.050 (0.074)	0.984 (0.083)
			400	200	0.038 (0.062)	0.892 (0.205)
			400	400	0.069 (0.085)	0.991 (0.084)
Experiment 2	FAR	0.1	200	200	0.070 (0.091)	0.915 (0.278)
			200	400	0.059 (0.086)	0.900 (0.294)
			400	200	0.071 (0.090)	0.938 (0.241)
			400	400	0.071 (0.097)	0.941 (0.234)
Experiment 2	FAR	0.2	200	200	0.146 (0.122)	0.978 (0.105)
			200	400	0.154 (0.145)	0.977 (0.093)
			400	200	0.137 (0.118)	0.977 (0.130)
			400	400	0.158 (0.145)	0.985 (0.092)
Normal errors	FAR	0.1	200	200	0.024 (0.051)	0.633 (0.478)
			200	400	0.036 (0.057)	0.941 (0.234)
			400	200	0.022 (0.049)	0.551 (0.495)
			400	400	0.041 (0.066)	0.970 (0.171)
Normal errors	FAR	0.2	200	200	0.039 (0.061)	0.922 (0.164)
			200	400	0.051 (0.067)	0.983 (0.106)
			400	200	0.038 (0.063)	0.882 (0.219)
			400	400	0.059 (0.074)	0.992 (0.069)
Skewed errors	FAR	0.1	200	200	0.027 (0.053)	0.591 (0.488)
			200	400	0.039 (0.060)	0.937 (0.241)
			400	200	0.029 (0.057)	0.591 (0.486)
			400	400	0.047 (0.071)	0.978 (0.147)
Skewed errors	FAR	0.2	200	200	0.043 (0.065)	0.900 (0.204)
			200	400	0.053 (0.071)	0.985 (0.088)
			400	200	0.046 (0.068)	0.888 (0.219)
			400	400	0.071 (0.087)	0.992 (0.078)

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Table 18 (continued)

Design	Method	q	N	T	FDR (SD)	Power (SD)
Non-factor design	FAR	0.1	200	200	0.045 (0.077)	0.668 (0.471)
			200	400	0.070 (0.103)	0.733 (0.442)
			400	200	0.050 (0.083)	0.687 (0.463)
			400	400	0.053 (0.089)	0.672 (0.470)
Non-factor design	FAR	0.2	200	200	0.099 (0.104)	0.871 (0.283)
			200	400	0.144 (0.137)	0.898 (0.259)
			400	200	0.104 (0.108)	0.885 (0.269)
			400	400	0.130 (0.139)	0.871 (0.285)
AR(0.9) errors	FAR	0.1	200	200	0.042 (0.076)	0.719 (0.449)
			200	400	0.078 (0.118)	0.769 (0.421)
			400	200	0.042 (0.077)	0.719 (0.449)
			400	400	0.076 (0.119)	0.893 (0.308)
AR(0.9) errors	FAR	0.2	200	200	0.095 (0.105)	0.892 (0.259)
			200	400	0.185 (0.175)	0.939 (0.177)
			400	200	0.096 (0.107)	0.893 (0.253)
			400	400	0.152 (0.162)	0.968 (0.138)
$\hat{r} = 2$	FAR	0.1	200	200	0.006 (0.028)	0.122 (0.325)
			200	400	0.018 (0.048)	0.443 (0.496)
			400	200	0.005 (0.027)	0.060 (0.233)
			400	400	0.018 (0.049)	0.445 (0.497)
$\hat{r} = 2$	FAR	0.2	200	200	0.029 (0.073)	0.453 (0.404)
			200	400	0.044 (0.072)	0.749 (0.354)
			400	200	0.021 (0.069)	0.226 (0.344)
			400	400	0.043 (0.069)	0.743 (0.373)
$\hat{r} = 4$	FAR	0.1	200	200	0.021 (0.051)	0.535 (0.497)
			200	400	0.037 (0.066)	0.929 (0.255)
			400	200	0.025 (0.052)	0.551 (0.492)
			400	400	0.043 (0.070)	0.961 (0.191)
$\hat{r} = 4$	FAR	0.2	200	200	0.036 (0.065)	0.887 (0.205)
			200	400	0.051 (0.079)	0.972 (0.139)
			400	200	0.039 (0.062)	0.858 (0.244)
			400	400	0.066 (0.087)	0.990 (0.080)

Table 19: Monte Carlo variability of pre-screened benchmarks

Design	Method	q	N	T	FDR (SD)	Power (SD)
Experiment 1	S+MX	0.1	200	200	0.030 (0.083)	0.197 (0.385)
			200	400	0.072 (0.115)	0.560 (0.477)
			400	200	0.025 (0.093)	0.092 (0.269)
			400	400	0.073 (0.141)	0.353 (0.447)
Experiment 1	S+MX	0.2	200	200	0.114 (0.147)	0.560 (0.405)
			200	400	0.176 (0.166)	0.807 (0.310)
			400	200	0.117 (0.197)	0.337 (0.367)
			400	400	0.229 (0.212)	0.665 (0.328)
Experiment 1	S+FX	0.1	200	200	0.037 (0.095)	0.223 (0.403)
			200	400	0.084 (0.128)	0.573 (0.475)
			400	200	0.025 (0.085)	0.099 (0.278)
			400	400	0.088 (0.154)	0.363 (0.445)
Experiment 1	S+FX	0.2	200	200	0.114 (0.156)	0.548 (0.413)
			200	400	0.191 (0.178)	0.828 (0.286)
			400	200	0.114 (0.186)	0.357 (0.373)
			400	400	0.241 (0.214)	0.673 (0.318)

Continued on next page

Table 19 (continued)

Design	Method	q	N	T	FDR (SD)	Power (SD)
Experiment 2	S+MX	0.1	200	200	0.006 (0.066)	0.002 (0.032)
			200	400	0.004 (0.060)	0.000 (0.004)
			400	200	0.006 (0.073)	0.000 (0.009)
			400	400	0.014 (0.110)	0.002 (0.026)
Experiment 2	S+MX	0.2	200	200	0.203 (0.379)	0.020 (0.070)
			200	400	0.252 (0.402)	0.024 (0.065)
			400	200	0.228 (0.401)	0.014 (0.055)
			400	400	0.360 (0.460)	0.019 (0.058)
Experiment 2	S+FX	0.1	200	200	0.004 (0.057)	0.000 (0.006)
			200	400	0.006 (0.075)	0.002 (0.036)
			400	200	0.008 (0.084)	0.002 (0.026)
			400	400	0.011 (0.104)	0.001 (0.008)
Experiment 2	S+FX	0.2	200	200	0.215 (0.382)	0.027 (0.087)
			200	400	0.271 (0.413)	0.028 (0.080)
			400	200	0.231 (0.404)	0.013 (0.048)
			400	400	0.381 (0.465)	0.018 (0.054)

S+MX and S+FX denote Screening+Model-X and Screening+Fixed-X, respectively. SD is the standard deviation of the replication-level FDP or power, not a standard error.

A.7 Conditional heteroskedasticity and imperfect screening

We add two stress tests targeted at the two departures covered by the revised theory. Both use $N = T = 400$, three latent factors, and 500 Monte Carlo replications. The first retains the ten signals of Experiment 1 but generates the regression error from a stationary GARCH(1, 1) process,

$$\varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = 0.05 + 0.10\varepsilon_{t-1}^2 + 0.85\sigma_{t-1}^2, \quad z_t \stackrel{\text{i.i.d.}}{\sim} N(0, 1).$$

This is a conditionally heteroskedastic martingale-difference special case of the causal-innovation framework in Appendix Assumption C.1. The second stress test retains the $t(5)$ error and deliberately appends three weak coefficients, 0.35, 0.50, and 0.65, to ten strong coefficients drawn from $\text{Unif}(1, 3)$. We use a more aggressive first-stage target of $0.10T_k$ to create imperfect screening through these weak signals. This calibration behaves as intended: among 697 signal omissions over the 500 replications, 695 (99.7%) are one of the three appended weak signals. The screen misses zero, one, two, and three weak signals in 53, 236, 174, and 37 replications, respectively. A strong signal is missed in only two isolated replications (0.4%), and no replication misses more than three signals in total.

The GARCH results show that conditional heteroskedasticity does not compromise FDR control or power in this design. The imperfect-screening results should be interpreted specifically as a weak-signal experiment: the strong signals are all retained in 99.6% of replications, and 99.7% of all signal omissions come from the three deliberately weak coefficients. Those weak omissions reduce attainable power, but they do not enter the null sign-symmetry argument, and FDR remains controlled. This is the finite-sample pattern predicted by the drift and power terms in Theorems 4.4 and 4.5. It is not a claim of robustness to omitted strong signals, which are excluded by the small missed-signal condition underlying the theory.

Table 20: FAR-Knockoff under GARCH errors and imperfect screening

Design	q	FDR	Power	Weak signals missed
GARCH(1, 1) errors	0.1	0.043 (0.072)	0.972 (0.165)	0.000 (0.000)
	0.2	0.065 (0.091)	0.996 (0.033)	0.000 (0.000)
Weak-signal screen	0.1	0.001 (0.009)	0.569 (0.368)	1.390 (0.774)
	0.2	0.002 (0.012)	0.745 (0.155)	1.390 (0.774)

Notes: Entries are Monte Carlo means with across-replication standard deviations in parentheses. Every row is based on 500 successful replications. FDR is the mean realized false-discovery proportion and power is the fraction of all 13 active variables selected. Thus the reported power deliberately counts an omitted weak signal as a failure to discover and does not conceal the first-stage loss. The final column counts only the three preassigned weak signals; the two isolated strong-signal omissions described in the text are excluded from that column.

A.8 Sensitivity to weak factors

To assess whether the results rely on strong pervasive factors, retain Experiment 1 but replace each loading λ_i by $\lambda_i^{(\alpha)} = N^{(\alpha-1)/2}\lambda_i$, with $\alpha \in \{1, 0.75, 0.5, 0\}$. The loading Gram eigenvalues then have order N^α : $\alpha = 1$ is the strong-factor baseline, $\alpha = 0.75, 0.5$ give weak-factor sequences, and $\alpha = 0$ is a bounded-strength boundary case. We keep three generated factors, ten coefficients drawn from $\text{Unif}(1, 3)$, idiosyncratic variance 0.25, and standardized $t(5)$ errors. Shared random draws across α hold the idiosyncratic signal-to-noise ratio $0.25\|\beta\|_2^2$ fixed; the total unconditional signal-to-noise ratio is not fixed because the common component changes. Thus this is a weak-factor experiment, not a weak-signal experiment.

For each $T = N \in \{200, 400\}$ and each strength, we use 200 replications and evaluate both $q = 0.1$ and $q = 0.2$ from the same fitted statistics. The half-sample FAR implementation and tuning in Appendix A.1 are unchanged across strengths. The true factor count is not supplied: each fold chooses from 1 to 8, so a zero-factor model is not a candidate. All 1,600 fits complete successfully. Empty selections remain in the averages, with FDP and power equal to zero.

Table 21 shows no evident FDR breakdown as loadings weaken. At $T = N = 200$ and $q = 0.1$, power changes from 0.642 to 0.627 and 0.619 at the two weak-factor levels. Their paired differences from the baseline have Monte Carlo standard errors 0.0268 and 0.0348, respectively; these small differences do not establish a systematic power loss. At $T = N = 400$, power remains above 0.97 at both weak-factor levels. Both folds still select three factors in every replication at $\alpha = 0.75, 0.5$. At the boundary $\alpha = 0$, both instead select one factor in every replication, while empirical FDR remains below target. This supports finite-sample robustness in the designs examined, but does not establish robustness to arbitrary factor misspecification. The experiment isolates FAR-Knockoff’s sensitivity; it does not replace a cross-method tuning comparison.

A.9 Pre-screened Model-X and Fixed-X

Tables 22–25 examine Model-X and Fixed-X after screening the raw covariates on the first half-sample and performing knockoff inference on the second. Unlike FAR-Knockoff, these screens do not remove common factors. Their Lasso-path size target is $\min(0.25T, 0.7N)$,

Table 21: FAR-Knockoff under weaker factor loadings

$T = N$	α	$q = 0.1$		$q = 0.2$	
		FDR	Power	FDR	Power
200	1	0.0239 (0.0035)	0.6420 (0.0338)	0.0338 (0.0039)	0.9260 (0.0116)
200	0.75	0.0250 (0.0037)	0.6270 (0.0341)	0.0362 (0.0043)	0.9305 (0.0108)
200	0.5	0.0248 (0.0038)	0.6190 (0.0340)	0.0409 (0.0046)	0.9350 (0.0078)
200	0	0.0349 (0.0041)	0.7500 (0.0303)	0.0534 (0.0046)	0.9535 (0.0086)
400	1	0.0386 (0.0046)	0.9600 (0.0139)	0.0534 (0.0053)	0.9975 (0.0013)
400	0.75	0.0372 (0.0047)	0.9850 (0.0086)	0.0615 (0.0062)	0.9990 (0.0007)
400	0.5	0.0360 (0.0044)	0.9750 (0.0111)	0.0538 (0.0053)	0.9955 (0.0027)
400	0	0.0450 (0.0050)	0.9745 (0.0111)	0.0681 (0.0061)	0.9915 (0.0054)

Notes: Entries are Monte Carlo means with Monte Carlo standard errors in parentheses, based on 200 replications per row. Standard errors equal the across-replication standard deviations divided by $\sqrt{200}$. The filter uses knockoff+ offset one. Power is the fraction of all ten active variables selected; neither empty selections nor screening omissions are excluded. The $\alpha = 0$ rows are boundary tests, not instances of consistent estimation of three factors.

compared with $\min(0.25\lfloor T/2 \rfloor, 0.7N)$ for FAR-Knockoff. All selections are evaluated in the original coordinate system. Thus these are additional implementations, not a controlled decomposition of FAR-Knockoff’s components.

Pre-screening alone is not sufficient for reliable selection. In the calibrated design, the raw-covariate screen can miss relevant predictors; at $q = 0.2$, the resulting procedures combine low power with substantial false-discovery rates in several configurations. At $q = 0.1$, few discoveries can instead produce small empirical FDR without useful power. These results caution against interpreting a small FDR in isolation. Monte Carlo standard deviations are reported in Appendix A.6.

Table 22: Simulation results for Experiment 1 with target FDR $q = 0.1$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
SCREENING+MODEL-X	0.030	0.197	0.187	0.958	0.002	SCREENING+MODEL-X	0.072	0.560	0.531	0.973	0.005
SCREENING+FIXED-X	0.037	0.223	0.211	0.959	0.002	SCREENING+FIXED-X	0.084	0.573	0.538	0.973	0.006
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
SCREENING+MODEL-X	0.025	0.092	0.087	0.977	0.001	SCREENING+MODEL-X	0.073	0.353	0.335	0.982	0.002
SCREENING+FIXED-X	0.025	0.099	0.095	0.977	0.001	SCREENING+FIXED-X	0.088	0.363	0.341	0.981	0.003

B Additional Empirical Results

For completeness, Table 26 reports the same comparison for four- and five-year bonds. The yield-factors-plus-CP benchmark has the lowest RMSE at both maturities, and the Clark–West tests provide no evidence of incremental predictive accuracy. Table 27 reports

Table 23: Simulation results for Experiment 1 with target FDR $q = 0.2$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
SCREENING+MODEL-X	0.114	0.560	0.551	0.971	0.008	SCREENING+MODEL-X	0.176	0.807	0.744	0.978	0.013
SCREENING+FIXED-X	0.114	0.548	0.535	0.970	0.008	SCREENING+FIXED-X	0.191	0.828	0.755	0.977	0.015
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
SCREENING+MODEL-X	0.117	0.337	0.344	0.980	0.004	SCREENING+MODEL-X	0.229	0.665	0.628	0.984	0.008
SCREENING+FIXED-X	0.114	0.357	0.368	0.981	0.003	SCREENING+FIXED-X	0.241	0.673	0.630	0.984	0.008

Table 24: Simulation results for Experiment 2 with target FDR $q = 0.1$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
SCREENING+MODEL-X	0.006	0.002	0.002	0.950	0.000	SCREENING+MODEL-X	0.004	0.000	0.000	0.950	0.000
SCREENING+FIXED-X	0.004	0.000	0.000	0.950	0.000	SCREENING+FIXED-X	0.006	0.002	0.002	0.950	0.000
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
SCREENING+MODEL-X	0.006	0.000	0.000	0.975	0.000	SCREENING+MODEL-X	0.014	0.002	0.002	0.975	0.000
SCREENING+FIXED-X	0.008	0.002	0.002	0.975	0.000	SCREENING+FIXED-X	0.011	0.001	0.001	0.975	0.000

Table 25: Simulation results for Experiment 2 with target FDR $q = 0.2$

(a) $N = 200, T = 200$						(b) $N = 200, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
SCREENING+MODEL-X	0.203	0.020	0.024	0.945	0.007	SCREENING+MODEL-X	0.252	0.024	0.030	0.943	0.008
SCREENING+FIXED-X	0.215	0.027	0.032	0.945	0.007	SCREENING+FIXED-X	0.271	0.028	0.033	0.943	0.009
(c) $N = 400, T = 200$						(d) $N = 400, T = 400$					
Method	FDR	Power	F_1	Accuracy	FPR	Method	FDR	Power	F_1	Accuracy	FPR
SCREENING+MODEL-X	0.228	0.014	0.017	0.972	0.004	SCREENING+MODEL-X	0.360	0.019	0.022	0.970	0.006
SCREENING+FIXED-X	0.231	0.013	0.015	0.972	0.004	SCREENING+FIXED-X	0.381	0.018	0.022	0.969	0.006

the corresponding selection frequencies. Uncertainty and labor-utilization signals recur in individual windows, but the resulting forecasts do not improve on the yield benchmark.

Table 26: Longer-maturity forecast performance

	4 years	5 years
Yield factors + CP	4.028 (–)	5.104 (–)
LASSO	6.927 (21.04)	7.876 (21.05)
FARMSELECT	7.464 (39.01)	9.253 (32.58)
MODEL-X	4.037 (3.51)	5.467 (3.36)
FIXED-X	4.149 (3.01)	5.530 (3.02)
FAR-KNOCKOFF	4.096 (1.97)	5.219 (1.90)
Clark–West statistic	–0.32	–0.94
One-sided p -value	0.6241	0.8253

Notes: Entries and tests are defined as in Table 4.

Table 27: Most frequently selected longer-maturity residual predictors

Maturity	FRED-MD series	Frequency	Incremental economic channel
4 years	VIXCLSx	26.3%	uncertainty and risk-bearing capacity
	T5YFFM	26.3%	policy-path / term-premium news
	CES0600000007	17.4%	goods-producing workweek
5 years	VIXCLSx	36.0%	uncertainty and risk-bearing capacity
	T5YFFM	21.2%	policy-path / term-premium news
	CES0600000007	15.7%	goods-producing workweek

Sensitivity to the factor cap. Table 28 retains the earlier cap-eight results alongside the updated cap-six specification, with all other FAR settings unchanged. Both caps give a two-year improvement and no longer-maturity gain; the three-year rejection occurs only with the lower cap. The three-year evidence is therefore specification-sensitive, rather than a uniform rejection across factor choices. With a cap of six, the estimation and inference PCA stages select six in all 944 window–maturity fits; recycling selects five in 10 fits and six otherwise. The cap is binding, so this exercise does not establish that six is the optimal factor count.

Table 28: FAR-Knockoff sensitivity to the macro factor cap

Maturity	RMSE		CW one-sided p -value	
	Cap 6	Cap 8	Cap 6	Cap 8
2 years	1.380	1.389	0.0065	0.0027
3 years	2.763	2.802	0.0377	0.1059
4 years	4.096	4.130	0.6241	0.8897
5 years	5.219	5.244	0.8253	0.9592

C Technical Results for the Main Theory

Causal-martingale conditions. The two FDR theorems in the main text use the same conditional-score reduction. The following assumptions are needed only for the causal martingale theorem, where the first arrow in the displayed proof pipeline is implemented by a score-level Gaussian coupling. To keep the main argument readable, the associated notation and threshold conditions are collected here.

Assumption C.1 (Causal martingale innovations). *The serially dependent error has the causal representation $\varepsilon = \Theta u$, with $\Theta\Theta' = \Sigma_\varepsilon^0$, where Θ is lower triangular and*

$$\max_s \sum_{t \geq s} |\Theta_{ts}| \leq C_\Theta.$$

Relative to a filtration $\{\mathcal{F}_s\}$, the innovations satisfy $\mathbb{E}(u_s \mid \mathcal{F}_{s-1}) = 0$, $\mathbb{E}(u_s^2) = 1$, and $\sup_s \mathbb{E}|u_s|^3 < \infty$. Their conditional variances may vary over time. Thus u_s , rather than the serially correlated error ε_t , is the martingale difference. Gaussianity and symmetry are not imposed.

The column-summability bound on Θ is a short-memory stability condition, not a distributional consequence of Gaussianity. It converts the realized score into a sum of predictable, nondominating martingale weights. It holds for finite-order stable filters and for many causal processes with geometrically decaying impulse responses.

Assumption C.2 (Score coupling). *There is a covariance-matched $\zeta^* \mid \mathcal{G} \sim N(0, \Xi)$ and deterministic sequences $\rho_{NT} \rightarrow 0$ and $\eta_{NT} \rightarrow 0$ such that*

$$\mathbb{P}(\|\zeta - \zeta^*\|_2 > \rho_{NT}) \leq \eta_{NT}.$$

Define $\Delta_{NT} = \sqrt{2\rho_{NT}}/\{\sigma_{\min}(\Psi)\sqrt{T}\}$. Section C.5 verifies this assumption from predictable martingale weights, a no-dominating-leverage bound, and control of conditional quadratic variation. Only the 2κ -dimensional sufficient score is coupled, not the original $T \times N$ data array.

Apply the same Lasso score-to-statistic map to $D + \zeta^*$ and denote the result by $W^* = (W_1^*, \dots, W_\kappa^*)$. Under P_D^* , let $V^{*,+}(t)$ and $V^{*,-}(t)$ count retained nulls with $W_j^* \geq t$ and $W_j^* \leq -t$, respectively, and put $M^*(t) = V^{*,+}(t) + V^{*,-}(t)$. Let $\tilde{N}_{2\Delta}^*(t)$ count retained nulls satisfying $||W_j^*| - t| \leq 2\Delta_{NT}$.

Assumption C.3 (Threshold stability). *There are positive integers m_{NT}, r_{NT} , a nonnegative integer e_{NT} , a sequence $v_{NT} \rightarrow \infty$, and $\gamma_{NT}, \varsigma_{NT} \rightarrow 0$ such that $v_{NT}\Delta_{NT} \rightarrow 0$, $m_{NT}/\ln(NT) \rightarrow \infty$, and $(m_{NT} + e_{NT})/r_{NT} \rightarrow 0$. Conditional on $\mathcal{G}$,*

$$P_D^* \left[\sup_{\substack{t \geq v_{NT}\Delta_{NT} \\ M^*(t) \geq m_{NT}}} \frac{\tilde{N}_{2\Delta}^*(t)}{M^*(t)} > \gamma_{NT} \text{ or } \sup_{t \geq v_{NT}\Delta_{NT}} \tilde{N}_{2\Delta}^*(t) > e_{NT} \right] \leq \varsigma_{NT}.$$

Moreover, $\mathbb{P}(|\hat{\mathcal{S}}| \geq r_{NT}) \rightarrow 1$ and $\mathbb{P}\{G \geq v_{NT}\Delta_{NT} \text{ or } \hat{\mathcal{S}} = \emptyset\} \rightarrow 1$.

The band count $\tilde{N}_{2\Delta}^*(t)$ records null statistics that a coupling perturbation could push across the threshold. The first bound rules out a local spike when the tail is populated; e_{NT} controls the sparse-tail case; and $(m_{NT} + e_{NT})/r_{NT} \rightarrow 0$ makes both errors negligible relative to discoveries. These conditions are unnecessary for the Gaussian theorem because that proof transfers the bounded FDP directly rather than perturbing the discontinuous selection rule.

Roadmap. The appendix follows the conditional-law pipeline displayed in Section 4.1. Section C.1 first converts the screening rule into bounds on the omitted coefficients and then controls the covariance estimator uniformly over the realized screened model. Section C.2 reduces the factor-augmented Lasso to its low-dimensional score and bounds the pairwise-antisymmetric drift created by missed signals. Section C.3 records the design-side bookkeeping for estimated factors, always-included controls, and data recycling. Section C.4 establishes the ideal null-sign bound and transfers from P_D^* to P_D^m and then to $P_D^\dagger$. Section C.5 constructs the martingale coupling from the actual score law P_D^a to P_D^* . Section C.6 assembles these ingredients into the finite-sample FDR inequality, and Section C.7 proves the power result on the retained active set. In short, C.1–C.5 control the individual arrows, whereas C.6–C.7 combine them into the two main conclusions.

C.1 Screening and covariance estimation

Lemma C.1 (Consequences of threshold screening). *Under Assumption 1, on the event (2.8),*

$$\kappa \leq K_{NT}, \quad \|\beta_{\mathcal{A}}\|_\infty \leq b_{NT}, \quad \|\beta_{\mathcal{A}}\|_1 \leq sb_{NT}, \quad \|\beta_{\mathcal{A}}\|_2 \leq \sqrt{s} b_{NT},$$

where $b_{NT} = \xi + Ca_{NT}$.

Proof. If $j \in \mathcal{A}$, then $|\tilde{\beta}_j| < \xi$, so $|\beta_j^M| \leq \xi + a_{NT}$. Also,

$$|\beta_j - \beta_j^M| = \frac{\left| \sum_{k \neq j} \mathbb{E}(U_{tj} U_{tk}) \beta_k \right|}{\mathbb{E}(U_{tj}^2)} \leq Ca_{NT}.$$

Hence $|\beta_j| \leq b_{NT}$ and the norm bounds follow from $|\mathcal{A}| \leq s$. Conversely, if $j \in \tilde{\mathcal{S}}$, then $|\beta_j^M| \geq \xi - a_{NT} \geq \xi/2$, which gives $\kappa \leq K_{NT}$. $\square$

Sure-screening specialization. If, in addition, $\min_{j \in \mathcal{S}_0} |\beta_j^M| \geq \xi + a_{NT}$, then

$$\mathbb{P}(\mathcal{S}_0 \subseteq \tilde{\mathcal{S}}) \geq 1 - \delta_{NT}^{\text{scr}}, \quad s \leq \kappa \leq K_{NT}$$

on the screening event. Equivalently, under the beta-weighted correlation bound in Assumption 1, it is sufficient that $\min_{j \in \mathcal{S}_0} |\beta_j| \geq \xi + (1 + C_0)a_{NT}$, where $\max_j |\beta_j^M - \beta_j| \leq C_0 a_{NT}$. In this special case $d_0 = 0$, $R_{\text{miss}} = 0$, and the covariance rate reduces to

$$\omega_{NT}^{\text{ss}} = C \left\{ \rho^m + mT^{-1/2} \ln T + m \frac{(K_{NT} + 1) \ln(NT)}{T} \right\}.$$

Indeed, for $j \in \mathcal{S}_0$, the screening event gives $|\tilde{\beta}_j| \geq |\beta_j^M| - a_{NT} \geq \xi$, proving inclusion. The probability statement follows from (2.8); the remaining claims follow from Lemma C.1 and $\mathcal{A} = \emptyset$.

The covariance proof first bounds the refit projection and the omitted-signal residual uniformly over screened models. It then decomposes each lag error into sampling, linear, and quadratic terms, bounds these separately, and sums over the retained band to obtain the operator-norm rate.

Proof of Theorem 2.2. We work on the intersection of the screening event (2.8) and the uniform events in Assumption 2; its probability tends to one. Put $P = P_{\tilde{\mathcal{S}}}$ and

$$r_0 = (I - P)d_0, \quad w = r_0 - P\varepsilon.$$

Because P is the orthogonal projection onto the refit design,

$$\hat{\varepsilon} = (I - P)(\varepsilon + d_0) = \varepsilon + w.$$

We prove the result in four steps.

Step 1. Write $Z = [\mathbf{1}, X_{\tilde{\mathcal{S}}}]$ and $d = \dim(Z) \leq K_{NT} + 1$. For any $v \in \mathbb{R}^T$, the uniform sparse-eigenvalue condition gives

$$\|Pv\|_2^2 = (Z'v)'(Z'Z)^{-1}(Z'v) \leq \frac{d}{cT} \max_l |Z'_l v|^2. \quad (\text{C.1})$$

Set $q_{NT} = \sqrt{\ln(NT)/T}$ and $r_{NT}^{\text{proj}} = (K_{NT} + 1) \ln(NT)/T$. We now show explicitly how the two projection bounds follow. The uniform error–design correlation condition states that

$$\max_l |Z'_l \varepsilon| = O_p(Tq_{NT}), \quad \max_{|h| \leq m} \max_l |Z'_l L'_h \varepsilon| = O_p(Tq_{NT}).$$

The first relation is the case $h = 0$; the second follows from $Z'_l L'_h \varepsilon = \langle \varepsilon, L_h Z_l \rangle$. These bounds remain valid after the random set $\tilde{\mathcal{S}}$ is substituted because Assumption 2 is uniform over all sets of size at most K_{NT} . Taking $v = \varepsilon$ in (C.1) gives

$$T^{-1} \|P\varepsilon\|_2^2 \leq \frac{K_{NT} + 1}{cT^2} \max_l |Z'_l \varepsilon|^2 = O_p\{(K_{NT} + 1)q_{NT}^2\} = O_p(r_{NT}^{\text{proj}}).$$

Likewise, taking $v = L'_h \varepsilon$ and maximizing over h gives

$$\max_{|h| \leq m} T^{-1} \|PL'_h \varepsilon\|_2^2 \leq \frac{K_{NT} + 1}{cT^2} \max_{|h| \leq m} \max_l |Z'_l L'_h \varepsilon|^2 = O_p(r_{NT}^{\text{proj}}).$$

Thus

$$\max_{|h| \leq m} T^{-1} \|PL'_h \varepsilon\|_2^2 = O_p(r_{NT}^{\text{proj}}), \quad T^{-1} \|P\varepsilon\|_2^2 = O_p(r_{NT}^{\text{proj}}). \quad (\text{C.2})$$

Because $P = P' = P^2$ and $P\varepsilon$ belongs to the refit space,

$$\langle \varepsilon, L_h P\varepsilon \rangle = \langle L'_h \varepsilon, P\varepsilon \rangle = \langle PL'_h \varepsilon, P\varepsilon \rangle.$$

Consequently, Cauchy–Schwarz and (C.2) give

$$\max_{|h| \leq m} T^{-1} |\langle \varepsilon, L_h P \varepsilon \rangle| \leq \left\{ \max_{|h| \leq m} T^{-1} \|P L'_h \varepsilon\|_2^2 \right\}^{1/2} \{T^{-1} \|P \varepsilon\|_2^2\}^{1/2} = O_p(r_{NT}^{\text{proj}}), \quad (\text{C.3})$$

and the same argument, using the opposite sign of h , controls $T^{-1} |\langle P \varepsilon, L_h \varepsilon \rangle|$.

Lemma C.1 and the upper sparse-eigenvalue bound also give

$$T^{-1/2} \|d_0\|_2 \leq \sum_{j \in \mathcal{A}} |\beta_j| T^{-1/2} \|X_j\|_2 \leq C \|\beta_{\mathcal{A}}\|_1. \quad (\text{C.4})$$

Since $d_0 = \sum_{j \in \mathcal{A}} X_j \beta_j$, the uniform error–design correlation bound yields

$$\begin{aligned} \max_{|h| \leq m} T^{-1} |\langle \varepsilon, L_h d_0 \rangle| &\leq \|\beta_{\mathcal{A}}\|_1 \max_{|h| \leq m} \max_{j \in \mathcal{A}} T^{-1} |\langle \varepsilon, L_h X_j \rangle| \\ &= O_p(\|\beta_{\mathcal{A}}\|_1 q_{NT}). \end{aligned} \quad (\text{C.5})$$

The opposite sign of h gives the same bound for $T^{-1} |\langle d_0, L_h \varepsilon \rangle|$.

For the projected missed-signal term, symmetry and idempotence of P give

$$\langle \varepsilon, L_h P d_0 \rangle = \langle P L'_h \varepsilon, P d_0 \rangle.$$

Since P is a contraction, $\|P d_0\|_2 \leq \|d_0\|_2$. Combining this fact with (C.2), (C.4), and Cauchy–Schwarz gives

$$\begin{aligned} \max_{|h| \leq m} T^{-1} |\langle \varepsilon, L_h P d_0 \rangle| &\leq \left\{ \max_{|h| \leq m} T^{-1} \|P L'_h \varepsilon\|_2^2 \right\}^{1/2} T^{-1/2} \|P d_0\|_2 \\ &= O_p(\|\beta_{\mathcal{A}}\|_1 \sqrt{r_{NT}^{\text{proj}}}). \end{aligned} \quad (\text{C.6})$$

The same calculation with $-h$ controls $T^{-1} |\langle P d_0, L_h \varepsilon \rangle|$.

Step 2. For $|h| \leq m$, let $n_h = T - |h|$ and interpret L_h as aligning the n_h overlapping observations. Since $m = o(T)$, $n_h \asymp T$ uniformly over these lags. Expanding $\hat{\varepsilon} = \varepsilon + w$ gives the exact decomposition

$$\hat{\gamma}(h) - \gamma(h) = A_h + B_h + C_h, \quad (\text{C.7})$$

where

$$\begin{aligned} A_h &= n_h^{-1} \langle \varepsilon, L_h \varepsilon \rangle - \gamma(h), \\ B_h &= n_h^{-1} \{ \langle \varepsilon, L_h w \rangle + \langle w, L_h \varepsilon \rangle \}, \quad C_h = n_h^{-1} \langle w, L_h w \rangle. \end{aligned}$$

Assumption 2 directly gives

$$\max_{|h| \leq m} |A_h| = O_p(T^{-1/2} \ln T). \quad (\text{C.8})$$

Step 3. Because $w = d_0 - Pd_0 - P\varepsilon$, the linear term in (C.7) expands into the following six terms:

$$\begin{aligned} n_h B_h &= \langle \varepsilon, L_h d_0 \rangle - \langle \varepsilon, L_h P d_0 \rangle - \langle \varepsilon, L_h P \varepsilon \rangle \\ &\quad + \langle d_0, L_h \varepsilon \rangle - \langle P d_0, L_h \varepsilon \rangle - \langle P \varepsilon, L_h \varepsilon \rangle. \end{aligned} \quad (\text{C.9})$$

The first and fourth terms are each $O_p(T\|\beta_{\mathcal{A}}\|_1 q_{NT})$ by (C.5). The second and fifth terms are each $O_p(T\|\beta_{\mathcal{A}}\|_1 \sqrt{r_{NT}^{\text{proj}}})$ by (C.6). The third and sixth terms are each $O_p(T r_{NT}^{\text{proj}})$ by (C.3). Since $n_h \asymp T$ uniformly for $|h| \leq m$, division by n_h yields

$$\max_{|h| \leq m} |B_h| = O_p\{r_{NT}^{\text{proj}} + \|\beta_{\mathcal{A}}\|_1 q_{NT} + \|\beta_{\mathcal{A}}\|_1 \sqrt{r_{NT}^{\text{proj}}}\}. \quad (\text{C.10})$$

It remains to control the quadratic term C_h . This term does not require a new empirical-process bound. Indeed, $(I - P)d_0$ lies in the orthogonal complement of the refit space, whereas $P\varepsilon$ lies in that space. Hence

$$\|w\|_2^2 = \|(I - P)d_0\|_2^2 + \|P\varepsilon\|_2^2.$$

Using (C.2), (C.4), and the contraction property of $I - P$,

$$\begin{aligned} T^{-1}\|w\|_2^2 &= T^{-1}\|(I - P)d_0\|_2^2 + T^{-1}\|P\varepsilon\|_2^2 \\ &\leq T^{-1}\|d_0\|_2^2 + T^{-1}\|P\varepsilon\|_2^2 \\ &= O_p(\|\beta_{\mathcal{A}}\|_1^2 + r_{NT}^{\text{proj}}). \end{aligned} \quad (\text{C.11})$$

Moreover, L_h is a contraction on the aligned observations. Therefore

$$\begin{aligned} \max_{|h| \leq m} |C_h| &\leq \max_{|h| \leq m} n_h^{-1} \|w\|_2 \|L_h w\|_2 \\ &\leq \max_{|h| \leq m} n_h^{-1} \|w\|_2^2 = O_p(\|\beta_{\mathcal{A}}\|_1^2 + r_{NT}^{\text{proj}}). \end{aligned} \quad (\text{C.12})$$

The last equality uses $n_h \asymp T$ and (C.11). Combining (C.8), (C.10), and (C.12),

$$\max_{|h| \leq m} |\hat{\gamma}(h) - \gamma(h)| = O_p\left\{T^{-1/2} \ln T + r_{NT}^{\text{proj}} + \|\beta_{\mathcal{A}}\|_1 q_{NT} + \|\beta_{\mathcal{A}}\|_1 \sqrt{r_{NT}^{\text{proj}}} + \|\beta_{\mathcal{A}}\|_1^2\right\}. \quad (\text{C.13})$$

Step 4. Let $\Sigma_{\varepsilon, m}^0$ be the population covariance matrix truncated outside the m th band. The largest absolute row sum of $\hat{\Sigma}_{\varepsilon} - \Sigma_{\varepsilon, m}^0$ is at most $(2m + 1)$ times the left-hand side of (C.13), and hence bounds its operator norm. Geometric autocovariance decay gives

$$\|\Sigma_{\varepsilon, m}^0 - \Sigma_{\varepsilon}^0\| \leq 2 \sum_{h > m} C \rho^h = O(\rho^m).$$

Substituting the definitions of q_{NT} and r_{NT}^{proj} into (C.13), multiplying by m , and adding the banding bias yields

$$\|\hat{\Sigma}_{\varepsilon} - \Sigma_{\varepsilon}^0\| = O_p(\omega_{NT}),$$

with ω_{NT} stated in the theorem. □

C.2 Score reduction and missed-signal drift

Let

$$\eta = T^{-1}\mathbb{U}'Y^\dagger, \quad Q_\eta(b) = \frac{1}{2}b'\Psi b - \eta'b + \lambda\|b\|_1.$$

Lemma C.2 (Conditional score reduction and drift). *On $\mathcal{B}_G$, the following statements hold.*

- (i) *The Lasso coefficients and W depend on the error only through ζ in (4.19). If η is changed to η' ,*

$$\begin{aligned} \|\widehat{b}(\eta) - \widehat{b}(\eta')\|_2 &\leq \sigma_{\min}(\Psi)^{-1}\|\eta - \eta'\|_2, \\ \max_{j \leq \kappa} |W_j(\eta) - W_j(\eta')| &\leq \sqrt{2}\sigma_{\min}(\Psi)^{-1}\|\eta - \eta'\|_2. \end{aligned}$$

- (ii) *For each missed column, write $X_l^\dagger = P_{\tilde{F}}X_l^\dagger + u_l^\dagger$. Then*

$$(\check{U}_j - \tilde{U}_j)'X_l^\dagger = (\check{U}_j - \tilde{U}_j)'u_l^\dagger.$$

- (iii) *Put $g_A^\dagger = M_{\tilde{F}}g = U_A^\dagger\beta_A$. For every retained null pair,*

$$D_j - D_{j+\kappa} = T^{-1/2}(\check{U}_j - \tilde{U}_j)'g_A^\dagger.$$

Consequently, Assumption 3 implies

$$\max_{j \in \tilde{\mathcal{S}} \cap \mathcal{S}_0^c} |D_j - D_{j+\kappa}| \leq \bar{\omega}_{NT}.$$

Profiling out the factors identifies the sufficient score. The proof then controls how the Lasso solution changes with that score and uses orthogonality to isolate the idiosyncratic missed-signal drift.

Proof of Lemmas 4.1 and C.2. The profiled objective is

$$Q_\eta(b) = \frac{1}{2}b'\Psi b - \eta'b + \lambda\|b\|_1$$

up to an additive constant. Let $\widehat{b} = \widehat{b}(\eta)$ and $\widehat{b}' = \widehat{b}(\eta')$. Strong convexity and optimality give

$$\begin{aligned} Q_\eta(\widehat{b}') - Q_\eta(\widehat{b}) &\geq \frac{1}{2}\sigma_{\min}(\Psi)\|\widehat{b} - \widehat{b}'\|_2^2, \\ Q_{\eta'}(\widehat{b}) - Q_{\eta'}(\widehat{b}') &\geq \frac{1}{2}\sigma_{\min}(\Psi)\|\widehat{b} - \widehat{b}'\|_2^2. \end{aligned}$$

Adding cancels the common quadratic and penalty terms and leaves

$$\sigma_{\min}(\Psi)\|\widehat{b} - \widehat{b}'\|_2^2 \leq (\eta - \eta')'\{\widehat{b} - \widehat{b}'\}.$$

Cauchy–Schwarz gives the coefficient bound. Since $W_j(\eta) = |\widehat{b}_j(\eta)| - |\widehat{b}_{j+\kappa}(\eta)|$, $\|x\| - \|y\| \leq \|x - y\|$ gives

$$|W_j(\eta) - W_j(\eta')| \leq |\widehat{b}_j - \widehat{b}'_j| + |\widehat{b}_{j+\kappa} - \widehat{b}'_{j+\kappa}| \leq \sqrt{2} \|\widehat{b} - \widehat{b}'\|_2,$$

which proves the statistic bound. These Lipschitz bounds are used below to translate score-level approximation errors into perturbations of the knockoff statistics and hence into the threshold band in Assumption C.3.

PCA gives $\check{U}'\check{F} = 0$. The construction

$$\widetilde{U} = \check{U}(I - \Sigma_{\check{U}}^{-1} \text{diag } \nu) + VO$$

and $V'\check{F} = 0$ give $\widetilde{U}'\check{F} = 0$, proving the exact factor cancellation. Hence $(\check{U}_j - \widetilde{U}_j)'g = (\check{U}_j - \widetilde{U}_j)'g_{\mathcal{A}}^{\dagger}$. Substitution into the definition $D = T^{-1/2}\mathbb{U}'g$ proves the identity in part (iii), and the weighted drift bound in Assumption 3, multiplied by $\sqrt{T}$, gives $\bar{\omega}_{NT} = \sqrt{T} \vartheta_{NT}$. $\square$

Proof of Lemma 4.2. Under P^{ind} , the estimation fold is independent of the inference block. Conditional on $\mathcal{G}$, the covariance estimator, the realized design, and the independent knock-off randomization are fixed, while $\varepsilon \mid \mathcal{G} \sim N(0, \Sigma_{\varepsilon}^0)$. Thus $\zeta = T^{-1/2}\mathbb{U}'\widehat{\Sigma}_{\varepsilon}^{-1/2}\varepsilon$ is a fixed linear transformation of a Gaussian vector, with mean zero and covariance Ξ . The transfer back to the actual fold-dependent law is accounted for separately by $2c_{NT}$. $\square$

Lemma C.3 (Weighted certification of the omitted-signal drift). *Let*

$$g_{\mathcal{A}}^{\dagger} = U_{\mathcal{A}}^{\dagger}\beta_{\mathcal{A}},$$

and define the two beta-weighted empirical scores

$$\tau_{U,NT} = \max_{k \in \widetilde{\mathcal{S}}} T^{-1} |\check{U}'_k g_{\mathcal{A}}^{\dagger}|, \quad \tau_{V,NT} = \max_{j \leq \kappa} T^{-1} |(VOe_j)'g_{\mathcal{A}}^{\dagger}|.$$

If $\max_{j \leq \kappa} \nu_j \|\Sigma_{\check{U}}^{-1} e_j\|_1 \leq C_1$, then

$$\max_{j \leq \kappa} T^{-1} |(\check{U}_j - \widetilde{U}_j)'g_{\mathcal{A}}^{\dagger}| \leq C_1 \tau_{U,NT} + \tau_{V,NT}. \quad (\text{C.14})$$

Proof. The knockoff construction gives

$$\check{U}_j - \widetilde{U}_j = \nu_j \check{U} \Sigma_{\check{U}}^{-1} e_j - VOe_j.$$

Hölder's inequality and the definition of C_1 yield

$$\begin{aligned} T^{-1} |(\check{U}_j - \widetilde{U}_j)'g_{\mathcal{A}}^{\dagger}| &\leq \nu_j \|\Sigma_{\check{U}}^{-1} e_j\|_1 \tau_{U,NT} + \tau_{V,NT} \\ &\leq C_1 \tau_{U,NT} + \tau_{V,NT}, \end{aligned}$$

which proves (C.14). $\square$

Primitive verification of the original-design score. Lemma C.3 reduces the drift bound to $\tau_{U,NT}$ and $\tau_{V,NT}$. We first give a low-level route for $\tau_{U,NT}$; Lemma C.4 below

treats $\tau_{V,NT}$. Let $u_l^{\dagger,0}$ be the oracle transformed idiosyncratic columns and define

$$g_{\mathcal{A}}^{\dagger,0} = U_{\mathcal{A}}^{\dagger,0} \beta_{\mathcal{A}}, \quad \tau_{U,NT}^0 = \max_{k \notin \mathcal{A}} T^{-1} |(u_k^{\dagger,0})' g_{\mathcal{A}}^{\dagger,0}|.$$

Suppose

$$\max_l T^{-1/2} \|u_l^{\dagger,0}\|_2 \leq C, \quad \max_l T^{-1/2} \|u_l^{\dagger} - u_l^{\dagger,0}\|_2 \leq r_{U,NT}.$$

Then, uniformly over the realized screened set,

$$\tau_{U,NT} \leq \tau_{U,NT}^0 + \|\beta_{\mathcal{A}}\|_1 \{2Cr_{U,NT} + r_{U,NT}^2\}. \quad (\text{C.15})$$

To verify this bound, write $a_l = u_l^{\dagger} - u_l^{\dagger,0}$ and $a_{\mathcal{A}} = \sum_{l \in \mathcal{A}} a_l \beta_l$. Then

$$g_{\mathcal{A}}^{\dagger} = g_{\mathcal{A}}^{\dagger,0} + a_{\mathcal{A}}, \quad T^{-1/2} \|a_{\mathcal{A}}\|_2 \leq r_{U,NT} \|\beta_{\mathcal{A}}\|_1.$$

The first oracle norm bound also implies $T^{-1/2} \|g_{\mathcal{A}}^{\dagger,0}\|_2 \leq C \|\beta_{\mathcal{A}}\|_1$. For $k \in \tilde{\mathcal{S}}$, we have $\check{U}_k = u_k^{\dagger,0} + a_k$ and $k \notin \mathcal{A}$. Expanding $T^{-1} \check{U}_k' g_{\mathcal{A}}^{\dagger}$ produces the oracle term plus

$$T^{-1} a_k' g_{\mathcal{A}}^{\dagger,0}, \quad T^{-1} (u_k^{\dagger,0})' a_{\mathcal{A}}, \quad T^{-1} a_k' a_{\mathcal{A}}.$$

Cauchy–Schwarz bounds these three terms by, respectively, $Cr_{U,NT} \|\beta_{\mathcal{A}}\|_1$, $Cr_{U,NT} \|\beta_{\mathcal{A}}\|_1$, and $r_{U,NT}^2 \|\beta_{\mathcal{A}}\|_1$, proving (C.15).

The remaining oracle score has the transparent expansion

$$T^{-1} (u_k^{\dagger,0})' g_{\mathcal{A}}^{\dagger,0} = \sum_{l \in \mathcal{A}} \left\{ T^{-1} (u_k^{\dagger,0})' u_l^{\dagger,0} \right\} \beta_l.$$

Thus the elementwise coherence condition

$$\max_{k \neq l} T^{-1} |(u_k^{\dagger,0})' u_l^{\dagger,0}| \leq \delta_{U,NT}$$

is one sufficient special case, giving $\tau_{U,NT}^0 \leq \delta_{U,NT} \|\beta_{\mathcal{A}}\|_1$, but it is not required. More generally, let $\ell_{U,NT} \geq 1$ collect the logarithmic loss in a uniform Bernstein bound for the centered cross-products. Under sub-Gaussian tails and geometric mixing it may be taken constant; under weaker sub-Weibull tails it may diverge slowly. The bound is

$$\tau_{U,NT}^0 \leq \mu_{\mathcal{A},NT}^0 + O_p \left\{ \|\beta_{\mathcal{A}}\|_1 \ell_{U,NT} \sqrt{\frac{\ln N}{T}} \right\}, \quad (\text{C.16})$$

where

$$\mu_{\mathcal{A},NT}^0 = \max_{k \notin \mathcal{A}} \left| \sum_{l \in \mathcal{A}} \mathbb{E} (u_{tk}^{\dagger,0} u_{tl}^{\dagger,0}) \beta_l \right|.$$

Hence individual oracle correlations may remain nonzero: what must be small is their population beta-weighted aggregate $\mu_{\mathcal{A},NT}^0$.

Lemma C.4 (Haar certification of the weighted complement score). *Conditionally on the data, suppose V is Haar-uniform on the set of orthonormal κ -frames in $\text{span}(\check{U}, \check{F})^\perp$, using*

independent auxiliary randomness. If $2(\kappa + r) \leq T$, $\max_j \nu_j \leq C_\nu$, and

$$T^{-1/2} \|g_{\mathcal{A}}^\dagger\|_2 \leq \sigma_{\mathcal{A}, NT},$$

then

$$\tau_{V, NT} \leq C \sigma_{\mathcal{A}, NT} \sqrt{\frac{\ln(\kappa NT)}{T}}$$

with probability at least $1 - (NT)^{-1} - \kappa e^{-T/32}$.

Intuitively, $\tau_{V, NT}$ measures whether the artificial orthogonal part of the Fixed-X knockoff aligns with the omitted signal. Haar randomization prevents systematic alignment, so this component does not amplify the omitted signal beyond the stated concentration rate. The proof represents a random direction by a normalized Gaussian vector, bounds its projection for one coordinate, and then takes a union bound over the knockoff columns.

Proof. Condition on the data, the screened model, and O , leaving only the Haar rotation V random. Put

$$\mathcal{L} = \text{span}(\check{U}, \check{F})^\perp, \quad d_V = \dim(\mathcal{L}) = T - \kappa - r \geq T/2.$$

For a fixed j , set $r_j = \|Oe_j\|_2$. If $r_j = 0$, the desired bound is immediate. Otherwise, Haar invariance implies that

$$q_j := \frac{VOe_j}{r_j}$$

is uniform on the unit sphere of $\mathcal{L}$. The directions q_j need not be independent; the union bound below uses only their marginal tail bounds.

This is the key geometric reduction. The matrix O determines only the length r_j , whereas the Haar frame V randomizes the direction q_j . We therefore control magnitude and alignment separately.

We first control the radius. The Fixed-X Gram identity gives

$$r_j^2 = T\{2\nu_j - \nu_j^2(\Sigma_{\check{V}}^{-1})_{jj}\} \leq 2TC_\nu, \quad (\text{C.17})$$

so $r_j \leq C\sqrt{T}$ uniformly in j .

Next project the omitted signal onto $\mathcal{L}$:

$$g_{\mathcal{L}} = P_{\mathcal{L}} g_{\mathcal{A}}^\dagger.$$

Because $q_j \in \mathcal{L}$, $q_j' g_{\mathcal{A}}^\dagger = q_j' g_{\mathcal{L}}$, and

$$\|g_{\mathcal{L}}\|_2 \leq \|g_{\mathcal{A}}^\dagger\|_2 \leq \sqrt{T} \sigma_{\mathcal{A}, NT}. \quad (\text{C.18})$$

Hence the part of the omitted signal lying in the realized design space is irrelevant for this term; only its projection onto the randomization space $\mathcal{L}$ can align with VOe_j .

To obtain the spherical tail bound, write $q_j \stackrel{d}{=} \gamma/\|\gamma\|_2$ with $\gamma \sim N(0, I_{d_V})$. For every $t > 0$,

$$\mathbb{P}(|\gamma' g_{\mathcal{L}}| > t\|g_{\mathcal{L}}\|_2) \leq 2e^{-t^2/2},$$

while the chi-square lower-tail inequality gives

$$\mathbb{P}\left(\|\gamma\|_2 < \sqrt{d_V/2}\right) \leq e^{-d_V/16}.$$

The numerator is a one-dimensional projection, while the denominator is of order $\sqrt{d_V}$. Their ratio is therefore of order $d_V^{-1/2}$, which is the high-dimensional near-orthogonality effect. On the complement of these two events,

$$|q'_j g_{\mathcal{A}}^\dagger| \leq \frac{\sqrt{2}t \|g_{\mathcal{L}}\|_2}{\sqrt{d_V}}.$$

Combining this inequality with (C.17), (C.18), and $d_V \geq T/2$ yields

$$T^{-1} |(V O e_j)' g_{\mathcal{A}}^\dagger| \leq C \sigma_{\mathcal{A}, NT} t / \sqrt{T}$$

outside an event of probability at most $2e^{-t^2/2} + e^{-d_V/16}$. Taking $t = \{2 \ln(2\kappa NT)\}^{1/2}$ gives

$$2\kappa e^{-t^2/2} = \frac{1}{NT}, \quad \kappa e^{-d_V/16} \leq \kappa e^{-T/32}.$$

A union bound over $j = 1, \dots, \kappa$ therefore proves the stated probability and rate. Consequently, the artificial knockoff directions cannot align systematically with the omitted idiosyncratic signal. The rate needed to make this contribution negligible in the FDR theorem is recorded immediately below; the nonrandom alignment with the original design is the separate $\tau_{U, NT}$ term handled above. $\square$

Implication for estimated-design regularity. Lemma C.4 by itself shows that the artificial component is of order $\sigma_{\mathcal{A}, NT} \sqrt{\ln(\kappa NT)/T}$. Assumption 3 requires the stronger aggregate drift rate $\sqrt{TK_{NT} \ln(NT)} \vartheta_{NT} = o(1)$. Hence the Haar component satisfies the rate required by the FDR theorem provided

$$\sigma_{\mathcal{A}, NT} \sqrt{K_{NT} \ln(NT) \ln(\kappa NT)} = o(1). \quad (\text{C.19})$$

This makes explicit that bounded $\sigma_{\mathcal{A}, NT}$ alone is not enough for the main FDR remainder when the screened dimension grows.

The condition has a direct screening interpretation. If the omitted idiosyncratic Gram matrix has bounded largest eigenvalue, one may take $\sigma_{\mathcal{A}, NT} = T^{-1/2} \|U_{\mathcal{A}}^\dagger \beta_{\mathcal{A}}\|_2$, and then

$$\begin{aligned} \sigma_{\mathcal{A}, NT}^2 &= T^{-1} \|U_{\mathcal{A}}^\dagger \beta_{\mathcal{A}}\|_2^2 \\ &\leq C \|\beta_{\mathcal{A}}\|_2^2 \leq C |\mathcal{A}| b_{NT}^2 \leq C s b_{NT}^2. \end{aligned}$$

Thus a lower-level sufficient condition for (C.19) is

$$b_{NT} \sqrt{s K_{NT} \ln(NT) \ln(\kappa NT)} = o(1). \quad (\text{C.20})$$

In words, sure screening is unnecessary, but the omitted signals must be few or collectively weak enough that their normalized aggregate size vanishes at the rate used by the mean-tilting argument.

Implication for power regularity. We now verify the full omitted-signal score needed for power, rather than only the original-knockoff contrast needed for FDR. Since both $\check{U}$ and $\tilde{U}$ are orthogonal to $\check{F}$, $\mathbb{U}'g = \mathbb{U}'g_{\mathcal{A}}^{\dagger}$. By the definition of $\tau_{U,NT}$ and (C.14),

$$\begin{aligned} \max_{j \leq \kappa} T^{-1} |\tilde{U}_j' g_{\mathcal{A}}^{\dagger}| &\leq \tau_{U,NT} + \max_{j \leq \kappa} T^{-1} |(\check{U}_j - \tilde{U}_j)' g_{\mathcal{A}}^{\dagger}| \\ &\leq (1 + C_1) \tau_{U,NT} + \tau_{V,NT}. \end{aligned}$$

The original half of the augmented score is bounded by $\tau_{U,NT}$. Consequently,

$$\chi_{NT} = \|T^{-1} \mathbb{U}' g\|_{\infty} \leq (1 + C_1) \tau_{U,NT} + \tau_{V,NT}. \quad (\text{C.21})$$

Combining (C.15), (C.16), and Lemma C.4 therefore gives, with

$$\begin{aligned} R_{\chi,NT} &:= \mu_{\mathcal{A},NT}^0 + \|\beta_{\mathcal{A}}\|_1 \left\{ \ell_{U,NT} \sqrt{\frac{\ln N}{T}} + r_{U,NT} + r_{U,NT}^2 \right\} \\ &\quad + \sigma_{\mathcal{A},NT} \sqrt{\frac{\ln(\kappa NT)}{T}}, \end{aligned}$$

the bound

$$\chi_{NT} = O_p(R_{\chi,NT}). \quad (\text{C.22})$$

Thus $R_{\chi,NT} = O(\lambda)$ is a direct lower-level sufficient condition for the final requirement in Assumption 4. If the omitted-signal Gram matrix has bounded largest eigenvalue, the screening bounds $\|\beta_{\mathcal{A}}\|_1 \leq sb_{NT}$ and $\sigma_{\mathcal{A},NT} \leq C\sqrt{s} b_{NT}$ reduce it to

$$\mu_{\mathcal{A},NT}^0 + sb_{NT} \left\{ \ell_{U,NT} \sqrt{\frac{\ln N}{T}} + r_{U,NT} + r_{U,NT}^2 \right\} + b_{NT} \sqrt{\frac{s \ln(\kappa NT)}{T}} = O(\lambda).$$

This derivation also explains why controlling only the pairwise contrast in Assumption 3 is not enough for power: the Lasso basic inequality requires the absolute score on both halves of the augmented design.

C.3 Estimated factors, controls, and recycling

Factor-estimation error affects FDR through the design and screening events, whitening, and the null-pair drift in (C.14)–(C.15). Power additionally requires the full omitted-signal score in (C.21)–(C.22) to be no larger than the tuning scale. The exact conditional identity below separates these rate requirements from the null-swap algebra.

One ingredient in a primitive verification of Assumption 3 is the common-component bound

$$\|\check{F} \check{\Lambda}'_{\check{\mathcal{S}}} - F^{\dagger} \Lambda'_{\check{\mathcal{S}}}\|_{\max} = O_p(\varrho_{NT}), \quad \varrho_{NT} = (\ln T)^{1/s_a} \sqrt{\frac{\ln N}{T}} + N^{-1/2} \sqrt{\ln T},$$

together with uniform lower and upper eigenvalue bounds for the relevant idiosyncratic Gram matrices. The omitted-score verification also requires control of the omitted columns, as made explicit next.

From factor estimation to the inference bounds. Write $A = \widehat{\Sigma}_\varepsilon^{-1/2}$ and $A_0 = (\Sigma_\varepsilon^0)^{-1/2}$. Extend the estimated common component to every relevant column using the same estimated factor projection. For $X_l = F\Lambda_l + U_l$, take the oracle column to be $u_l^{\dagger,0} = A_0 U_l$. Then

$$u_l^\dagger - u_l^{\dagger,0} = (AF\Lambda_l - \check{F}\check{\Lambda}_l) + (A - A_0)U_l.$$

If the normalized columnwise norm $T^{-1/2}\|AF\Lambda_l - \check{F}\check{\Lambda}_l\|_2$ is at most $r_{C,NT}$ uniformly over retained and omitted columns, and $\max_l T^{-1/2}\|U_l\|_2 \leq C$, the triangle inequality gives $r_{U,NT} \leq r_{C,NT} + C\|A - A_0\|$. With covariance eigenvalues bounded away from zero, the inverse-square-root perturbation bound gives $\|A - A_0\| = O_p(\omega_{NT})$. Thus a common-component bound of order ϱ_{NT} uniform over these columns yields $r_{U,NT} = O_p(\varrho_{NT} + \omega_{NT})$. The displayed bound on retained columns alone does not establish this extension to omitted columns.

Substitution into (C.15) shows that the additional original-design omitted score is of order $\|\beta_A\|_1\{r_{U,NT} + r_{U,NT}^2\}$. For FDR, (C.14) converts it into ϑ_{NT} , and $\bar{\omega}_{NT} = \sqrt{T}\vartheta_{NT}$ determines $\epsilon_{NT}^{\text{mean}}$ in the main theorem; design-event failures enter separately. For power, (C.22) requires this contribution, the oracle omitted score, and the complement score together to be $O_p(\lambda_{NT})$. The restricted-eigenvalue and retained-signal conditions must also hold. These implications explain the propagation of estimation error without asserting that arbitrary factor estimates satisfy the required rates.

Lemma C.5 (Conditional PCA bookkeeping). *Fix a realization of $\mathcal{G}$, including $\widehat{r}$, $\check{F}$, $\check{U}$, $\widetilde{U}$, and the screened set. On $\mathcal{B}_{\mathcal{G}}$:*

(i)

$$X_{\check{\mathcal{S}}}^\dagger \beta_{\check{\mathcal{S}}} = \check{F}\check{\Lambda}'_{\check{\mathcal{S}}} \beta_{\check{\mathcal{S}}} + \check{U}_{\check{\mathcal{S}}} \beta_{\check{\mathcal{S}}}$$

holds exactly for the realized PCA decomposition;

(ii) *profiling out $\check{F}$ leaves the penalized objective determined by Ψ and $\eta = T^{-1}\mathbb{U}'Y^\dagger$;*

(iii) *under the ideal law $N(\bar{D}, \Psi)$, every null pair is swap invariant for each realized value of $\widehat{r}$.*

Thus PCA and factor-number estimation enter FDR only through the probability of $\mathcal{B}_{\mathcal{G}}$, covariance and design rates, and the missed-signal drift; there is no additional PCA remainder in the conditional null-sign argument.

Proof. By definition, $\check{U}_{\check{\mathcal{S}}} = X_{\check{\mathcal{S}}}^\dagger - \check{F}\check{\Lambda}'_{\check{\mathcal{S}}}$, which proves (i) without invoking a convergence rate. Both $\check{U}'\check{F} = 0$ and $\widetilde{U}'\check{F} = 0$ hold by PCA and (3.14). The Frisch–Waugh–Lovell projection on $\check{F}$ therefore leaves the Gram matrix of $[\check{U}, \widetilde{U}]$ unchanged and gives the profiled objective Q_η in Lemma C.2. This proves (ii).

For (iii), the Fixed-X construction makes Ψ invariant under a swap of the two columns in any pair. The symmetrized mean $\bar{D}$ has the same invariance for null pairs. Hence the Gaussian law $N(\bar{D}, \Psi)$ and every antisymmetric statistic W_j have the usual null-swap property conditional on the realized PCA output. Changing $\widehat{r}$ changes the conditioned-on matrices and their rates, but not this algebraic argument. $\square$

Lemma C.6 (Partialling out low-dimensional controls). *Let $q_0 = O(1)$ and suppose the response and screened covariates are partialled out with respect to always-included controls*

Q before PCA. Under strict exogeneity, take the ordinary whitened residualizer $M_{Q^\dagger}$ and impose $V'Q^\dagger = 0$. Then the score and pair-swap argument above apply with the nuisance rank increased from r to $r + q_0$.

If Q_t is predetermined rather than strictly exogenous, the same conclusion holds for a cross-fitted or one-sided residualizer whenever its score weights satisfy the primitive conditions in Appendix C.5 and hence Assumption C.2. Ordinary full-sample projection on a lagged dependent variable is not covered automatically.

Proof. Under strict exogeneity, $\check{F}$, $\check{U}$, and $\tilde{U}$ all lie in $\text{span}(Q^\dagger)^\perp$. Profiling out $Q^\dagger$ hence does not change the penalized Gram matrix or $T^{-1}\mathbb{U}'Y^\dagger$, and conditioning on Q is absorbed into $\mathcal{G}$. Since q_0 is fixed, only the nuisance rank changes.

For predetermined Q_t , estimate the nuisance projection on an earlier fold or use a one-sided residualizer whose time- t coefficients are $\mathcal{F}_{t-1}$ -measurable. With the lower-triangular whitening root, the projected score is then still a sum $\sum_s w_s u_s$ with predictable w_s . The primitive conditions in Appendix C.5 supply the coupling in Assumption C.2, and Assumption C.3 supplies indicator stability. A full-sample projection can depend on future Q values and need not have this property, which explains the qualification in Remark 3.3. $\square$

Lemma C.7 (Recycling bookkeeping). *Let folds 1 and 3 be the estimation and inference folds. On fold 1 set the knockoff copy equal to the original column and define*

$$\overline{U} = \begin{bmatrix} \check{U}^{(1)} \\ \check{U}^{(3)} \end{bmatrix}, \quad \overline{U}^{\text{ko}} = \begin{bmatrix} \check{U}^{(1)} \\ \tilde{U}^{(3)} \end{bmatrix}.$$

Conditional on fold 1, its contribution to every original–knockoff score pair is identical. The stacked Gram matrix is pair-swap invariant, while the covariance of the random score is generated only by fold 3. Consequently, the FDR proof continues to apply when Lasso stability is verified with the stacked Gram matrix and Assumptions C.2–C.3 are verified for the inference-fold score.

Proof. We first give the explicit recycling algorithm and then verify the two invariances used by the FDR proof.

Recycling algorithm. After Fold 1 has produced $\tilde{\mathcal{S}}$ and $\hat{\Sigma}_\varepsilon$, form its factor-adjusted design $\check{U}^{(1)}$ on the retained variables. Do not generate a new knockoff on this fold; instead set

$$\tilde{U}^{(1)} = \check{U}^{(1)}.$$

Fold 2 remains the separating block and is not used in the final fit. On Fold 3, use the Fold-1 estimates to whiten and factor-adjust the data, obtaining $\check{U}^{(3)}$, and construct the ordinary Fixed-X knockoff $\tilde{U}^{(3)}$. With $T_\bullet = T_1 + T_3$, stack

$$\overline{Y}^\dagger = \begin{bmatrix} Y^{\dagger(1)} \\ Y^{\dagger(3)} \end{bmatrix}, \quad \overline{U} = \begin{bmatrix} \check{U}^{(1)} & \check{U}^{(1)} \\ \check{U}^{(3)} & \tilde{U}^{(3)} \end{bmatrix}.$$

The response rows are not duplicated: each recycled observation appears once, but its original and knockoff predictor columns are identical. The final coefficients are obtained

from the single stacked Lasso

$$(\hat{\theta}, \hat{b}, \hat{\tilde{b}}) = \arg \min_{\theta, b, \tilde{b}} \left\{ \frac{1}{2T_{\bullet}} \|\bar{Y}^{\dagger} - \bar{F}\theta - \bar{U}b - \bar{U}^{\text{ko}}\tilde{b}\|_2^2 + \lambda(\|b\|_1 + \|\tilde{b}\|_1) \right\},$$

followed by $W_j = |\hat{b}_j| - |\hat{\tilde{b}}_j|$ and the usual knockoff+ threshold.

What the recycled rows estimate. Introduce the common and contrast coordinates

$$c = \frac{b + \tilde{b}}{2}, \quad d = \frac{b - \tilde{b}}{2}, \quad b = c + d, \quad \tilde{b} = c - d.$$

The recycled Fold-1 linear predictor is

$$\check{U}^{(1)}b + \check{U}^{(1)}\tilde{b} = 2\check{U}^{(1)}c,$$

so it contains information about the importance of the pair through c but no information about the original–knockoff contrast d . In Fold 3, by contrast,

$$\check{U}^{(3)}b + \check{U}^{(3)}\tilde{b} = (\check{U}^{(3)} + \check{U}^{(3)})c + (\check{U}^{(3)} - \check{U}^{(3)})d.$$

Thus Fold 1 can help a true pair cross the Lasso entry threshold, while Fold 3 alone determines which member of the pair wins. The LCD statistic has the exact representation (suppressing hats on the fitted coordinates)

$$W_j = |c_j + d_j| - |c_j - d_j|.$$

In particular, if $c_j d_j > 0$, then

$$W_j = 2 \min\{|c_j|, |d_j|\} > 0;$$

if $c_j d_j < 0$, the same expression has the opposite sign. Recycling may therefore improve power by stabilizing the common coordinate c_j , but it does not manufacture directional evidence in d_j and does not guarantee a power gain in every design.

Score symmetry. The score of the stacked Lasso is

$$\bar{\eta} = \frac{1}{T_{\bullet}} \left[\check{U}^{(1)'} Y^{\dagger(1)} \right] + \frac{1}{T_{\bullet}} \left[\check{U}^{(3)'} Y^{\dagger(3)} \right].$$

Conditional on Fold 1, the first vector is fixed and has equal coordinates within every pair. Even if screening makes this common shift large, a pair swap leaves it unchanged. All remaining score randomness is in the second vector, which is the inference-fold score up to the deterministic scale $T_3/T_{\bullet}$.

Gram symmetry and the two matrices. Let $G_1 = \check{U}^{(1)'} \check{U}^{(1)}$. The Fold-1 contribution to the stacked Gram matrix is

$$\begin{bmatrix} G_1 & G_1 \\ G_1 & G_1 \end{bmatrix},$$

which is invariant under swapping either member of any pair. The Fold-3 Gram matrix has the same pair-swap invariance by the Fixed-X construction; hence their sum, the Hessian of the stacked Lasso, is swap invariant.

Two matrices must nevertheless be kept distinct. Lasso stability and the map from score perturbations to W use the full stacked Gram matrix. The martingale coupling and likelihood-ratio calculations condition on Fold 1 and therefore use only the covariance of the random Fold-3 score, with the scale $T_3/T_\bullet$ included. Applying the former matrix in the optimization argument and the latter in Assumptions C.2–C.3 proves the stated recycling result. $\square$

C.4 Reference-law bounds

Lemma C.8 (Uniform null-sign ratio). *Under the ideal swap-invariant Gaussian law with mean $\bar{D}$, define, for $t > 0$,*

$$V^+(t) = \sum_{j \in \tilde{S} \cap S_0^c} \mathbf{1}\{W_j \geq t\}, \quad V^-(t) = \sum_{j \in \tilde{S} \cap S_0^c} \mathbf{1}\{W_j \leq -t\},$$

and $M(t) = V^+(t) + V^-(t)$. Thus $V^+(t)$ and $V^-(t)$ are the numbers of positive and negative null statistics beyond the threshold, and $M(t)$ is the number of null statistics with $|W_j| \geq t$. Let $\bar{M}$ be the total number of nonzero null statistics. If $\bar{M} = 0$, the conclusions below are vacuous. Otherwise, order these statistics by decreasing absolute value and let $s_1, \dots, s_{\bar{M}}$ be their signs. Conditional on the absolute null statistics and all nonnull statistics, these signs are independent Rademacher variables. For $\alpha \in (0, 1)$, put $Z_m = \sum_{\ell \leq m} s_\ell$ and

$$L_\alpha = \ln \left[\frac{4\{\log_2(2\bar{M}) + 1\}}{\alpha} \right].$$

With conditional probability at least $1 - \alpha$,

$$|Z_m| \leq \sqrt{8mL_\alpha}, \quad 1 \leq m \leq \bar{M}.$$

Denote this simultaneous event by

$$\mathcal{C}_\alpha := \left\{ |Z_m| \leq \sqrt{8mL_\alpha} \text{ for every } 1 \leq m \leq \bar{M} \right\}.$$

If $a^\circ \geq 1$ and $M(t) \geq 32L_\alpha a^\circ$, then, uniformly over t ,

$$\frac{V^+(t)}{1 + V^-(t)} \leq 1 + \frac{4}{\sqrt{a^\circ}}, \quad 1 + V^-(t) \geq \frac{M(t)}{4}.$$

The proof views the ordered null signs as a simple martingale. A maximal inequality controls each dyadic block, and a union bound makes the control simultaneous over all

thresholds; the count ratio then follows by algebra.

Proof. Condition on the absolute null statistics and all nonnull statistics. With respect to the filtration generated successively by $s_1, s_2, \dots$, the partial sums Z_m form a martingale with increments bounded by one. The maximal Azuma–Hoeffding inequality therefore gives, for every $n \geq 1$ and $x > 0$,

$$\mathbb{P} \left(\max_{\ell \leq n} |Z_\ell| > x \mid \{|W_j| : j \in \tilde{\mathcal{S}} \cap \mathcal{S}_0^c\}, \{W_j : j \in \tilde{\mathcal{S}} \cap \mathcal{S}_0\} \right) \leq 2 \exp \left(-\frac{x^2}{2n} \right).$$

To make the bound simultaneous in m , partition $\{1, \dots, \bar{M}\}$ into the dyadic blocks $(2^{k-1}, 2^k] \cap \mathbb{N}$. If some $m \in (2^{k-1}, 2^k]$ satisfies $|Z_m| > \sqrt{8mL_\alpha}$, then $m > 2^{k-1}$ implies

$$\max_{\ell \leq 2^k} |Z_\ell| > \sqrt{4 \cdot 2^k L_\alpha}.$$

Taking $n = 2^k$ and $x = \sqrt{4 \cdot 2^k L_\alpha}$ in the preceding inequality bounds the probability for each block by $2e^{-2L_\alpha}$. There are at most $K_{\bar{M}} := \log_2(2\bar{M}) + 1$ such blocks, and hence a union bound gives

$$\mathbb{P} \left(\max_{1 \leq m \leq \bar{M}} \frac{|Z_m|}{\sqrt{8mL_\alpha}} > 1 \mid \{|W_j| : j \in \tilde{\mathcal{S}} \cap \mathcal{S}_0^c\}, \{W_j : j \in \tilde{\mathcal{S}} \cap \mathcal{S}_0\} \right) \leq 2K_{\bar{M}}e^{-2L_\alpha} \leq \alpha.$$

It remains to translate the sign imbalance into counts. At a threshold t with $m = M(t)$, ordering by absolute value makes the relevant nulls exactly the first m statistics. Therefore

$$V^+(t) + V^-(t) = m, \quad V^+(t) - V^-(t) = Z_m,$$

and hence

$$V^+(t) = \frac{m + Z_m}{2}, \quad V^-(t) = \frac{m - Z_m}{2}.$$

If $m \geq 32L_\alpha a^\circ$, the simultaneous bound gives

$$\delta_m := \frac{|Z_m|}{m} \leq \sqrt{\frac{8L_\alpha}{m}} \leq \frac{1}{2\sqrt{a^\circ}} \leq \frac{1}{2}.$$

Consequently,

$$\frac{V^+(t)}{1 + V^-(t)} = \frac{m + Z_m}{2 + m - Z_m} \leq \frac{m + |Z_m|}{m - |Z_m|} = \frac{1 + \delta_m}{1 - \delta_m} \leq 1 + 4\delta_m \leq 1 + \frac{4}{\sqrt{a^\circ}},$$

where the last constant is deliberately loose. Similarly,

$$1 + V^-(t) \geq \frac{m - |Z_m|}{2} = \frac{m}{2}(1 - \delta_m) \geq \frac{m}{4}.$$

Because the sign event holds simultaneously for every m , both bounds remain valid when t is a data-dependent threshold. $\square$

Lemma C.9 (Mean and covariance transfer). *Let $P_D^* = N(D, \Xi)$ and $P_{\bar{D}}^\dagger = N(\bar{D}, \Psi)$. If $\lambda_{\min}(\Psi) \geq c_0$ and $\|\Xi - \Psi\| \leq C\omega_{NT} \leq c_0/2$, then for every score-measurable $0 \leq f \leq 1$ and*

$u > 0$,

$$\mathbb{E}_D^*[f] \leq \exp\{\epsilon_{NT}^{\text{cov}}(u) + \epsilon_{NT}^{\text{mean}}(u)\} \mathbb{E}_D^\dagger[f] + 2\{1 + \exp(\epsilon_{NT}^{\text{mean}}(u))\}e^{-u},$$

where

$$\begin{aligned}\epsilon_{NT}^{\text{mean}}(u) &\leq C\{\|D - \bar{D}\|_2\sqrt{u} + \|D - \bar{D}\|_2^2\}, \\ \epsilon_{NT}^{\text{cov}}(u) &\leq C\{K_{NT}\omega_{NT} + \omega_{NT}\sqrt{K_{NT}u} + \omega_{NT}u\}.\end{aligned}$$

The comparison changes the mean first and the covariance second. We bound each log likelihood ratio outside a small exceptional set, then split the expectation over the good and exceptional sets to transfer any $[0, 1]$ -valued score functional.

Proof. Insert the intermediate law $P^m = N(\bar{D}, \Xi)$. Let Z denote a generic vector on the score space. We first change only the mean. Write $\Delta_D = D - \bar{D}$ and let

$$L_{\text{mean}}(Z) := \log \frac{dP_D^*}{dP^m}(Z) = \Delta_D' \Xi^{-1}(Z - \bar{D}) - \frac{1}{2} \Delta_D' \Xi^{-1} \Delta_D.$$

Put $q_D = \Delta_D' \Xi^{-1} \Delta_D$. Under the receiving law P_D^* , L_{mean} is Gaussian with mean $q_D/2$ and variance q_D . Since $\lambda_{\min}(\Xi) \geq c_0/2$, we have $q_D \leq C\|\Delta_D\|_2^2$. A Gaussian tail bound therefore gives

$$L_{\text{mean}}(Z) \leq C\{\|\Delta_D\|_2\sqrt{u} + \|\Delta_D\|_2^2\} =: \epsilon_{NT}^{\text{mean}}(u)$$

with P_D^* -probability at least $1 - 2e^{-u}$. Let $\mathcal{H}_{\text{mean}} = \{L_{\text{mean}} \leq \epsilon_{NT}^{\text{mean}}(u)\}$. For $0 \leq f \leq 1$, splitting the likelihood-ratio identity over this event gives

$$\begin{aligned}\mathbb{E}_D^*[f] &= \mathbb{E}^m[e^{L_{\text{mean}}} f 1_{\mathcal{H}_{\text{mean}}}] + \mathbb{E}_D^*[f 1_{\mathcal{H}_{\text{mean}}^c}] \\ &\leq e^{\epsilon_{NT}^{\text{mean}}(u)} \mathbb{E}^m[f] + P_D^*(\mathcal{H}_{\text{mean}}^c) \\ &\leq e^{\epsilon_{NT}^{\text{mean}}(u)} \mathbb{E}^m[f] + 2e^{-u}.\end{aligned}$$

We next change only the covariance. Under the receiving law $P^m = N(\bar{D}, \Xi)$, put $G_\Xi = \Xi^{-1/2}(Z - \bar{D}) \sim N(0, I_d)$, where $d = 2\kappa \leq 2K_{NT}$. The covariance likelihood ratio is

$$L_{\text{cov}}(Z) := \log \frac{dP^m}{dP_D^\dagger}(Z) = -\frac{1}{2} \log \det(I_d + B) + \frac{1}{2} G_\Xi' B G_\Xi,$$

where $B = \Xi^{1/2} \Psi^{-1} \Xi^{1/2} - I_d$. The eigenvalue bound $\lambda_{\min}(\Psi) \geq c_0$ and $\|\Xi - \Psi\| \leq C\omega_{NT} \leq c_0/2$ imply, by writing $R = \Xi^{-1/2}(\Psi - \Xi)\Xi^{-1/2}$ and $B = (I_d + R)^{-1} - I_d$, that

$$\|B\| \leq C\omega_{NT}, \quad |\log \det(I_d + B)| \leq CK_{NT}\omega_{NT}.$$

Moreover, the Gaussian quadratic-form inequality gives, with P^m -probability at least $1 - 2e^{-u}$,

$$|G_\Xi' B G_\Xi - \text{tr}(B)| \leq C\{\|B\|_F\sqrt{u} + \|B\|u\} \leq C\omega_{NT}\{\sqrt{K_{NT}u} + u\}.$$

Since $|\text{tr}(B)| \leq d\|B\| \leq CK_{NT}\omega_{NT}$, combining this bound with the determinant term yields

$$L_{\text{cov}}(Z) \leq C\{K_{NT}\omega_{NT} + \omega_{NT}\sqrt{K_{NT}u} + \omega_{NT}u\} =: \epsilon_{NT}^{\text{cov}}(u)$$

with P^m -probability at least $1 - 2e^{-u}$. Let $\mathcal{H}_{\text{cov}} = \{L_{\text{cov}} \leq \epsilon_{NT}^{\text{cov}}(u)\}$. Splitting under the

receiving law once more gives

$$\begin{aligned}\mathbb{E}^m[f] &= \mathbb{E}_D^\dagger[e^{L_{\text{cov}}} f 1_{\mathcal{H}_{\text{cov}}}] + \mathbb{E}^m[f 1_{\mathcal{H}_{\text{cov}}}] \\ &\leq e^{\epsilon_{NT}^{\text{cov}}(u)} \mathbb{E}_D^\dagger[f] + 2e^{-u}.\end{aligned}$$

Substituting the covariance-transfer bound into the mean-transfer bound proves

$$\mathbb{E}_D^*[f] \leq \exp\{\epsilon_{NT}^{\text{mean}}(u) + \epsilon_{NT}^{\text{cov}}(u)\} \mathbb{E}_D^\dagger[f] + 2\{1 + \exp(\epsilon_{NT}^{\text{mean}}(u))\}e^{-u},$$

as claimed. $\square$

Remark C.10 (What Lemma C.9 transfers). *The intermediate law $P^m = N(\bar{D}, \Xi)$ separates two logically distinct errors. The first likelihood ratio changes D to the null-symmetric $\bar{D}$ and therefore measures only the effect of missed signals. The second changes Ξ to the knockoff Gram matrix Ψ and therefore measures only whitening and covariance-estimation error. The lemma applies to any bounded score-measurable functional, including the FDP itself. Threshold stability is needed only when a preceding coupling perturbs the score before this transfer is invoked. Taking $u = \ln(NT)$ makes the truncation failure probability of order $(NT)^{-1}$; no total-variation approximation of the original data array is required.*

For the Gaussian FDR theorem, we first apply the knockoff+ bound under the swap-invariant law, then transfer the bounded FDP by Lemma C.9, and finally restore the screening, design, and fold-separation errors.

Proof of Theorem 4.3. Work first conditional on $\mathcal{G}$, the screening event in Assumption 1, and $\mathcal{B}_{\mathcal{G}}$. Under $P_D^\dagger = N(\bar{D}, \Psi)$, both the mean and covariance are invariant under swapping the two coordinates of any null pair. The flip-sign property therefore makes the null signs conditionally independent Rademacher variables given their absolute statistics and the nonnull statistics. The knockoff+ optional-stopping argument of Barber and Candès (2015) gives

$$\mathbb{E}_D^\dagger(\text{FDP} \mid \mathcal{G}) \leq q.$$

The FDP, with value zero when the selected set is empty, is a score-measurable functional taking values in $[0, 1]$. Under the Gaussian condition in Theorem 4.3, the conditional score law is exactly $P_D^* = N(D, \Xi)$, Lemma C.9, applied directly with $f = \text{FDP}$ and $u = \ln(NT)$, yields

$$\mathbb{E}_D^*(\text{FDP} \mid \mathcal{G}) \leq q \exp(\bar{\epsilon}_{NT}) + \frac{2\{1 + \exp(\epsilon_{NT}^{\text{mean}})\}}{NT}.$$

Integrating over $\mathcal{G}$ and adding the screening, design, and fold-separation failure probabilities proves the displayed finite-sample bound in Theorem 4.3. The assumed vanishing rates then give $\text{FDR} \leq q + o(1)$. No coupling or threshold-stability argument is used. $\square$

C.5 Martingale score coupling

Let $H = \widehat{\Sigma}_\varepsilon^{-1/2} \mathbb{U}$ and let h'_t be its t th row. Assumption C.1 gives

$$\zeta = T^{-1/2} \mathbb{U}' \widehat{\Sigma}_\varepsilon^{-1/2} \Theta u = \sum_{s=1}^T w_s u_s, \quad w_s = T^{-1/2} \sum_{t \geq s} \Theta_{ts} h_t.$$

The following conditions are a primitive, readily checkable route to the high-level coupling in Assumption C.2. Relative to a filtration $\{\mathcal{H}_s\}$ with $\mathcal{H}_0 = \mathcal{G}$, suppose w_s is $\mathcal{H}_{s-1}$ -measurable, $\max_t \|h_t\|_2 \leq B_{NT}$, $B_{NT}^2 \lesssim K_{NT} \ln T$, and

$$\max_s \|w_s\|_2 \leq C_\Theta B_{NT} T^{-1/2}, \quad \max_{j \leq 2\kappa} \sum_s w_{sj}^2 \leq C, \quad \sum_s \|w_s\|_2^2 \leq C K_{NT},$$

Also suppose

$$\mathbb{E} \left[\left\| \sum_s w_s w'_s \{ \mathbb{E}(u_s^2 \mid \mathcal{H}_{s-1}) - 1 \} \right\| \middle| \mathcal{G} \right] \leq \varrho_{NT}^{\text{qv}}.$$

The first display controls leverage and total score energy; the second prices random conditional quadratic variation.

Let $d = 2\kappa$, take the filtration $\{\mathcal{H}_s\}$ from the preceding display, and set

$$\xi_s = w_s u_s, \quad V_s = \mathbb{E}(\xi_s \xi'_s \mid \mathcal{H}_{s-1}).$$

For independent $g_s \sim N(0, I_d)$, also independent of $\mathcal{H}_T$, define the second-order Yurinskii moment budget

$$\beta_{2,2} := \sum_{s=1}^T \mathbb{E} [\|\xi_s\|_2^3 + \|V_s^{1/2} g_s\|_2^3]. \quad (\text{C.23})$$

Thus $\beta_{2,2}$ includes both the martingale increments and the Gaussian increments used to replace them; it is not an additional covariance error.

The coupling result used. Corollary 2.2 of Cattaneo et al. (2025) applies to a sequence $\xi_s \in \mathbb{R}^d$ of martingale differences relative to $\{\mathcal{H}_s\}$. Given the conditional covariance increments V_s , an $\mathcal{H}_0$ -measurable target covariance Ξ , and

$$\Omega = \sum_s V_s - \Xi,$$

it constructs, on an enlarged space, a vector $\zeta^\star \mid \mathcal{H}_0 \sim N(0, \Xi)$ such that, for every $\rho > 0$,

$$\mathbb{P}(\|\sum_s \xi_s - \zeta^\star\|_p > \rho) \leq 24 \left\{ \frac{\beta_{p,2} \phi_p(d)^2}{\rho^3} \right\}^{1/3} + 17 \left\{ \frac{\mathbb{E}\|\Omega\| \phi_p(d)^2}{\rho^2} \right\}^{1/3}.$$

Here $\beta_{p,2}$ is the corresponding p -norm moment budget and $\phi_p(d)$ is the dimension factor in the cited corollary. Only $p = 2$ is used below: its moment budget is $\beta_{2,2}$ in (C.23) and $\phi_2(d)^2 = 2d$. The phrase “second order” means that each martingale increment is replaced by a Gaussian increment with the same first two conditional moments; the remaining

Lindeberg error is therefore measured by third moments in $\beta_{p,2}$. We take $p = 2$ because the score perturbation is controlled in Euclidean norm. Since $d = 2\kappa$, the dimension factor satisfies $\phi_2(d)^2 = 4\kappa \leq 4K_{NT}$. Thus the first term below prices non-Gaussian increments, whereas the second prices random quadratic variation.

Lemma C.11 (Martingale Yurinskii coupling). *Under Assumption C.1 and the primitive weight and quadratic-variation conditions above, suppose the weighted third moments satisfy*

$$\sum_{s=1}^T \mathbb{E} [\|w_s\|_2^3 \mathbb{E}(|u_s|^3 \mid \mathcal{H}_{s-1})] \leq CB_{NT}K_{NT}T^{-1/2}. \quad (\text{C.24})$$

Then

$$\beta_{2,2} \leq CB_{NT}K_{NT}T^{-1/2}.$$

Condition (C.24) follows from the displayed weight bounds if $\sup_s \mathbb{E}(|u_s|^3 \mid \mathcal{H}_{s-1}) \leq C$ almost surely. The weighted formulation is weaker and permits time-varying conditional third moments.

Define the natural coupling scale

$$r_{NT}^{\text{cpl}} := (4K_{NT}\beta_{2,2})^{1/3} + (4K_{NT}\varrho_{NT}^{\text{qv}})^{1/2}.$$

If $r_{NT}^{\text{cpl}} \rightarrow 0$, take any slowly diverging $\ell_{NT} \rightarrow \infty$ such that $\ell_{NT}r_{NT}^{\text{cpl}} \rightarrow 0$, and set

$$\rho_{NT} = \ell_{NT}r_{NT}^{\text{cpl}}, \quad \eta_{NT}^{\text{cpl}} = 24\ell_{NT}^{-1}, \quad \eta_{NT}^{\text{qv}} = 17\ell_{NT}^{-2/3}.$$

Put $\eta_{NT} := \eta_{NT}^{\text{cpl}} + \eta_{NT}^{\text{qv}}$. Then

$$\rho_{NT} \rightarrow 0, \quad \eta_{NT} \rightarrow 0,$$

and on an enlarged probability space there is a $\zeta^\star \mid \mathcal{G} \sim N(0, \Xi)$ such that

$$\mathbb{P}(\|\zeta - \zeta^\star\|_2 > \rho_{NT}) \leq \eta_{NT}.$$

The coupling proof bounds the third-moment budget and the quadratic-variation error separately. Substitution into the Yurinskii inequality then gives a choice of coupling radius and failure probability that both vanish.

Proof. We verify the ingredients of Corollary 2.2 of Cattaneo et al. (2025). Because w_s is $\mathcal{H}_{s-1}$ -measurable and $\mathbb{E}(u_s \mid \mathcal{H}_{s-1}) = 0$, the vectors $\xi_s = w_s u_s$ are martingale differences and $\zeta = \sum_s \xi_s$. Causality and column summability give

$$\begin{aligned} \max_s \|w_s\|_2 &\leq T^{-1/2} \max_s \sum_{t \geq s} |\Theta_{ts}| \|h_t\|_2 \\ &\leq C_\Theta B_{NT} T^{-1/2}. \end{aligned}$$

The score-covariance trace bound gives

$$\sum_s \|w_s\|_2^2 = \text{tr}(\Xi) \leq CK_{NT}.$$

We next bound the two components of (C.23). Iterated expectations and (C.24) give

$$\begin{aligned}\sum_s \mathbb{E} \|\xi_s\|_2^3 &= \sum_s \mathbb{E} [\|w_s\|_2^3 \mathbb{E}(|u_s|^3 \mid \mathcal{H}_{s-1})] \\ &\leq CB_{NT}K_{NT}T^{-1/2}.\end{aligned}$$

Write $\sigma_s^2 = \mathbb{E}(u_s^2 \mid \mathcal{H}_{s-1})$. Since $V_s = \sigma_s^2 w_s w_s'$ has rank one, $\|V_s^{1/2} g_s\|_2$ has the same conditional distribution as $\sigma_s \|w_s\|_2 |z_s|$, where $z_s \sim N(0, 1)$. Conditional Lyapunov gives

$$\sigma_s^3 = \{\mathbb{E}(u_s^2 \mid \mathcal{H}_{s-1})\}^{3/2} \leq \mathbb{E}(|u_s|^3 \mid \mathcal{H}_{s-1}).$$

Consequently,

$$\begin{aligned}\sum_s \mathbb{E} \|V_s^{1/2} g_s\|_2^3 &= \mathbb{E} |z_1|^3 \sum_s \mathbb{E}(\sigma_s^3 \|w_s\|_2^3) \\ &\leq CB_{NT}K_{NT}T^{-1/2}.\end{aligned}$$

Adding the last two displays proves the bound for $\beta_{2,2}$.

It remains to identify the quadratic-variation error. We have

$$\sum_s V_s = \sum_s w_s w_s' \mathbb{E}(u_s^2 \mid \mathcal{H}_{s-1}), \quad \Xi = \sum_s w_s w_s',$$

and hence

$$\Omega := \sum_s V_s - \Xi = \sum_s w_s w_s' \{\mathbb{E}(u_s^2 \mid \mathcal{H}_{s-1}) - 1\}.$$

The primitive quadratic-variation condition above therefore implies $\mathbb{E} \|\Omega\| \leq \varrho_{NT}^{\text{qv}}$ after integrating over $\mathcal{G}$.

Now apply the second-order martingale Yurinskii inequality with $p = 2$. Its dimension factor satisfies

$$\phi_2(d)^2 = 2d = 4\kappa \leq 4K_{NT}.$$

It constructs, on an enlarged space, a vector $\zeta^\star \mid \mathcal{G} \sim N(0, \Xi)$ satisfying, for every $\rho > 0$,

$$\mathbb{P}(\|\zeta - \zeta^\star\|_2 > \rho) \leq 24 \left(\frac{4K_{NT}\beta_{2,2}}{\rho^3} \right)^{1/3} + 17 \left(\frac{4K_{NT}\varrho_{NT}^{\text{qv}}}{\rho^2} \right)^{1/3}.$$

Finally, set $\rho = \rho_{NT} = \ell_{NT} r_{NT}^{\text{cpl}}$. Since r_{NT}^{cpl} dominates both terms in its definition,

$$24 \frac{(4K_{NT}\beta_{2,2})^{1/3}}{\rho_{NT}} \leq 24\ell_{NT}^{-1}, \quad 17 \left(\frac{4K_{NT}\varrho_{NT}^{\text{qv}}}{\rho_{NT}^2} \right)^{1/3} \leq 17\ell_{NT}^{-2/3}.$$

Thus the two failure probabilities vanish, while $\rho_{NT} = \ell_{NT} r_{NT}^{\text{cpl}} \rightarrow 0$ by construction. This proves the claim. The approximating vector is Gaussian conditional on $\mathcal{G}$, while the coupling-failure probability is unconditional, exactly as in the cited martingale coupling theorem. $\square$

C.6 Proof of FDR control

For the null-sign argument only, define the auxiliary quantities

$$\alpha_{NT} = e^{-\bar{\epsilon}_{NT}}(NT)^{-1}, \quad L_{NT} = \ln \left[\frac{4\{\log_2(2K_{NT}) + 1\}}{\alpha_{NT}} \right], \quad a_{NT}^\circ = \frac{m_{NT}}{32L_{NT}}.$$

Under Theorem 4.4, $L_{NT} = O\{\ln(NT)\}$; hence Assumption C.3 implies $a_{NT}^\circ \rightarrow \infty$. These are proof devices, not additional regularity conditions.

Proposition C.12 (Finite-sample FDR bound). *Under the assumptions of Theorem 4.4,*

$$\begin{aligned} \text{FDR} \leq & q \left(1 + \frac{4}{\sqrt{a_{NT}^\circ}} + 4\gamma_{NT} \right) + \frac{m_{NT} + e_{NT}}{r_{NT}} \\ & + \eta_{NT} + \varsigma_{NT} + 6(NT)^{-1} + 2c_{NT} + \delta_{NT}^{\text{scr}} + \mathbb{P}(\mathcal{B}_G^c) \\ & + \mathbb{P} \left(\{|\hat{\mathcal{S}}| < r_{NT}\} \cup \{0 < G < v_{NT}\Delta_{NT}\} \right) + o(1). \end{aligned}$$

The proof has five steps. Steps 1–2 bound the actual FDP by a Gaussian null-sign ratio and a threshold-band count. Step 3 transfers the null-sign event using Lemmas C.8–C.9 and invokes Assumption C.3 for the band count. Step 4 treats populated and sparse null tails separately; Step 5 combines the bounds and restores all failure probabilities.

Proof. Work first conditional on the screening event, $\mathcal{G}$, and $\mathcal{B}_G$, under the conditionally separated-fold law. The screening, design, and fold-separation errors are restored in Step 5.

Step 1. For the actual statistics define

$$\begin{aligned} R(G) &= \sum_{j \in \tilde{\mathcal{S}}} \mathbf{1}\{W_j \geq G\}, & N^-(G) &= \sum_{j \in \tilde{\mathcal{S}}} \mathbf{1}\{W_j \leq -G\}, \\ V^{a,+}(G) &= \sum_{j \in \tilde{\mathcal{S}} \cap \mathcal{S}_0^c} \mathbf{1}\{W_j \geq G\}, & V^{a,-}(G) &= \sum_{j \in \tilde{\mathcal{S}} \cap \mathcal{S}_0^c} \mathbf{1}\{W_j \leq -G\}. \end{aligned}$$

Here $R(G) = |\hat{\mathcal{S}}|$. If the selected set is empty, the FDP is zero. Otherwise, the definition of the knockoff+ threshold gives

$$\frac{1 + N^-(G)}{R(G)} \leq q.$$

Consequently,

$$\text{FDP} = \frac{V^{a,+}(G)}{R(G)} \leq q \frac{V^{a,+}(G)}{1 + N^-(G)} \leq q \frac{V^{a,+}(G)}{1 + V^{a,-}(G)}. \quad (\text{C.25})$$

The last inequality uses $N^-(G) \geq V^{a,-}(G)$; negative nonnull statistics can only enlarge the denominator.

Step 2. Recall that $V^{*,+}(t)$ and $V^{*,-}(t)$ count, respectively, the retained nulls satisfying $W_j^* \geq t$ and $W_j^* \leq -t$, and $M^*(t) = V^{*,+}(t) + V^{*,-}(t)$. Lemma C.11 constructs the Gaussian

score jointly with the actual score. The two Lasso problems have the same design, signal, and missed-signal drift D ; they differ only by replacing ζ with ζ^* . Hence

$$\eta - \eta^* = T^{-1/2}(\zeta - \zeta^*), \quad \|\eta - \eta^*\|_2 \leq T^{-1/2} \rho_{NT}$$

on the coupling event. Applying the stability bound in Lemma C.2(i) now gives

$$\max_j |W_j - W_j^*| \leq \frac{\sqrt{2}}{\sigma_{\min}(\Psi)} \|\eta - \eta^*\|_2 \leq \frac{\sqrt{2} \rho_{NT}}{\sigma_{\min}(\Psi) \sqrt{T}} = \Delta_{NT}.$$

Suppose also that $G \geq v_{NT} \Delta_{NT}$. Since $v_{NT} \rightarrow \infty$, we may take $v_{NT} > 1$, and put $t = G + \Delta_{NT}$. For every retained null coordinate,

$$\{W_j \geq G\} \subseteq \{W_j^* \geq t\} \cup \{|W_j^*| - t \leq 2\Delta_{NT}\},$$

because a statistic selected under the actual score either remains above t under the Gaussian score or lies within the intervening $2\Delta_{NT}$ band. In the opposite direction,

$$\{W_j^* \leq -t\} \subseteq \{W_j \leq -G\}.$$

After summing these inclusions over the null coordinates,

$$V^{a,+}(G) \leq V^{*,+}(t) + \tilde{N}_{2\Delta}^*(t), \quad V^{a,-}(G) \geq V^{*,-}(t).$$

Substitution into (C.25) yields

$$\text{FDP} \leq q \frac{V^{*,+}(t)}{1 + V^{*,-}(t)} + q \frac{\tilde{N}_{2\Delta}^*(t)}{1 + V^{*,-}(t)}. \quad (\text{C.26})$$

This display isolates the two effects: null-sign imbalance under the Gaussian reference law and threshold crossings caused by coupling.

Step 3. The two events introduced here control the two terms in (C.26): $\mathcal{C}_{\alpha_{NT}}$ controls Gaussian null-sign imbalance, whereas $\mathcal{H}_{NT}$ controls threshold crossings caused by the coupling. Let $\mathcal{C}_{\alpha_{NT}}$ be the simultaneous null-sign event in Lemma C.8. Under the ideal law, conditional on the null absolute statistics, the nonnull statistics, and $\mathcal{G}$, the null signs are independent Rademacher variables. If the nonzero null statistics are ordered by decreasing absolute value and Z_m is the sum of the first m signs, then, at any threshold t with $m = M^*(t)$,

$$Z_m = V^{*,+}(t) - V^{*,-}(t).$$

Lemma C.8 controls Z_m simultaneously for every m and hence for every t . It therefore bounds the conditional failure probability of $\mathcal{C}_{\alpha_{NT}}$ by α_{NT} ; integrating out the null absolute statistics and the nonnull statistics gives

$$P_D^\dagger(\mathcal{C}_{\alpha_{NT}}^c \mid \mathcal{G}) \leq \alpha_{NT}.$$

To transfer this event to the law of $W^\star$, apply Lemma C.9 to the score-measurable bounded functional $f = \mathbf{1}\{\mathcal{C}_{\alpha_{NT}}^c\}$, taking $u = \ln(NT)$ and $\alpha_{NT} = e^{-\bar{\epsilon}_{NT}}(NT)^{-1}$. This gives

$$\begin{aligned} P_D^\star(\mathcal{C}_{\alpha_{NT}}^c \mid \mathcal{G}) &\leq e^{\bar{\epsilon}_{NT}} \alpha_{NT} + 2\{1 + e^{\epsilon_{NT}^{\text{mean}}}\}(NT)^{-1} \\ &\leq 6(NT)^{-1} + o(1). \end{aligned}$$

Here the definition of α_{NT} makes the first term exactly $e^{\bar{\epsilon}_{NT}} \alpha_{NT} = (NT)^{-1}$; the remaining truncation term is of the same order because $\epsilon_{NT}^{\text{mean}} = o(1)$. Let $\mathcal{H}_{NT}$ denote the band-count event in Assumption C.3, on which both displayed suprema in that assumption satisfy their stated bounds. Conditional on $\mathcal{G}$,

$$P_D^\star(\mathcal{H}_{NT}^c \mid \mathcal{G}) \leq \varsigma_{NT}.$$

Both events hold uniformly over all eligible t , so they remain valid at the random value $t = G + \Delta_{NT}$.

Step 4. Relative sign and band-count bounds are useful only when the Gaussian null tail contains sufficiently many observations. We therefore use relative bounds when $M^\star(t)$ is large and count false discoveries directly when it is small. First suppose $M^\star(t) \geq m_{NT}$. The total number $\bar{M}$ of nonzero retained null statistics is at most K_{NT} , so the quantity $L_{\alpha_{NT}}$ in Lemma C.8 is bounded above by

$$\ln \left[\frac{4\{\log_2(2K_{NT}) + 1\}}{\alpha_{NT}} \right].$$

The definitions of L_{NT} and a_{NT}° thus ensure $M^\star(t) \geq m_{NT} = 32a_{NT}^\circ L_{NT}$, which is precisely the sample-size requirement in that lemma. On $\mathcal{C}_{\alpha_{NT}}$, it gives

$$\frac{V^{\star,+}(t)}{1 + V^{\star,-}(t)} \leq 1 + \frac{4}{\sqrt{a_{NT}^\circ}}, \quad 1 + V^{\star,-}(t) \geq \frac{M^\star(t)}{4}.$$

On $\mathcal{H}_{NT}$, the relative band-count bound therefore gives

$$\frac{\tilde{N}_{2\Delta}^\star(t)}{1 + V^{\star,-}(t)} \leq 4 \frac{\tilde{N}_{2\Delta}^\star(t)}{M^\star(t)} \leq 4\gamma_{NT}.$$

Equation (C.26) is consequently bounded by

$$q \left(1 + \frac{4}{\sqrt{a_{NT}^\circ}} + 4\gamma_{NT} \right).$$

Now suppose $M^\star(t) < m_{NT}$. In this sparse-tail case the ratio bound may be unstable, so we instead count false discoveries directly. The inclusions in Step 2 and the absolute band cap on $\mathcal{H}_{NT}$ give

$$V^{\mathbf{a},+}(G) \leq V^{\star,+}(t) + \tilde{N}_{2\Delta}^\star(t) \leq m_{NT} + e_{NT}.$$

On $|\widehat{\mathcal{S}}| \geq r_{NT}$ this implies

$$\text{FDP} \leq \frac{m_{NT} + e_{NT}}{r_{NT}}.$$

Step 5. On the intersection of the events used above, one of the two bounds in Step 4 always applies. Since $0 \leq \text{FDP} \leq 1$, the expectation is bounded by the sum of those two deterministic bounds plus the probability of the complement of the good event. The Gaussian reference events have the correct marginal law under the joint coupling, so their failure probabilities are exactly those bounded in Step 3. Adding the unconditional coupling failure η_{NT} , the screening failure δ_{NT}^{scr} , the design failure $\mathbb{P}(\mathcal{B}_G^c)$, the fold-separation cost $2c_{NT}$, and the discovery and threshold-localization failure gives the displayed finite-sample inequality. This proves the proposition. Theorem 4.4 follows because every remainder in that inequality vanishes. $\square$

C.7 Proof of the power theorem

The power argument separates score concentration from signal counting. The Gaussian and martingale cases supply different concentration bounds, but use the same Lasso argument on the retained active set. The missed fraction π_{NT} is added only at the final counting step.

Assumption C.4 (Martingale innovation-score concentration). *At the tuning scale*

$$\lambda_{NT}^M := \{T^{-1}K_{NT} + T^{-1/2}\}m,$$

the stochastic score satisfies $\|T^{-1}\mathbb{U}'\varepsilon^\dagger\|_\infty = O_p(\lambda_{NT}^M)$.

This condition concerns only the maximum stochastic score over the 2κ augmented coordinates. It is not used for FDR control, and the next lemma supplies primitive tail and leverage conditions under which it holds.

Lemma C.13 (Primitive innovation-score concentration). *Suppose Assumption C.1 and the primitive weight conditions in Appendix C.5 hold. Assume either that the innovations are conditionally sub-Weibull of order $\gamma > 0$ with uniformly bounded conditional Orlicz norms and*

$$B_{NT}^2(\ln \kappa)^{(2+\gamma)/\gamma} = o(T),$$

or that, for some $q_0 > 2$, $\text{ess sup}_s \mathbb{E}(|u_s|^{q_0} \mid \mathcal{H}_{s-1}) \leq C$ and

$$\kappa B_{NT}^{q_0-2} = o\{T^{q_0/2-1}(\ln \kappa)^{q_0/2}\}.$$

Then

$$\|T^{-1}\mathbb{U}'\varepsilon^\dagger\|_\infty = O_p\left(\sqrt{\frac{\ln \kappa}{T}}\right).$$

Consequently, if $\sqrt{\ln \kappa} = O(m)$, Appendix Assumption C.4 holds.

We first express each score coordinate as a weighted innovation sum. Bounds on the weights then allow a tail inequality and a union bound over the augmented coordinates, with separate sub-Weibull and finite-moment cases.

Proof. Write $\mathbb{U}_j$ for the j th column of the augmented original-knockoff design, and let h_{tj} denote entry (t, j) of $H = \widehat{\Sigma}_\varepsilon^{-1/2} \mathbb{U}$. Since $\varepsilon^\dagger = \widehat{\Sigma}_\varepsilon^{-1/2} \varepsilon$ and $\varepsilon = \Theta u$, lower triangularity of Θ gives

$$\begin{aligned} T^{-1} \mathbb{U}'_j \varepsilon^\dagger &= T^{-1} \sum_{t=1}^T h_{tj} \sum_{s \leq t} \Theta_{ts} u_s = \sum_{s=1}^T a_{sj} u_s, \\ a_{sj} &:= T^{-1} \sum_{t \geq s} \Theta_{ts} h_{tj} = T^{-1/2} w_{sj}. \end{aligned}$$

Consequently, the primitive coordinatewise energy bound gives

$$\sum_s a_{sj}^2 = T^{-1} \sum_s w_{sj}^2 \leq C/T.$$

Moreover, $\max_t \|h_t\|_2 \leq B_{NT}$ and the column-summability of Θ imply

$$\max_s |a_{sj}| \leq T^{-1} \max_s \sum_{t \geq s} |\Theta_{ts}| \max_t |h_{tj}| \leq C_\Theta B_{NT}/T.$$

There are 2κ such coordinates, which has the same logarithmic order as κ .

Suppose first that the innovations are uniformly sub-Weibull of order γ . A weighted martingale Bernstein inequality gives, uniformly in $j \leq 2\kappa$,

$$\mathbb{P} \left(\left| \sum_s a_{sj} u_s \right| > x \mid \mathcal{G} \right) \leq 2 \exp \left[-c \min \left\{ \frac{x^2}{\sum_s a_{sj}^2}, \left(\frac{x}{\max_s |a_{sj}|} \right)^{\gamma/(1+\gamma)} \right\} \right].$$

At $x = C_0 \sqrt{\ln \kappa / T}$, the two exponents are bounded below by constant multiples of

$$C_0^2 \ln \kappa \quad \text{and} \quad \left(\frac{C_0 \sqrt{T \ln \kappa}}{B_{NT}} \right)^{\gamma/(1+\gamma)},$$

respectively. The sub-Weibull rate in the lemma implies that the second quantity is larger than $\ln \kappa$ by a diverging factor. Choosing C_0 large and taking a union bound over the 2κ coordinates therefore makes the resulting probability tend to zero.

If instead the innovations have uniformly bounded q_0 th moments, the martingale Fuk-Nagaev inequality and $\sum_s |a_{sj}|^{q_0} \leq (\max_s |a_{sj}|)^{q_0-2} \sum_s a_{sj}^2$ give, at the same value of x ,

$$\begin{aligned} \mathbb{P} \left(\left| \sum_s a_{sj} u_s \right| > x \mid \mathcal{G} \right) &\leq C x^{-q_0} \sum_s |a_{sj}|^{q_0} + 2 \exp \left(-\frac{c x^2}{\sum_s a_{sj}^2} \right) \\ &\leq C B_{NT}^{q_0-2} T^{1-q_0/2} (\ln \kappa)^{-q_0/2} + 2\kappa^{-c C_0^2}. \end{aligned}$$

After the union bound, the first term is

$$C \kappa B_{NT}^{q_0-2} T^{1-q_0/2} (\ln \kappa)^{-q_0/2} = o(1)$$

by the finite-moment rate in the lemma; the second also vanishes when C_0 is chosen suffi-

ciently large. Thus either branch yields

$$\|T^{-1}\mathbb{U}'\varepsilon^\dagger\|_\infty = O_p\left(\sqrt{\frac{\ln \kappa}{T}}\right).$$

The final assertion follows from $\sqrt{\ln \kappa} = O(m)$ and the definition of λ_{NT}^M . $\square$

Lemma C.14 (Power from score concentration). *Suppose Assumptions 1, 3, and 4 hold for a tuning scale λ_{NT} , and*

$$\|T^{-1}\mathbb{U}'\varepsilon^\dagger\|_\infty = O_p(\lambda_{NT}).$$

With $\lambda = C_2\lambda_{NT}$ for a sufficiently large constant C_2 ,

$$\frac{|\hat{\mathcal{S}} \cap \mathcal{S}_0|}{s \vee 1} \geq 1 - \pi_{NT} - C(\varphi + 1)\varkappa_T^{-1} - o_p(1), \quad \varphi = \frac{1 + \sqrt{5}}{2}.$$

The proof combines the stochastic and omitted-signal scores to obtain Lasso error bounds. These bounds control missed retained signals; the screened-out signals are counted separately, so sure screening is not needed.

Proof. The proof is carried out on the retained active set $\mathcal{S}_R = \mathcal{S}_0 \cap \tilde{\mathcal{S}}$. The assumed stochastic-score bound and $\lambda = C_2\lambda_{NT}$ give $\|T^{-1}\mathbb{U}'\varepsilon^\dagger\|_\infty = O_p(\lambda)$.

For completeness, (C.21) writes the other part of the Lasso score as

$$\|T^{-1}\mathbb{U}'g\|_\infty \leq (1 + C_1)\tau_{U,NT} + \tau_{V,NT}.$$

The lower-level bound (C.22) gives primitive sufficient rates for this quantity, while Assumption 4 records their high-level consequence $\|T^{-1}\mathbb{U}'g\|_\infty = O_p(\lambda)$. Consequently,

$$\|T^{-1}\mathbb{U}'(\varepsilon^\dagger + g)\|_\infty = O_p(\lambda).$$

Together with the restricted-eigenvalue condition in Assumption 4, this is the score event required by the standard Lasso basic inequality, which yields

$$\|\hat{b} - b_R\|_1 \leq C\kappa\lambda, \quad \|\hat{b} - b_R\|_2 \leq C\sqrt{\kappa}\lambda,$$

where $b_R = (\beta'_{\mathcal{S}_R}, 0)'$.

We spell out how these bounds enter the knockoff counting argument. Let

$$\mathcal{S}_R^{\text{str}} = \{j \in \mathcal{S}_R : |\beta_j| \gg \sqrt{\kappa}\lambda\}.$$

Assumption 4 gives $|\mathcal{S}_R^{\text{str}}| \geq c\kappa$ and $\min_{j \in \mathcal{S}_R} |\beta_j| \geq \varkappa_T\lambda$. For a retained active pair to be missed by the LCD threshold, either its original coefficient must be attenuated or its knockoff coefficient must be inflated by an amount comparable with its signal; otherwise its statistic is positive and exceeds the data-dependent threshold. Summing these required pairwise distortions over the missed retained signals and using the displayed ℓ_1 and ℓ_2 error bounds limits their number. The nondegenerate knockoff conditional variance prevents an original–knockoff pair from collapsing, while the retained strong subset keeps the threshold from being driven by weak pairs. These are exactly the deterministic ingredients in the

LCD counting argument of Fan et al. (2020b); applying that argument to $\mathcal{S}_R$ rather than to the full active set gives

$$\frac{|\mathcal{S}_R \setminus \widehat{\mathcal{S}}|}{s \vee 1} \leq C(\varphi + 1)\varkappa_T^{-1} + o_p(1).$$

The omitted set is not used in this counting step. It is added only at the last inclusion: since $\mathcal{S}_0 \setminus \widehat{\mathcal{S}} \subseteq \mathcal{A} \cup (\mathcal{S}_R \setminus \widehat{\mathcal{S}})$, we obtain

$$\frac{|\widehat{\mathcal{S}} \cap \mathcal{S}_0|}{s \vee 1} \geq 1 - \pi_{NT} - C(\varphi + 1)\varkappa_T^{-1} - o_p(1).$$

This proves the claimed probability bound. $\square$

Proof of Theorem 4.5. On $\mathcal{B}_G$, the diagonal entries of Ξ are uniformly bounded. Lemma 4.2 and the conditional Gaussian maximal inequality give

$$\|T^{-1}\mathbb{U}'\varepsilon^\dagger\|_\infty = T^{-1/2}\|\zeta\|_\infty = O_p\left(\sqrt{\frac{\ln(2\kappa \vee 2)}{T}}\right) = O_p(\lambda_{NT}).$$

Lemma C.14 gives the probability bound. Since the discovered fraction and π_{NT} both lie in $[0, 1]$, taking expectations gives the asserted power bound. The screening, design, and fold-separation failure probabilities vanish and are absorbed into the remainder. $\square$

Theorem C.15 (Power with causal martingale innovations). *Suppose Assumptions 1, 2, 3, and 4 hold, together with Appendix Assumptions C.1 and C.4. In Assumption 4, take $\lambda_{NT} = \lambda_{NT}^M$ and use $\lambda = C_2\lambda_{NT}^M$. Then*

$$\frac{|\widehat{\mathcal{S}} \cap \mathcal{S}_0|}{s \vee 1} \geq 1 - \pi_{NT} - C(\varphi + 1)\varkappa_T^{-1} - o_p(1), \quad \text{Power} \geq 1 - \mathbb{E}(\pi_{NT}) - o(1).$$

Proof. Assumption C.4 supplies the stochastic-score bound in Lemma C.14. Applying that lemma and then taking expectations proves the result. Lemma C.13 gives lower-level sufficient conditions for the score assumption. $\square$